

DE RHAM COHOMOLOGY

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1. Introduction

In this chapter we start with a discussion of the de Rham complex of a morphism of schemes and we end with a proof that de Rham cohomology defines a Weil cohomology theory when the base field has characteristic zero.

2. The de Rham complex

Let $p : X \rightarrow S$ be a morphism of schemes. There is a complex

$$\Omega_{X/S}^\bullet = \mathcal{O}_{X/S} \rightarrow \Omega_{X/S}^1 \rightarrow \Omega_{X/S}^2 \rightarrow \dots$$

of $p^{-1}\mathcal{O}_S$ -modules with $\Omega_{X/S}^i = \wedge^i(\Omega_{X/S})$ placed in degree i and differential determined by the rule $d(g_0 dg_1 \wedge \dots \wedge dg_p) = dg_0 \wedge dg_1 \wedge \dots \wedge dg_p$ on local sections. See Modules, Section ??.

Given a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\quad f \quad} & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

of schemes, there are canonical maps of complexes $f^{-1}\Omega_{X/S}^\bullet \rightarrow \Omega_{X'/S'}^\bullet$ and $\Omega_{X/S}^\bullet \rightarrow f_*\Omega_{X'/S'}^\bullet$. See Modules, Section ??. Linearizing, for every p we obtain a linear map $f^*\Omega_{X/S}^p \rightarrow \Omega_{X'/S'}^p$.

In particular, if $f : Y \rightarrow X$ is a morphism of schemes over a base scheme S , then there is a map of complexes

$$\Omega_{X/S}^\bullet \longrightarrow f_*\Omega_{Y/S}^\bullet$$

Linearizing, we see that for every $p \geq 0$ we obtain a canonical map

$$\Omega_{X/S}^p \otimes_{\mathcal{O}_X} f_*\mathcal{O}_Y \longrightarrow f_*\Omega_{Y/S}^p$$

Lemma 2.1. *Let*

$$\begin{array}{ccc} X' & \xrightarrow{\quad f \quad} & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

be a cartesian diagram of schemes. Then the maps discussed above induce isomorphisms $f^\Omega_{X/S}^p \rightarrow \Omega_{X'/S'}^p$.*

Proof. Combine Morphisms, Lemma ?? with the fact that formation of exterior power commutes with base change. $\square$

Lemma 2.2. *Consider a commutative diagram of schemes*

$$\begin{array}{ccc} X' & \xrightarrow{\quad f \quad} & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

If $X' \rightarrow X$ and $S' \rightarrow S$ are étale, then the maps discussed above induce isomorphisms $f^\Omega_{X/S}^p \rightarrow \Omega_{X'/S'}^p$.*

Proof. We have $\Omega_{S'/S} = 0$ and $\Omega_{X'/X} = 0$, see for example Morphisms, Lemma ??. Then by the short exact sequences of Morphisms, Lemmas ?? and ?? we see that $\Omega_{X'/S'} = \Omega_{X'/S} = f^*\Omega_{X/S}$. Taking exterior powers we conclude. $\square$

3. de Rham cohomology

Let $p : X \rightarrow S$ be a morphism of schemes. We define the *de Rham cohomology of X over S* to be the cohomology groups

$$H_{dR}^i(X/S) = H^i(R\Gamma(X, \Omega_{X/S}^\bullet))$$

Since $\Omega_{X/S}^\bullet$ is a complex of $p^{-1}\mathcal{O}_S$ -modules, these cohomology groups are naturally modules over $H^0(S, \mathcal{O}_S)$.

Given a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\quad f \quad} & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

of schemes, using the canonical maps of Section 2 we obtain pullback maps

$$f^* : R\Gamma(X, \Omega_{X/S}^\bullet) \longrightarrow R\Gamma(X', \Omega_{X'/S'}^\bullet)$$

and

$$f^* : H_{dR}^i(X/S) \longrightarrow H_{dR}^i(X'/S')$$

These pullbacks satisfy an obvious composition law. In particular, if we work over a fixed base scheme S , then de Rham cohomology is a contravariant functor on the category of schemes over S .

Lemma 3.1. *Let $X \rightarrow S$ be a morphism of affine schemes given by the ring map $R \rightarrow A$. Then $R\Gamma(X, \Omega_{X/S}^\bullet) = \Omega_{A/R}^\bullet$ in $D(R)$ and $H_{dR}^i(X/S) = H^i(\Omega_{A/R}^\bullet)$.*

Proof. This follows from Cohomology of Schemes, Lemma ?? and Leray's acyclicity lemma (Derived Categories, Lemma ??). $\square$

Lemma 3.2. *Let $p : X \rightarrow S$ be a morphism of schemes. If p is quasi-compact and quasi-separated, then $Rp_*\Omega_{X/S}^\bullet$ is an object of $D_{QCoh}(\mathcal{O}_S)$.*

Proof. There is a spectral sequence with first page $E_1^{a,b} = R^b p_* \Omega_{X/S}^a$ converging to the cohomology of $Rp_*\Omega_{X/S}^\bullet$ (see Derived Categories, Lemma ??). Hence by Homology, Lemma ?? it suffices to show that $R^b p_* \Omega_{X/S}^a$ is quasi-coherent. This follows from Cohomology of Schemes, Lemma ??). $\square$

Lemma 3.3. *Let $p : X \rightarrow S$ be a proper morphism of schemes with S locally Noetherian. Then $Rp_*\Omega_{X/S}^\bullet$ is an object of $D_{Coh}(\mathcal{O}_S)$.*

Proof. In this case by Morphisms, Lemma ?? the modules $\Omega_{X/S}^i$ are coherent. Hence we can use exactly the same argument as in the proof of Lemma 3.2 using Cohomology of Schemes, Proposition ??). $\square$

Lemma 3.4. *Let A be a Noetherian ring. Let X be a proper scheme over $S = \text{Spec}(A)$. Then $H_{dR}^i(X/S)$ is a finite A -module for all i .*

Proof. This is a special case of Lemma 3.3. $\square$

Lemma 3.5. *Let $f : X \rightarrow S$ be a proper smooth morphism of schemes. Then $Rf_*\Omega_{X/S}^p$, $p \geq 0$ and $Rf_*\Omega_{X/S}^\bullet$ are perfect objects of $D(\mathcal{O}_S)$ whose formation commutes with arbitrary change of base.*

Proof. Since f is smooth the modules $\Omega_{X/S}^p$ are finite locally free $\mathcal{O}_X$ -modules, see Morphisms, Lemma ??). Their formation commutes with arbitrary change of base by Lemma 2.1. Hence $Rf_*\Omega_{X/S}^p$ is a perfect object of $D(\mathcal{O}_S)$ whose formation commutes with arbitrary base change, see Derived Categories of Schemes, Lemma ??). This proves the first assertion of the lemma.

To prove that $Rf_*\Omega_{X/S}^\bullet$ is perfect on S we may work locally on S . Thus we may assume S is quasi-compact. This means we may assume that $\Omega_{X/S}^n$ is zero for n large enough. For every $p \geq 0$ we claim that $Rf_*\sigma_{\geq p}\Omega_{X/S}^\bullet$ is a perfect object of $D(\mathcal{O}_S)$ whose formation commutes with arbitrary change of base. By the above we see that this is true for $p \gg 0$. Suppose the claim holds for p and consider the distinguished triangle

$$\sigma_{\geq p}\Omega_{X/S}^\bullet \rightarrow \sigma_{\geq p-1}\Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^{p-1}[-(p-1)] \rightarrow (\sigma_{\geq p}\Omega_{X/S}^\bullet)[1]$$

in $D(f^{-1}\mathcal{O}_S)$. Applying the exact functor Rf_* we obtain a distinguished triangle in $D(\mathcal{O}_S)$. Since we have the 2-out-of-3 property for being perfect (Cohomology, Lemma ??) we conclude $Rf_*\sigma_{\geq p-1}\Omega_{X/S}^\bullet$ is a perfect object of $D(\mathcal{O}_S)$. Similarly for the commutation with arbitrary base change. $\square$

4. Cup product

Consider the maps $\Omega_{X/S}^p \times \Omega_{X/S}^q \rightarrow \Omega_{X/S}^{p+q}$ given by $(\omega, \eta) \mapsto \omega \wedge \eta$. Using the formula for d given in Section 2 and the Leibniz rule for $d : \mathcal{O}_X \rightarrow \Omega_{X/S}$ we see that $d(\omega \wedge \eta) = d(\omega) \wedge \eta + (-1)^{\deg(\omega)} \omega \wedge d(\eta)$. This means that $\wedge$ defines a morphism

$$(4.0.1) \quad \wedge : \text{Tot}(\Omega_{X/S}^\bullet \otimes_{p^{-1}\mathcal{O}_S} \Omega_{X/S}^\bullet) \longrightarrow \Omega_{X/S}^\bullet$$

of complexes of $p^{-1}\mathcal{O}_S$ -modules.

Combining the cup product of Cohomology, Section ?? with (4.0.1) we find a $H^0(S, \mathcal{O}_S)$ -bilinear cup product map

$$\cup : H_{dR}^i(X/S) \times H_{dR}^j(X/S) \longrightarrow H_{dR}^{i+j}(X/S)$$

For example, if $\omega \in \Gamma(X, \Omega_{X/S}^i)$ and $\eta \in \Gamma(X, \Omega_{X/S}^j)$ are closed, then the cup product of the de Rham cohomology classes of ω and η is the de Rham cohomology class of $\omega \wedge \eta$, see discussion in Cohomology, Section ??.

Given a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\quad f \quad} & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

of schemes, the pullback maps $f^* : R\Gamma(X, \Omega_{X/S}^\bullet) \rightarrow R\Gamma(X', \Omega_{X'/S'}^\bullet)$ and $f^* : H_{dR}^i(X/S) \rightarrow H_{dR}^i(X'/S')$ are compatible with the cup product defined above.

Lemma 4.1. *Let $p : X \rightarrow S$ be a morphism of schemes. The cup product on $H_{dR}^*(X/S)$ is associative and graded commutative.*

Proof. This follows from Cohomology, Lemmas ?? and ?? and the fact that $\wedge$ is associative and graded commutative. $\square$

Remark 4.2. Let $p : X \rightarrow S$ be a morphism of schemes. Then we can think of $\Omega_{X/S}^\bullet$ as a sheaf of differential graded $p^{-1}\mathcal{O}_S$ -algebras, see Differential Graded

Sheaves, Definition ???. In particular, the discussion in Differential Graded Sheaves, Section ?? applies. For example, this means that for any commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ p \downarrow & & \downarrow q \\ S & \xrightarrow{h} & T \end{array}$$

of schemes there is a canonical relative cup product

$$\mu : Rf_*\Omega_{X/S}^\bullet \otimes_{q^{-1}\mathcal{O}_T}^{\mathbf{L}} Rf_*\Omega_{X/S}^\bullet \longrightarrow Rf_*\Omega_{X/S}^\bullet$$

in $D(Y, q^{-1}\mathcal{O}_T)$ which is associative and which on cohomology reproduces the cup product discussed above.

Remark 4.3. Let $f : X \rightarrow S$ be a morphism of schemes. Let $\xi \in H_{dR}^n(X/S)$. According to the discussion Differential Graded Sheaves, Section ?? there exists a canonical morphism

$$\xi' : \Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^\bullet[n]$$

in $D(f^{-1}\mathcal{O}_S)$ uniquely characterized by (1) and (2) of the following list of properties:

- (1) ξ' can be lifted to a map in the derived category of right differential graded $\Omega_{X/S}^\bullet$ -modules, and
- (2) $\xi'(1) = \xi$ in $H^0(X, \Omega_{X/S}^\bullet[n]) = H_{dR}^n(X/S)$,
- (3) the map ξ' sends $\eta \in H_{dR}^m(X/S)$ to $\xi \cup \eta$ in $H_{dR}^{n+m}(X/S)$,
- (4) the construction of ξ' commutes with restrictions to opens: for $U \subset X$ open the restriction $\xi'|_U$ is the map corresponding to the image $\xi|_U \in H_{dR}^n(U/S)$,
- (5) for any diagram as in Remark 4.2 we obtain a commutative diagram

$$\begin{array}{ccc} Rf_*\Omega_{X/S}^\bullet \otimes_{q^{-1}\mathcal{O}_T}^{\mathbf{L}} Rf_*\Omega_{X/S}^\bullet & \xrightarrow{\mu} & Rf_*\Omega_{X/S}^\bullet \\ \xi' \otimes \text{id} \downarrow & & \downarrow \xi' \\ Rf_*\Omega_{X/S}^\bullet[n] \otimes_{q^{-1}\mathcal{O}_T}^{\mathbf{L}} Rf_*\Omega_{X/S}^\bullet & \xrightarrow{\mu} & Rf_*\Omega_{X/S}^\bullet[n] \end{array}$$

in $D(Y, q^{-1}\mathcal{O}_T)$.

5. Hodge cohomology

Let $p : X \rightarrow S$ be a morphism of schemes. We define the *Hodge cohomology* of X over S to be the cohomology groups

$$H_{Hodge}^n(X/S) = \bigoplus_{n=p+q} H^q(X, \Omega_{X/S}^p)$$

viewed as a graded $H^0(X, \mathcal{O}_X)$ -module. The wedge product of forms combined with the cup product of Cohomology, Section ?? defines a $H^0(X, \mathcal{O}_X)$ -bilinear cup product

$$\cup : H_{Hodge}^i(X/S) \times H_{Hodge}^j(X/S) \longrightarrow H_{Hodge}^{i+j}(X/S)$$

Of course if $\xi \in H^q(X, \Omega_{X/S}^p)$ and $\xi' \in H^{q'}(X, \Omega_{X/S}^{p'})$ then $\xi \cup \xi' \in H^{q+q'}(X, \Omega_{X/S}^{p+p'})$.

Lemma 5.1. *Let $p : X \rightarrow S$ be a morphism of schemes. The cup product on $H_{Hodge}^*(X/S)$ is associative and graded commutative.*

Proof. The proof is identical to the proof of Lemma 4.1. $\square$

Given a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\quad} & X \\ \downarrow & \searrow f & \downarrow \\ S' & \longrightarrow & S \end{array}$$

of schemes, there are pullback maps $f^* : H_{Hodge}^i(X/S) \rightarrow H_{Hodge}^i(X'/S')$ compatible with gradings and with the cup product defined above.

6. Two spectral sequences

Let $p : X \rightarrow S$ be a morphism of schemes. Since the category of $p^{-1}\mathcal{O}_S$ -modules on X has enough injectives there exist a Cartan-Eilenberg resolution for $\Omega_{X/S}^\bullet$. See Derived Categories, Lemma ???. Hence we can apply Derived Categories, Lemma ??? to get two spectral sequences both converging to the de Rham cohomology of X over S .

The first is customarily called *the Hodge-to-de Rham spectral sequence*. The first page of this spectral sequence has

$$E_1^{p,q} = H^q(X, \Omega_{X/S}^p)$$

which are the Hodge cohomology groups of X/S (whence the name). The differential d_1 on this page is given by the maps $d_1^{p,q} : H^q(X, \Omega_{X/S}^p) \rightarrow H^q(X, \Omega_{X/S}^{p+1})$ induced by the differential $d : \Omega_{X/S}^p \rightarrow \Omega_{X/S}^{p+1}$. Here is a picture

$$\begin{array}{ccccccc} H^2(X, \mathcal{O}_X) & \longrightarrow & H^2(X, \Omega_{X/S}^1) & \longrightarrow & H^2(X, \Omega_{X/S}^2) & \longrightarrow & H^2(X, \Omega_{X/S}^3) \\ & \searrow \text{dashed} & & \searrow \text{dashed} & & \searrow \text{dashed} & \\ H^1(X, \mathcal{O}_X) & \longrightarrow & H^1(X, \Omega_{X/S}^1) & \longrightarrow & H^1(X, \Omega_{X/S}^2) & \longrightarrow & H^1(X, \Omega_{X/S}^3) \\ & \searrow \text{dashed} & & \searrow \text{dashed} & & \searrow \text{dashed} & \\ H^0(X, \mathcal{O}_X) & \longrightarrow & H^0(X, \Omega_{X/S}^1) & \longrightarrow & H^0(X, \Omega_{X/S}^2) & \longrightarrow & H^0(X, \Omega_{X/S}^3) \end{array}$$

where we have drawn striped arrows to indicate the source and target of the differentials on the E_2 page and a dotted arrow for a differential on the E_3 page. Looking in degree 0 we conclude that

$$H_{dR}^0(X/S) = \text{Ker}(d : H^0(X, \mathcal{O}_X) \rightarrow H^0(X, \Omega_{X/S}^1))$$

Of course, this is also immediately clear from the fact that the de Rham complex starts in degree 0 with $\mathcal{O}_X \rightarrow \Omega_{X/S}^1$.

The second spectral sequence is usually called *the conjugate spectral sequence*. The second page of this spectral sequence has

$$E_2^{p,q} = H^p(X, H^q(\Omega_{X/S}^\bullet)) = H^p(X, \mathcal{H}^q)$$

where $\mathcal{H}^q = H^q(\Omega_{X/S}^\bullet)$ is the q th cohomology sheaf of the de Rham complex of X/S . The differentials on this page are given by $E_2^{p,q} \rightarrow E_2^{p+2,q-1}$. Here is a

picture

$$\begin{array}{ccccccc}
 H^0(X, \mathcal{H}^2) & & H^1(X, \mathcal{H}^2) & & H^2(X, \mathcal{H}^2) & & H^3(X, \mathcal{H}^2) \\
 & \searrow & \searrow & \searrow & \searrow & \searrow & \searrow \\
 H^0(X, \mathcal{H}^1) & & H^1(X, \mathcal{H}^1) & & H^2(X, \mathcal{H}^1) & & H^3(X, \mathcal{H}^1) \\
 & \searrow & \searrow & \searrow & \searrow & \searrow & \searrow \\
 H^0(X, \mathcal{H}^0) & & H^1(X, \mathcal{H}^0) & & H^2(X, \mathcal{H}^0) & & H^3(X, \mathcal{H}^0)
 \end{array}$$

Looking in degree 0 we conclude that

$$H_{dR}^0(X/S) = H^0(X, \mathcal{H}^0)$$

which is obvious if you think about it. In degree 1 we get an exact sequence

$$0 \rightarrow H^1(X, \mathcal{H}^0) \rightarrow H_{dR}^1(X/S) \rightarrow H^0(X, \mathcal{H}^1) \rightarrow H^2(X, \mathcal{H}^0) \rightarrow H_{dR}^2(X/S)$$

It turns out that if $X \rightarrow S$ is smooth and S lives in characteristic p , then the sheaves $\mathcal{H}^q$ are computable (in terms of a certain sheaves of differentials) and the conjugate spectral sequence is a valuable tool (insert future reference here).

7. The Hodge filtration

Let $X \rightarrow S$ be a morphism of schemes. The Hodge filtration on $H_{dR}^n(X/S)$ is the filtration induced by the Hodge-to-de Rham spectral sequence (Homology, Definition ??). To avoid misunderstanding, we explicitly define it as follows.

Definition 7.1. Let $X \rightarrow S$ be a morphism of schemes. The *Hodge filtration* on $H_{dR}^n(X/S)$ is the filtration with terms

$$F^p H_{dR}^n(X/S) = \text{Im} \left(H^n(X, \sigma_{\geq p} \Omega_{X/S}^\bullet) \rightarrow H_{dR}^n(X/S) \right)$$

where $\sigma_{\geq p} \Omega_{X/S}^\bullet$ is as in Homology, Section ??.

Of course $\sigma_{\geq p} \Omega_{X/S}^\bullet$ is a subcomplex of the relative de Rham complex and we obtain a filtration

$$\Omega_{X/S}^\bullet = \sigma_{\geq 0} \Omega_{X/S}^\bullet \supset \sigma_{\geq 1} \Omega_{X/S}^\bullet \supset \sigma_{\geq 2} \Omega_{X/S}^\bullet \supset \sigma_{\geq 3} \Omega_{X/S}^\bullet \supset \dots$$

of the relative de Rham complex with $\text{gr}^p(\Omega_{X/S}^\bullet) = \Omega_{X/S}^p[-p]$. The spectral sequence constructed in Cohomology, Lemma ?? for $\Omega_{X/S}^\bullet$ viewed as a filtered complex of sheaves is the same as the Hodge-to-de Rham spectral sequence constructed in Section 6 by Cohomology, Example ??. Further the wedge product (4.0.1) sends $\text{Tot}(\sigma_{\geq i} \Omega_{X/S}^\bullet \otimes_{p^{-1}\mathcal{O}_S} \sigma_{\geq j} \Omega_{X/S}^\bullet)$ into $\sigma_{\geq i+j} \Omega_{X/S}^\bullet$. Hence we get commutative diagrams

$$\begin{array}{ccc}
 H^n(X, \sigma_{\geq i} \Omega_{X/S}^\bullet) \times H^m(X, \sigma_{\geq j} \Omega_{X/S}^\bullet) & \longrightarrow & H^{n+m}(X, \sigma_{\geq i+j} \Omega_{X/S}^\bullet) \\
 \downarrow & & \downarrow \\
 H_{dR}^n(X/S) \times H_{dR}^m(X/S) & \xrightarrow{\cup} & H_{dR}^{n+m}(X/S)
 \end{array}$$

In particular we find that

$$F^i H_{dR}^n(X/S) \cup F^j H_{dR}^m(X/S) \subset F^{i+j} H_{dR}^{n+m}(X/S)$$

8. Künneth formula

An important feature of de Rham cohomology is that there is a Künneth formula.

Let $a : X \rightarrow S$ and $b : Y \rightarrow S$ be morphisms of schemes with the same target. Let $p : X \times_S Y \rightarrow X$ and $q : X \times_S Y \rightarrow Y$ be the projection morphisms and $f = a \circ p = b \circ q$. Here is a picture

$$\begin{array}{ccccc}
 & & X \times_S Y & & \\
 & \swarrow p & \downarrow f & \searrow q & \\
 X & & & & Y \\
 & \searrow a & & \swarrow b & \\
 & & S & &
 \end{array}$$

In this section, given an $\mathcal{O}_X$ -module $\mathcal{F}$ and an $\mathcal{O}_Y$ -module $\mathcal{G}$ let us set

$$\mathcal{F} \boxtimes \mathcal{G} = p^* \mathcal{F} \otimes_{\mathcal{O}_{X \times_S Y}} q^* \mathcal{G}$$

The bifunctor $(\mathcal{F}, \mathcal{G}) \mapsto \mathcal{F} \boxtimes \mathcal{G}$ on quasi-coherent modules extends to a bifunctor on quasi-coherent modules and differential operators of finite order over S , see Morphisms, Remark ???. The differentials of the de Rham complexes $\Omega_{X/S}^\bullet$ and $\Omega_{Y/S}^\bullet$ are differential operators of order 1 over S by Modules, Lemma ???. Thus it makes sense to consider the complex

$$\mathrm{Tot}(\Omega_{X/S}^\bullet \boxtimes \Omega_{Y/S}^\bullet)$$

Please see the discussion in Derived Categories of Schemes, Section ??.

Lemma 8.1. *In the situation above there is a canonical isomorphism*

$$\mathrm{Tot}(\Omega_{X/S}^\bullet \boxtimes \Omega_{Y/S}^\bullet) \longrightarrow \Omega_{X \times_S Y/S}^\bullet$$

of complexes of $f^{-1}\mathcal{O}_S$ -modules.

Proof. We know that $\Omega_{X \times_S Y/S} = p^* \Omega_{X/S} \oplus q^* \Omega_{Y/S}$ by Morphisms, Lemma ???. Taking exterior powers we obtain

$$\Omega_{X \times_S Y/S}^n = \bigoplus_{i+j=n} p^* \Omega_{X/S}^i \otimes_{\mathcal{O}_{X \times_S Y}} q^* \Omega_{Y/S}^j = \bigoplus_{i+j=n} \Omega_{X/S}^i \boxtimes \Omega_{Y/S}^j$$

by elementary properties of exterior powers. These identifications determine isomorphisms between the terms of the complexes on the left and the right of the arrow in the lemma. We omit the verification that these maps are compatible with differentials. $\square$

Set $A = \Gamma(S, \mathcal{O}_S)$. Combining the result of Lemma 8.1 with the map Derived Categories of Schemes, Equation (??) we obtain a cup product

$$R\Gamma(X, \Omega_{X/S}^\bullet) \otimes_A^{\mathbf{L}} R\Gamma(Y, \Omega_{Y/S}^\bullet) \longrightarrow R\Gamma(X \times_S Y, \Omega_{X \times_S Y/S}^\bullet)$$

On the level of cohomology, using the discussion in More on Algebra, Section ??, we obtain a canonical map

$$H_{dR}^i(X/S) \otimes_A H_{dR}^j(Y/S) \longrightarrow H_{dR}^{i+j}(X \times_S Y/S), \quad (\xi, \zeta) \longmapsto p^* \xi \cup q^* \zeta$$

We note that the construction above indeed proceeds by first pulling back and then taking the cup product.

Lemma 8.2. *Assume X and Y are smooth, quasi-compact, with affine diagonal over $S = \text{Spec}(A)$. Then the map*

$$R\Gamma(X, \Omega_{X/S}^\bullet) \otimes_A^L R\Gamma(Y, \Omega_{Y/S}^\bullet) \longrightarrow R\Gamma(X \times_S Y, \Omega_{X \times_S Y/S}^\bullet)$$

is an isomorphism in $D(A)$.

Proof. By Morphisms, Lemma ?? the sheaves $\Omega_{X/S}^n$ and $\Omega_{Y/S}^m$ are finite locally free $\mathcal{O}_X$ and $\mathcal{O}_Y$ -modules. On the other hand, X and Y are flat over S (Morphisms, Lemma ??) and hence we find that $\Omega_{X/S}^n$ and $\Omega_{Y/S}^m$ are flat over S . Also, observe that $\Omega_{X/S}^\bullet$ is a locally bounded. Thus the result by Lemma 8.1 and Derived Categories of Schemes, Lemma ??.

There is a relative version of the cup product, namely a map

$$Ra_*\Omega_{X/S}^\bullet \otimes_{\mathcal{O}_S}^L Rb_*\Omega_{Y/S}^\bullet \longrightarrow Rf_*\Omega_{X \times_S Y/S}^\bullet$$

in $D(\mathcal{O}_S)$. The construction combines Lemma 8.1 with the map Derived Categories of Schemes, Equation (??). The construction shows that this map is given by the diagram

$$\begin{array}{ccc} Ra_*\Omega_{X/S}^\bullet \otimes_{\mathcal{O}_S}^L Rb_*\Omega_{Y/S}^\bullet & & \\ \downarrow \text{units of adjunction} & & \\ Rf_*(p^{-1}\Omega_{X/S}^\bullet) \otimes_{\mathcal{O}_S}^L Rf_*(q^{-1}\Omega_{Y/S}^\bullet) & \longrightarrow & Rf_*(\Omega_{X \times_S Y/S}^\bullet) \otimes_{\mathcal{O}_S}^L Rf_*(\Omega_{X \times_S Y/S}^\bullet) \\ \downarrow \text{relative cup product} & & \downarrow \text{relative cup product} \\ Rf_*(p^{-1}\Omega_{X/S}^\bullet) \otimes_{f^{-1}\mathcal{O}_S}^L q^{-1}\Omega_{Y/S}^\bullet & \longrightarrow & Rf_*(\Omega_{X \times_S Y/S}^\bullet) \otimes_{f^{-1}\mathcal{O}_S}^L \Omega_{X \times_S Y/S}^\bullet \\ \downarrow \text{from derived to usual} & & \downarrow \text{from derived to usual} \\ Rf_*\text{Tot}(p^{-1}\Omega_{X/S}^\bullet \otimes_{f^{-1}\mathcal{O}_S} q^{-1}\Omega_{Y/S}^\bullet) & \longrightarrow & Rf_*\text{Tot}(\Omega_{X \times_S Y/S}^\bullet \otimes_{f^{-1}\mathcal{O}_S} \Omega_{X \times_S Y/S}^\bullet) \\ \downarrow \text{canonical map} & & \downarrow \eta \otimes \omega \mapsto \eta \wedge \omega \\ Rf_*\text{Tot}(\Omega_{X/S}^\bullet \boxtimes \Omega_{Y/S}^\bullet) & \xlongequal{\quad\quad\quad} & Rf_*\Omega_{X \times_S Y/S}^\bullet \end{array}$$

Here the first arrow uses the units $\text{id} \rightarrow Rp_*p^{-1}$ and $\text{id} \rightarrow Rq_*q^{-1}$ of adjunction as well as the identifications $Rf_*p^{-1} = Ra_*Rp_*p^{-1}$ and $Rf_*q^{-1} = Rb_*Rq_*q^{-1}$. The second arrow is the relative cup product of Cohomology, Remark ???. The third arrow is the map sending a derived tensor product of complexes to the totalization of the tensor product of complexes. The final equality is Lemma 8.1. This construction recovers on global section the construction given earlier.

Lemma 8.3. *Assume $X \rightarrow S$ and $Y \rightarrow S$ are smooth and quasi-compact and the morphisms $X \rightarrow X \times_S X$ and $Y \rightarrow Y \times_S Y$ are affine. Then the relative cup product*

$$Ra_*\Omega_{X/S}^\bullet \otimes_{\mathcal{O}_S}^L Rb_*\Omega_{Y/S}^\bullet \longrightarrow Rf_*\Omega_{X \times_S Y/S}^\bullet$$

is an isomorphism in $D(\mathcal{O}_S)$.

Proof. Immediate consequence of Lemma 8.2.

9. First Chern class in de Rham cohomology

Let $X \rightarrow S$ be a morphism of schemes. There is a map of complexes

$$d \log : \mathcal{O}_X^*[-1] \longrightarrow \Omega_{X/S}^\bullet$$

which sends the section $g \in \mathcal{O}_X^*(U)$ to the section $d \log(g) = g^{-1}dg$ of $\Omega_{X/S}^1(U)$. Thus we can consider the map

$$\text{Pic}(X) = H^1(X, \mathcal{O}_X^*) = H^2(X, \mathcal{O}_X^*[-1]) \longrightarrow H_{dR}^2(X/S)$$

where the first equality is Cohomology, Lemma ???. The image of the isomorphism class of the invertible module $\mathcal{L}$ is denoted $c_1^{dR}(\mathcal{L}) \in H_{dR}^2(X/S)$.

We can also use the map $d \log : \mathcal{O}_X^* \rightarrow \Omega_{X/S}^1$ to define a Chern class in Hodge cohomology

$$c_1^{Hodge} : \text{Pic}(X) \longrightarrow H^1(X, \Omega_{X/S}^1) \subset H_{Hodge}^2(X/S)$$

These constructions are compatible with pullbacks.

Lemma 9.1. *Given a commutative diagram*

$$\begin{array}{ccc} X' & \xrightarrow{\quad} & X \\ \downarrow & f & \downarrow \\ S' & \longrightarrow & S \end{array}$$

of schemes the diagrams

$$\begin{array}{ccc} \text{Pic}(X') & \xleftarrow{f^*} & \text{Pic}(X) \\ c_1^{dR} \downarrow & & \downarrow c_1^{dR} \\ H_{dR}^2(X'/S') & \xleftarrow{f^*} & H_{dR}^2(X/S) \end{array} \quad \begin{array}{ccc} \text{Pic}(X') & \xleftarrow{f^*} & \text{Pic}(X) \\ c_1^{Hodge} \downarrow & & \downarrow c_1^{Hodge} \\ H^1(X', \Omega_{X'/S'}^1) & \xleftarrow{f^*} & H^1(X, \Omega_{X/S}^1) \end{array}$$

commute.

Proof. Omitted. □

Let us “compute” the element $c_1^{dR}(\mathcal{L})$ in Čech cohomology (with sign rules for Čech differentials as in Cohomology, Section ??). Namely, choose an open covering $\mathcal{U} : X = \bigcup_{i \in I} U_i$ such that we have a trivializing section s_i of $\mathcal{L}|_{U_i}$ for all i . On the overlaps $U_{i_0 i_1} = U_{i_0} \cap U_{i_1}$ we have an invertible function $f_{i_0 i_1}$ such that $f_{i_0 i_1} = s_{i_1}|_{U_{i_0 i_1}} s_{i_0}^{-1}|_{U_{i_0 i_1}}$.¹ Of course we have

$$f_{i_1 i_2}|_{U_{i_0 i_1 i_2}} f_{i_0 i_2}^{-1}|_{U_{i_0 i_1 i_2}} f_{i_0 i_1}|_{U_{i_0 i_1 i_2}} = 1$$

The cohomology class of $\mathcal{L}$ in $H^1(X, \mathcal{O}_X^*)$ is the image of the Čech cohomology class of the cocycle $\{f_{i_0 i_1}\}$ in $\check{C}^\bullet(\mathcal{U}, \mathcal{O}_X^*)$. Therefore we see that $c_1^{dR}(\mathcal{L})$ is the image of the cohomology class associated to the Čech cocycle $\{\alpha_{i_0 \dots i_p}\}$ in $\text{Tot}(\check{C}^\bullet(\mathcal{U}, \Omega_{X/S}^\bullet))$ of degree 2 given by

- (1) $\alpha_{i_0} = 0$ in $\Omega_{X/S}^2(U_{i_0})$,
- (2) $\alpha_{i_0 i_1} = f_{i_0 i_1}^{-1} df_{i_0 i_1}$ in $\Omega_{X/S}^1(U_{i_0 i_1})$, and
- (3) $\alpha_{i_0 i_1 i_2} = 0$ in $\mathcal{O}_{X/S}(U_{i_0 i_1 i_2})$.

¹The Čech differential of a 0-cycle $\{a_{i_0}\}$ has $a_{i_1} - a_{i_0}$ over $U_{i_0 i_1}$.

Suppose we have invertible modules $\mathcal{L}_k$, $k = 1, \dots, a$ each trivialized over U_i for all $i \in I$ giving rise to cocycles $f_{k,i_0 i_1}$ and $\alpha_k = \{\alpha_{k,i_0 \dots i_p}\}$ as above. Using the rule in Cohomology, Section ?? we can compute

$$\beta = \alpha_1 \cup \alpha_2 \cup \dots \cup \alpha_a$$

to be given by the cocycle $\beta = \{\beta_{i_0 \dots i_p}\}$ described as follows

- (1) $\beta_{i_0 \dots i_p} = 0$ in $\Omega_{X/S}^{2a-p}(U_{i_0 \dots i_p})$ unless $p = a$, and
- (2) $\beta_{i_0 \dots i_a} = (-1)^{a(a-1)/2} \alpha_{1,i_0 i_1} \wedge \alpha_{2,i_1 i_2} \wedge \dots \wedge \alpha_{a,i_{a-1} i_a}$ in $\Omega_{X/S}^a(U_{i_0 \dots i_a})$.

Thus this is a cocycle representing $c_1^{dR}(\mathcal{L}_1) \cup \dots \cup c_1^{dR}(\mathcal{L}_a)$. Of course, the same computation shows that the cocycle $\{\beta_{i_0 \dots i_a}\}$ in $\check{C}^a(\mathcal{U}, \Omega_{X/S}^a)$ represents the cohomology class $c_1^{Hodge}(\mathcal{L}_1) \cup \dots \cup c_1^{Hodge}(\mathcal{L}_a)$

Remark 9.2. Here is a reformulation of the calculations above in more abstract terms. Let $p : X \rightarrow S$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module. If we view $d \log$ as a map

$$\mathcal{O}_X^*[-1] \rightarrow \sigma_{\geq 1} \Omega_{X/S}^\bullet$$

then using $\text{Pic}(X) = H^1(X, \mathcal{O}_X^*)$ as above we find a cohomology class

$$\gamma_1(\mathcal{L}) \in H^2(X, \sigma_{\geq 1} \Omega_{X/S}^\bullet)$$

The image of $\gamma_1(\mathcal{L})$ under the map $\sigma_{\geq 1} \Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^\bullet$ recovers $c_1^{dR}(\mathcal{L})$. In particular we see that $c_1^{dR}(\mathcal{L}) \in F^1 H_{dR}^2(X/S)$, see Section 7. The image of $\gamma_1(\mathcal{L})$ under the map $\sigma_{\geq 1} \Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^1[-1]$ recovers $c_1^{Hodge}(\mathcal{L})$. Taking the cup product (see Section 7) we obtain

$$\xi = \gamma_1(\mathcal{L}_1) \cup \dots \cup \gamma_1(\mathcal{L}_a) \in H^{2a}(X, \sigma_{\geq a} \Omega_{X/S}^\bullet)$$

The commutative diagrams in Section 7 show that ξ is mapped to $c_1^{dR}(\mathcal{L}_1) \cup \dots \cup c_1^{dR}(\mathcal{L}_a)$ in $H_{dR}^{2a}(X/S)$ by the map $\sigma_{\geq a} \Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^\bullet$. Also, it follows $c_1^{dR}(\mathcal{L}_1) \cup \dots \cup c_1^{dR}(\mathcal{L}_a)$ is contained in $F^a H_{dR}^{2a}(X/S)$. Similarly, the map $\sigma_{\geq a} \Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^a[-a]$ sends ξ to $c_1^{Hodge}(\mathcal{L}_1) \cup \dots \cup c_1^{Hodge}(\mathcal{L}_a)$ in $H^a(X, \Omega_{X/S}^a)$.

Remark 9.3. Let $p : X \rightarrow S$ be a morphism of schemes. For $i > 0$ denote $\Omega_{X/S, \log}^i \subset \Omega_{X/S}^i$ the abelian subsheaf generated by local sections of the form

$$d \log(u_1) \wedge \dots \wedge d \log(u_i)$$

where $u_1, \dots, u_n$ are invertible local sections of $\mathcal{O}_X$. For $i = 0$ the subsheaf $\Omega_{X/S, \log}^0 \subset \mathcal{O}_X$ is the image of $\mathbf{Z} \rightarrow \mathcal{O}_X$. For every $i \geq 0$ we have a map of complexes

$$\Omega_{X/S, \log}^i[-i] \longrightarrow \Omega_{X/S}^\bullet$$

because the derivative of a logarithmic form is zero. Moreover, wedging logarithmic forms gives another, hence we find bilinear maps

$$\wedge : \Omega_{X/S, \log}^i \times \Omega_{X/S, \log}^j \longrightarrow \Omega_{X/S, \log}^{i+j}$$

compatible with (4.0.1) and the maps above. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module. Using the map of abelian sheaves $d \log : \mathcal{O}_X^* \rightarrow \Omega_{X/S, \log}^1$ and the identification $\text{Pic}(X) = H^1(X, \mathcal{O}_X^*)$ we find a canonical cohomology class

$$\tilde{\gamma}_1(\mathcal{L}) \in H^1(X, \Omega_{X/S, \log}^1)$$

These classes have the following properties

- (1) the image of $\tilde{\gamma}_1(\mathcal{L})$ under the canonical map $\Omega_{X/S, \log}^1[-1] \rightarrow \sigma_{\geq 1}\Omega_{X/S}^\bullet$ sends $\tilde{\gamma}_1(\mathcal{L})$ to the class $\gamma_1(\mathcal{L}) \in H^2(X, \sigma_{\geq 1}\Omega_{X/S}^\bullet)$ of Remark 9.2,
- (2) the image of $\tilde{\gamma}_1(\mathcal{L})$ under the canonical map $\Omega_{X/S, \log}^1[-1] \rightarrow \Omega_{X/S}^\bullet$ sends $\tilde{\gamma}_1(\mathcal{L})$ to $c_1^{dR}(\mathcal{L})$ in $H_{dR}^2(X/S)$,
- (3) the image of $\tilde{\gamma}_1(\mathcal{L})$ under the canonical map $\Omega_{X/S, \log}^1 \rightarrow \Omega_{X/S}^1$ sends $\tilde{\gamma}_1(\mathcal{L})$ to $c_1^{Hodge}(\mathcal{L})$ in $H^1(X, \Omega_{X/S}^1)$,
- (4) the construction of these classes is compatible with pullbacks,
- (5) add more here.

10. de Rham cohomology of a line bundle

A line bundle is a special case of a vector bundle, which in turn is a cone endowed with some extra structure. To intelligently talk about the de Rham complex of these, it makes sense to discuss the de Rham complex of a graded ring.

Remark 10.1 (de Rham complex of a graded ring). Let G be an abelian monoid written additively with neutral element 0. Let $R \rightarrow A$ be a ring map and assume A comes with a grading $A = \bigoplus_{g \in G} A_g$ by R -modules such that R maps into A_0 and $A_g \cdot A_{g'} \subset A_{g+g'}$. Then the module of differentials comes with a grading

$$\Omega_{A/R} = \bigoplus_{g \in G} \Omega_{A/R, g}$$

where $\Omega_{A/R, g}$ is the R -submodule of $\Omega_{A/R}$ generated by $a_0 da_1$ with $a_i \in A_{g_i}$ such that $g = g_0 + g_1$. Similarly, we obtain

$$\Omega_{A/R}^p = \bigoplus_{g \in G} \Omega_{A/R, g}^p$$

where $\Omega_{A/R, g}^p$ is the R -submodule of $\Omega_{A/R}^p$ generated by $a_0 da_1 \wedge \dots \wedge da_p$ with $a_i \in A_{g_i}$ such that $g = g_0 + g_1 + \dots + g_p$. Of course the differentials preserve the grading and the wedge product is compatible with the gradings in the obvious manner.

Let $f : X \rightarrow S$ be a morphism of schemes. Let $\pi : C \rightarrow X$ be a cone, see Constructions, Definition ???. Recall that this means π is affine and we have a grading $\pi_* \mathcal{O}_C = \bigoplus_{n \geq 0} \mathcal{A}_n$ with $\mathcal{A}_0 = \mathcal{O}_X$. Using the discussion in Remark 10.1 over affine opens we find that²

$$\pi_*(\Omega_{C/S}^\bullet) = \bigoplus_{n \geq 0} \Omega_{C/S, n}^\bullet$$

is canonically a direct sum of subcomplexes. Moreover, we have a factorization

$$\Omega_{X/S}^\bullet \rightarrow \Omega_{C/S, 0}^\bullet \rightarrow \pi_*(\Omega_{C/S}^\bullet)$$

and we know that $\omega \wedge \eta \in \Omega_{C/S, n+m}^{p+q}$ if $\omega \in \Omega_{C/S, n}^p$ and $\eta \in \Omega_{C/S, m}^q$.

Let $f : X \rightarrow S$ be a morphism of schemes. Let $\pi : L \rightarrow X$ be the line bundle associated to the invertible $\mathcal{O}_X$ -module $\mathcal{L}$. This means that π is the unique affine morphism such that

$$\pi_* \mathcal{O}_L = \bigoplus_{n \geq 0} \mathcal{L}^{\otimes n}$$

²With excuses for the notation!

as $\mathcal{O}_X$ -algebras. Thus L is a cone over X . By the discussion above we find a canonical direct sum decomposition

$$\pi_*(\Omega_{L/S}^\bullet) = \bigoplus_{n \geq 0} \Omega_{L/S,n}^\bullet$$

compatible with wedge product, compatible with the decomposition of $\pi_*\mathcal{O}_L$ above, and such that $\Omega_{X/S}$ maps into the part $\Omega_{L/S,0}$ of degree 0.

There is another case which will be useful to us. Namely, consider the complement³ $L^* \subset L$ of the zero section $o : X \rightarrow L$ in our line bundle L . A local computation shows we have a canonical isomorphism

$$(L^* \rightarrow X)_*\mathcal{O}_{L^*} = \bigoplus_{n \in \mathbf{Z}} \mathcal{L}^{\otimes n}$$

of $\mathcal{O}_X$ -algebras. The right hand side is a $\mathbf{Z}$ -graded quasi-coherent $\mathcal{O}_X$ -algebra. Using the discussion in Remark 10.1 over affine opens we find that

$$(L^* \rightarrow X)_*(\Omega_{L^*/S}^\bullet) = \bigoplus_{n \in \mathbf{Z}} \Omega_{L^*/S,n}^\bullet$$

compatible with wedge product, compatible with the decomposition of $(L^* \rightarrow X)_*\mathcal{O}_{L^*}$ above, and such that $\Omega_{X/S}$ maps into the part $\Omega_{L^*/S,0}$ of degree 0. The complex $\Omega_{L^*/S,0}^\bullet$ will be of particular interest to us.

Lemma 10.2. *With notation as above, there is a short exact sequence of complexes*

$$0 \rightarrow \Omega_{X/S}^\bullet \rightarrow \Omega_{L^*/S,0}^\bullet \rightarrow \Omega_{X/S}^\bullet[-1] \rightarrow 0$$

Proof. We have constructed the map $\Omega_{X/S}^\bullet \rightarrow \Omega_{L^*/S,0}^\bullet$ above.

Construction of $\text{Res} : \Omega_{L^*/S,0}^\bullet \rightarrow \Omega_{X/S}^\bullet[-1]$. Let $U \subset X$ be an open and let $s \in \mathcal{L}(U)$ and $s' \in \mathcal{L}^{\otimes -1}(U)$ be sections such that $s's = 1$. Then s gives an invertible section of the sheaf of algebras $(L^* \rightarrow X)_*\mathcal{O}_{L^*}$ over U with inverse $s' = s^{-1}$. Then we can consider the 1-form $d \log(s) = s'd(s)$ which is an element of $\Omega_{L^*/S,0}^1(U)$ by our construction of the grading on $\Omega_{L^*/S}^\bullet$. Our computations on affines given below will show that 1 and $d \log(s)$ freely generate $\Omega_{L^*/S,0}^\bullet|_U$ as a right module over $\Omega_{X/S}^\bullet|_U$. Thus we can define Res over U by the rule

$$\text{Res}(\omega' + d \log(s) \wedge \omega) = \omega$$

for all $\omega', \omega \in \Omega_{X/S}^\bullet(U)$. This map is independent of the choice of local generator s and hence glues to give a global map. Namely, another choice of s would be of the form gs for some invertible $g \in \mathcal{O}_X(U)$ and we would get $d \log(gs) = g^{-1}d(g) + d \log(s)$ from which the independence easily follows. Finally, observe that our rule for Res is compatible with differentials as $d(\omega' + d \log(s) \wedge \omega) = d(\omega') - d \log(s) \wedge d(\omega)$ and because the differential on $\Omega_{X/S}^\bullet[-1]$ sends ω' to $-d(\omega')$ by our sign convention in Homology, Definition ??.

Local computation. We can cover X by affine opens $U \subset X$ such that $\mathcal{L}|_U \cong \mathcal{O}_U$ which moreover map into an affine open $V \subset S$. Write $U = \text{Spec}(A)$, $V = \text{Spec}(R)$ and choose a generator s of $\mathcal{L}$. We find that we have

$$L^* \times_X U = \text{Spec}(A[s, s^{-1}])$$

³The scheme L^* is the $\mathbf{G}_m$ -torsor over X associated to L . This is why the grading we get below is a $\mathbf{Z}$ -grading, compare with Groupoids, Example ?? and Lemmas ?? and ??.

Computing differentials we see that

$$\Omega_{A[s,s^{-1}]/R}^1 = A[s, s^{-1}] \otimes_A \Omega_{A/R}^1 \oplus A[s, s^{-1}] d \log(s)$$

and therefore taking exterior powers we obtain

$$\Omega_{A[s,s^{-1}]/R}^p = A[s, s^{-1}] \otimes_A \Omega_{A/R}^p \oplus A[s, s^{-1}] d \log(s) \otimes_A \Omega_{A/R}^{p-1}$$

Taking degree 0 parts we find

$$\Omega_{A[s,s^{-1}]/R,0}^p = \Omega_{A/R}^p \oplus d \log(s) \otimes_A \Omega_{A/R}^{p-1}$$

and the proof of the lemma is complete. $\square$

Lemma 10.3. *The “boundary” map $\delta : \Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^\bullet[2]$ in $D(X, f^{-1}\mathcal{O}_S)$ coming from the short exact sequence in Lemma 10.2 is the map of Remark 4.3 for $\xi = c_1^{dR}(\mathcal{L})$.*

Proof. To be precise we consider the shift

$$0 \rightarrow \Omega_{X/S}^\bullet[1] \rightarrow \Omega_{L^*/S,0}^\bullet[1] \rightarrow \Omega_{X/S}^\bullet \rightarrow 0$$

of the short exact sequence of Lemma 10.2. As the degree zero part of a grading on $(L^* \rightarrow X)_* \Omega_{L^*/S}^\bullet$ we see that $\Omega_{L^*/S,0}^\bullet$ is a differential graded $\mathcal{O}_X$ -algebra and that the map $\Omega_{X/S}^\bullet \rightarrow \Omega_{L^*/S,0}^\bullet$ is a homomorphism of differential graded $\mathcal{O}_X$ -algebras. Hence we may view $\Omega_{X/S}^\bullet[1] \rightarrow \Omega_{L^*/S,0}^\bullet[1]$ as a map of right differential graded $\Omega_{X/S}^\bullet$ -modules on X . The map $\text{Res} : \Omega_{L^*/S,0}^\bullet[1] \rightarrow \Omega_{X/S}^\bullet$ is a map of right differential graded $\Omega_{X/S}^\bullet$ -modules since it is locally defined by the rule $\text{Res}(\omega' + d \log(s) \wedge \omega) = \omega$, see proof of Lemma 10.2. Thus by the discussion in Differential Graded Sheaves, Section ?? we see that δ comes from a map $\delta' : \Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^\bullet[2]$ in the derived category $D(\Omega_{X/S}^\bullet, d)$ of right differential graded modules over the de Rham complex. The uniqueness asserted in Remark 4.3 shows it suffices to prove that $\delta(1) = c_1^{dR}(\mathcal{L})$.

We claim that there is a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{O}_X^* & \longrightarrow & E & \longrightarrow & \mathbf{Z} \longrightarrow 0 \\ & & \downarrow d \log & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \Omega_{X/S}^\bullet[1] & \longrightarrow & \Omega_{L^*/S,0}^\bullet[1] & \longrightarrow & \Omega_{X/S}^\bullet \longrightarrow 0 \end{array}$$

where the top row is a short exact sequence of abelian sheaves whose boundary map sends 1 to the class of $\mathcal{L}$ in $H^1(X, \mathcal{O}_X^*)$. It suffices to prove the claim by the compatibility of boundary maps with maps between short exact sequences. We define E as the sheafification of the rule

$$U \mapsto \{(s, n) \mid n \in \mathbf{Z}, s \in \mathcal{L}^{\otimes n}(U) \text{ generator}\}$$

with group structure given by $(s, n) \cdot (t, m) = (s \otimes t, n + m)$. The middle vertical map sends (s, n) to $d \log(s)$. This produces a map of short exact sequences because the map $\text{Res} : \Omega_{L^*/S,0}^\bullet \rightarrow \mathcal{O}_X$ constructed in the proof of Lemma 10.2 sends $d \log(s)$ to 1 if s is a local generator of $\mathcal{L}$. To calculate the boundary of 1 in the top row, choose local trivializations s_i of $\mathcal{L}$ over opens U_i as in Section 9. On the overlaps $U_{i_0 i_1} = U_{i_0} \cap U_{i_1}$ we have an invertible function $f_{i_0 i_1}$ such that

$f_{i_0 i_1} = s_{i_1}|_{U_{i_0 i_1}} s_{i_0}|_{U_{i_0 i_1}}^{-1}$ and the cohomology class of $\mathcal{L}$ is given by the Čech cocycle $\{f_{i_0 i_1}\}$. Then of course we have

$$(f_{i_0 i_1}, 0) = (s_{i_1}, 1)|_{U_{i_0 i_1}} \cdot (s_{i_0}, 1)|_{U_{i_0 i_1}}^{-1}$$

as sections of E which finishes the proof. $\square$

Lemma 10.4. *With notation as above we have*

- (1) $\Omega_{L^*/S, n}^p = \Omega_{L^*/S, 0}^p \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n}$ for all $n \in \mathbf{Z}$ as quasi-coherent $\mathcal{O}_X$ -modules,
- (2) $\Omega_{X/S}^\bullet = \Omega_{L/X, 0}^\bullet$ as complexes, and
- (3) for $n > 0$ and $p \geq 0$ we have $\Omega_{L/X, n}^p = \Omega_{L^*/S, n}^p$.

Proof. In each case there is a globally defined canonical map which is an isomorphism by local calculations which we omit. $\square$

Lemma 10.5. *In the situation above, assume there is a morphism $S \rightarrow \operatorname{Spec}(\mathbf{Q})$. Then $\Omega_{X/S}^\bullet \rightarrow \pi_* \Omega_{L/S}^\bullet$ is a quasi-isomorphism and $H_{dR}^*(X/S) = H_{dR}^*(L/S)$.*

Proof. Let R be a $\mathbf{Q}$ -algebra. Let A be an R -algebra. The affine local statement is that the map

$$\Omega_{A/R}^\bullet \longrightarrow \Omega_{A[t]/R}^\bullet$$

is a quasi-isomorphism of complexes of R -modules. In fact it is a homotopy equivalence with homotopy inverse given by the map sending $g\omega + g'dt \wedge \omega'$ to $g(0)\omega$ for $g, g' \in A[t]$ and $\omega, \omega' \in \Omega_{A/R}^\bullet$. The homotopy sends $g\omega + g'dt \wedge \omega'$ to $(\int g')\omega'$ where $\int g' \in A[t]$ is the polynomial with vanishing constant term whose derivative with respect to t is g' . Of course, here we use that R contains $\mathbf{Q}$ as $\int t^n = (1/n)t^{n+1}$. $\square$

Example 10.6. Lemma 10.5 is false in positive characteristic. The de Rham complex of $\mathbf{A}_k^1 = \operatorname{Spec}(k[x])$ over a field k looks like a direct sum

$$k \oplus \bigoplus_{n \geq 1} (k \cdot t^n \xrightarrow{n} k \cdot t^{n-1} dt)$$

Hence if the characteristic of k is $p > 0$, then we see that both $H_{dR}^0(\mathbf{A}_k^1/k)$ and $H_{dR}^1(\mathbf{A}_k^1/k)$ are infinite dimensional over k .

11. de Rham cohomology of projective space

Let A be a ring. Let $n \geq 1$. The structure morphism $\mathbf{P}_A^n \rightarrow \operatorname{Spec}(A)$ is a proper smooth of relative dimension n . It is smooth of relative dimension n and of finite type as $\mathbf{P}_A^n$ has a finite affine open covering by schemes each isomorphic to $\mathbf{A}_A^n$, see Constructions, Lemma ???. It is proper because it is also separated and universally closed by Constructions, Lemma ???. Let us denote $\mathcal{O}$ and $\mathcal{O}(d)$ the structure sheaf $\mathcal{O}_{\mathbf{P}_A^n}$ and the Serre twists $\mathcal{O}_{\mathbf{P}_A^n}(d)$. Let us denote $\Omega = \Omega_{\mathbf{P}_A^n/A}$ the sheaf of relative differentials and Ω^p its exterior powers.

Lemma 11.1. *There exists a short exact sequence*

$$0 \rightarrow \Omega \rightarrow \mathcal{O}(-1)^{\oplus n+1} \rightarrow \mathcal{O} \rightarrow 0$$

Proof. To explain this, we recall that $\mathbf{P}_A^n = \operatorname{Proj}(A[T_0, \dots, T_n])$, and we write symbolically

$$\mathcal{O}(-1)^{\oplus n+1} = \bigoplus_{j=0, \dots, n} \mathcal{O}(-1) dT_j$$

The first arrow

$$\Omega \rightarrow \bigoplus_{j=0, \dots, n} \mathcal{O}(-1) dT_j$$

in the short exact sequence above is given on each of the standard opens $D_+(T_i) = \text{Spec}(A[T_0/T_i, \dots, T_n/T_i])$ mentioned above by the rule

$$\sum_{j \neq i} g_j d(T_j/T_i) \mapsto \sum_{j \neq i} g_j/T_i dT_j - (\sum_{j \neq i} g_j T_j/T_i^2) dT_i$$

This makes sense because $1/T_i$ is a section of $\mathcal{O}(-1)$ over $D_+(T_i)$. The map

$$\bigoplus_{j=0, \dots, n} \mathcal{O}(-1) dT_j \rightarrow \mathcal{O}$$

is given by sending dT_j to T_j , more precisely, on $D_+(T_i)$ we send the section $\sum g_j dT_j$ to $\sum T_j g_j$. We omit the verification that this produces a short exact sequence. $\square$

Given an integer $k \in \mathbf{Z}$ and a quasi-coherent $\mathcal{O}_{\mathbf{P}_A^n}$ -module $\mathcal{F}$ denote as usual $\mathcal{F}(k)$ the k th Serre twist of $\mathcal{F}$. See Constructions, Definition ??.

Lemma 11.2. *In the situation above we have the following cohomology groups*

- (1) $H^q(\mathbf{P}_A^n, \Omega^p) = 0$ unless $0 \leq p = q \leq n$,
- (2) for $0 \leq p \leq n$ the A -module $H^p(\mathbf{P}_A^n, \Omega^p)$ free of rank 1.
- (3) for $q > 0$, $k > 0$, and p arbitrary we have $H^q(\mathbf{P}_A^n, \Omega^p(k)) = 0$, and
- (4) add more here.

Proof. We are going to use the results of Cohomology of Schemes, Lemma ?? without further mention. In particular, the statements are true for $H^q(\mathbf{P}_A^n, \mathcal{O}(k))$.

Proof for $p = 1$. Consider the short exact sequence

$$0 \rightarrow \Omega \rightarrow \mathcal{O}(-1)^{\oplus n+1} \rightarrow \mathcal{O} \rightarrow 0$$

of Lemma 11.1. Since $\mathcal{O}(-1)$ has vanishing cohomology in all degrees, this gives that $H^q(\mathbf{P}_A^n, \Omega)$ is zero except in degree 1 where it is freely generated by the boundary of 1 in $H^0(\mathbf{P}_A^n, \mathcal{O})$.

Assume $p > 1$. Let us think of the short exact sequence above as defining a 2 step filtration on $\mathcal{O}(-1)^{\oplus n+1}$. The induced filtration on $\wedge^p \mathcal{O}(-1)^{\oplus n+1}$ looks like this

$$0 \rightarrow \Omega^p \rightarrow \wedge^p (\mathcal{O}(-1)^{\oplus n+1}) \rightarrow \Omega^{p-1} \rightarrow 0$$

Observe that $\wedge^p \mathcal{O}(-1)^{\oplus n+1}$ is isomorphic to a direct sum of $n+1$ choose p copies of $\mathcal{O}(-p)$ and hence has vanishing cohomology in all degrees. By induction hypothesis, this shows that $H^q(\mathbf{P}_A^n, \Omega^p)$ is zero unless $q = p$ and $H^p(\mathbf{P}_A^n, \Omega^p)$ is free of rank 1 with generator the boundary of the generator in $H^{p-1}(\mathbf{P}_A^n, \Omega^{p-1})$.

Let $k > 0$. Observe that $\Omega^n = \mathcal{O}(-n-1)$ for example by the short exact sequence above for $p = n+1$. Hence $\Omega^n(k)$ has vanishing cohomology in positive degrees. Using the short exact sequences

$$0 \rightarrow \Omega^p(k) \rightarrow \wedge^p (\mathcal{O}(-1)^{\oplus n+1})(k) \rightarrow \Omega^{p-1}(k) \rightarrow 0$$

and *descending* induction on p we get the vanishing of cohomology of $\Omega^p(k)$ in positive degrees for all p . $\square$

Lemma 11.3. *We have $H^q(\mathbf{P}_A^n, \Omega^p) = 0$ unless $0 \leq p = q \leq n$. For $0 \leq p \leq n$ the A -module $H^p(\mathbf{P}_A^n, \Omega^p)$ free of rank 1 with basis element $c_1^{\text{Hodge}}(\mathcal{O}(1))^p$.*

Proof. We have the vanishing and freeness by Lemma 11.2. For $p = 0$ it is certainly true that $1 \in H^0(\mathbf{P}_A^n, \mathcal{O})$ is a generator.

Proof for $p = 1$. Consider the short exact sequence

$$0 \rightarrow \Omega \rightarrow \mathcal{O}(-1)^{\oplus n+1} \rightarrow \mathcal{O} \rightarrow 0$$

of Lemma 11.1. In the proof of Lemma 11.2 we have seen that the generator of $H^1(\mathbf{P}_A^n, \Omega)$ is the boundary ξ of $1 \in H^0(\mathbf{P}_A^n, \mathcal{O})$. As in the proof of Lemma 11.1 we will identify $\mathcal{O}(-1)^{\oplus n+1}$ with $\bigoplus_{j=0, \dots, n} \mathcal{O}(-1)dT_j$. Consider the open covering

$$\mathcal{U} : \mathbf{P}_A^n = \bigcup_{i=0, \dots, n} D_+(T_i)$$

We can lift the restriction of the global section 1 of $\mathcal{O}$ to $U_i = D_+(T_i)$ by the section $T_i^{-1}dT_i$ of $\bigoplus \mathcal{O}(-1)dT_j$ over U_i . Thus the cocycle representing ξ is given by

$$T_{i_1}^{-1}dT_{i_1} - T_{i_0}^{-1}dT_{i_0} = d \log(T_{i_1}/T_{i_0}) \in \Omega(U_{i_0 i_1})$$

On the other hand, for each i the section T_i is a trivializing section of $\mathcal{O}(1)$ over U_i . Hence we see that $f_{i_0 i_1} = T_{i_1}/T_{i_0} \in \mathcal{O}^*(U_{i_0 i_1})$ is the cocycle representing $\mathcal{O}(1)$ in $\text{Pic}(\mathbf{P}_A^n)$, see Section 9. Hence $c_1^{\text{Hodge}}(\mathcal{O}(1))$ is given by the cocycle $d \log(T_{i_1}/T_{i_0})$ which agrees with what we got for ξ above.

Proof for general p by induction. The base cases $p = 0, 1$ were handled above. Assume $p > 1$. In the proof of Lemma 11.2 we have seen that the generator of $H^p(\mathbf{P}_A^n, \Omega^p)$ is the boundary of $c_1^{\text{Hodge}}(\mathcal{O}(1))^{p-1}$ in the long exact cohomology sequence associated to

$$0 \rightarrow \Omega^p \rightarrow \wedge^p(\mathcal{O}(-1)^{\oplus n+1}) \rightarrow \Omega^{p-1} \rightarrow 0$$

By the calculation in Section 9 the cohomology class $c_1^{\text{Hodge}}(\mathcal{O}(1))^{p-1}$ is, up to a sign, represented by the cocycle with terms

$$\beta_{i_0 \dots i_{p-1}} = d \log(T_{i_1}/T_{i_0}) \wedge d \log(T_{i_2}/T_{i_1}) \wedge \dots \wedge d \log(T_{i_{p-1}}/T_{i_{p-2}})$$

in $\Omega^{p-1}(U_{i_0 \dots i_{p-1}})$. These $\beta_{i_0 \dots i_{p-1}}$ can be lifted to the sections $\tilde{\beta}_{i_0 \dots i_{p-1}} = T_{i_0}^{-1}dT_{i_0} \wedge \beta_{i_0 \dots i_{p-1}}$ of $\wedge^p(\bigoplus \mathcal{O}(-1)dT_j)$ over $U_{i_0 \dots i_{p-1}}$. We conclude that the generator of $H^p(\mathbf{P}_A^n, \Omega^p)$ is given by the cocycle whose components are

$$\begin{aligned} \sum_{a=0}^p (-1)^a \tilde{\beta}_{i_0 \dots \hat{i}_a \dots i_p} &= T_{i_1}^{-1}dT_{i_1} \wedge \beta_{i_1 \dots i_p} + \sum_{a=1}^p (-1)^a T_{i_0}^{-1}dT_{i_0} \wedge \beta_{i_0 \dots \hat{i}_a \dots i_p} \\ &= (T_{i_1}^{-1}dT_{i_1} - T_{i_0}^{-1}dT_{i_0}) \wedge \beta_{i_1 \dots i_p} + T_{i_0}^{-1}dT_{i_0} \wedge d(\beta)_{i_0 \dots i_p} \\ &= d \log(T_{i_1}/T_{i_0}) \wedge \beta_{i_1 \dots i_p} \end{aligned}$$

viewed as a section of Ω^p over $U_{i_0 \dots i_p}$. This is up to sign the same as the cocycle representing $c_1^{\text{Hodge}}(\mathcal{O}(1))^p$ and the proof is complete. $\square$

Lemma 11.4. For $0 \leq i \leq n$ the de Rham cohomology $H_{dR}^{2i}(\mathbf{P}_A^n/A)$ is a free A -module of rank 1 with basis element $c_1^{dR}(\mathcal{O}(1))^i$. In all other degrees the de Rham cohomology of $\mathbf{P}_A^n$ over A is zero.

Proof. Consider the Hodge-to-de Rham spectral sequence of Section 6. By the computation of the Hodge cohomology of $\mathbf{P}_A^n$ over A done in Lemma 11.3 we see that the spectral sequence degenerates on the E_1 page. In this way we see that $H_{dR}^{2i}(\mathbf{P}_A^n/A)$ is a free A -module of rank 1 for $0 \leq i \leq n$ and zero else. Observe that

$c_1^{dR}(\mathcal{O}(1))^i \in H_{dR}^{2i}(\mathbf{P}_A^n/A)$ for $i = 0, \dots, n$ and that for $i = n$ this element is the image of $c_1^{Hodge}(\mathcal{L})^n$ by the map of complexes

$$\Omega_{\mathbf{P}_A^n/A}^n[-n] \longrightarrow \Omega_{\mathbf{P}_A^n/A}^\bullet$$

This follows for example from the discussion in Remark 9.2 or from the explicit description of cocycles representing these classes in Section 9. The spectral sequence shows that the induced map

$$H^n(\mathbf{P}_A^n, \Omega_{\mathbf{P}_A^n/A}^n) \longrightarrow H_{dR}^{2n}(\mathbf{P}_A^n/A)$$

is an isomorphism and since $c_1^{Hodge}(\mathcal{L})^n$ is a generator of the source (Lemma 11.3), we conclude that $c_1^{dR}(\mathcal{L})^n$ is a generator of the target. By the A -bilinearity of the cup products, it follows that also $c_1^{dR}(\mathcal{L})^i$ is a generator of $H_{dR}^{2i}(\mathbf{P}_A^n/A)$ for $0 \leq i \leq n$. $\square$

12. The spectral sequence for a smooth morphism

Consider a commutative diagram of schemes

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & \searrow p \quad \swarrow q & \\ & S & \end{array}$$

where f is a smooth morphism. Then we obtain a locally split short exact sequence

$$0 \rightarrow f^*\Omega_{Y/S} \rightarrow \Omega_{X/S} \rightarrow \Omega_{X/Y} \rightarrow 0$$

by Morphisms, Lemma ?? . Let us think of this as a descending filtration F on $\Omega_{X/S}$ with $F^0\Omega_{X/S} = \Omega_{X/S}$, $F^1\Omega_{X/S} = f^*\Omega_{Y/S}$, and $F^2\Omega_{X/S} = 0$. Applying the functor $\wedge^p$ we obtain for every p an induced filtration

$$\Omega_{X/S}^p = F^0\Omega_{X/S}^p \supset F^1\Omega_{X/S}^p \supset F^2\Omega_{X/S}^p \supset \dots \supset F^{p+1}\Omega_{X/S}^p = 0$$

whose successive quotients are

$$\mathrm{gr}^k\Omega_{X/S}^p = F^k\Omega_{X/S}^p / F^{k+1}\Omega_{X/S}^p = f^*\Omega_{Y/S}^k \otimes_{\mathcal{O}_X} \Omega_{X/Y}^{p-k} = f^{-1}\Omega_{Y/S}^k \otimes_{f^{-1}\mathcal{O}_Y} \Omega_{X/Y}^{p-k}$$

for $k = 0, \dots, p$. In fact, the reader can check using the Leibniz rule that $F^k\Omega_{X/S}^\bullet$ is a subcomplex of $\Omega_{X/S}^\bullet$. In this way $\Omega_{X/S}^\bullet$ has the structure of a filtered complex. We can also see this by observing that

$$F^k\Omega_{X/S}^\bullet = \mathrm{Im} \left(\wedge : \mathrm{Tot}(f^{-1}\sigma_{\geq k}\Omega_{Y/S}^\bullet \otimes_{p^{-1}\mathcal{O}_S} \Omega_{X/S}^\bullet) \longrightarrow \Omega_{X/S}^\bullet \right)$$

is the image of a map of complexes on X . The filtered complex

$$\Omega_{X/S}^\bullet = F^0\Omega_{X/S}^\bullet \supset F^1\Omega_{X/S}^\bullet \supset F^2\Omega_{X/S}^\bullet \supset \dots$$

has the following associated graded parts

$$\mathrm{gr}^k\Omega_{X/S}^\bullet = f^{-1}\Omega_{Y/S}^k[-k] \otimes_{f^{-1}\mathcal{O}_Y} \Omega_{X/Y}^\bullet$$

by what was said above.

Lemma 12.1. *Let $f : X \rightarrow Y$ be a quasi-compact, quasi-separated, and smooth morphism of schemes over a base scheme S . There is a bounded spectral sequence with first page*

$$E_1^{p,q} = H^q(\Omega_{Y/S}^p \otimes_{\mathcal{O}_Y}^{\mathbf{L}} Rf_* \Omega_{X/Y}^\bullet)$$

converging to $R^{p+q}f_ \Omega_{X/S}^\bullet$.*

Proof. Consider $\Omega_{X/S}^\bullet$ as a filtered complex with the filtration introduced above. The spectral sequence is the spectral sequence of Cohomology, Lemma ???. By Derived Categories of Schemes, Lemma ?? we have

$$Rf_* \mathrm{gr}^k \Omega_{X/S}^\bullet = \Omega_{Y/S}^k[-k] \otimes_{\mathcal{O}_Y}^{\mathbf{L}} Rf_* \Omega_{X/Y}^\bullet$$

and thus we conclude. $\square$

Remark 12.2. In Lemma 12.1 consider the cohomology sheaves

$$\mathcal{H}_{dR}^q(X/Y) = H^q(Rf_* \Omega_{X/Y}^\bullet)$$

If f is proper in addition to being smooth and S is a scheme over $\mathbf{Q}$ then $\mathcal{H}_{dR}^q(X/Y)$ is finite locally free (insert future reference here). If we only assume $\mathcal{H}_{dR}^q(X/Y)$ are flat $\mathcal{O}_Y$ -modules, then we obtain (tiny argument omitted)

$$E_1^{p,q} = \Omega_{Y/S}^p \otimes_{\mathcal{O}_Y} \mathcal{H}_{dR}^q(X/Y)$$

and the differentials in the spectral sequence are maps

$$d_1^{p,q} : \Omega_{Y/S}^p \otimes_{\mathcal{O}_Y} \mathcal{H}_{dR}^q(X/Y) \longrightarrow \Omega_{Y/S}^{p+1} \otimes_{\mathcal{O}_Y} \mathcal{H}_{dR}^q(X/Y)$$

In particular, for $p = 0$ we obtain a map $d_1^{0,q} : \mathcal{H}_{dR}^q(X/Y) \rightarrow \Omega_{Y/S}^1 \otimes_{\mathcal{O}_Y} \mathcal{H}_{dR}^q(X/Y)$ which turns out to be an integrable connection ∇ (insert future reference here) and the complex

$$\mathcal{H}_{dR}^q(X/Y) \rightarrow \Omega_{Y/S}^1 \otimes_{\mathcal{O}_Y} \mathcal{H}_{dR}^q(X/Y) \rightarrow \Omega_{Y/S}^2 \otimes_{\mathcal{O}_Y} \mathcal{H}_{dR}^q(X/Y) \rightarrow \dots$$

with differentials given by $d_1^{\bullet,q}$ is the de Rham complex of ∇ . The connection ∇ is known as the *Gauss-Manin connection*.

13. Leray-Hirsch type theorems

In this section we prove that for a smooth proper morphism one can sometimes express the de Rham cohomology upstairs in terms of the de Rham cohomology downstairs.

Lemma 13.1. *Let $f : X \rightarrow Y$ be a smooth proper morphism of schemes. Let N and $n_1, \dots, n_N \geq 0$ be integers and let $\xi_i \in H_{dR}^{n_i}(X/Y)$, $1 \leq i \leq N$. Assume for all points $y \in Y$ the images of $\xi_1, \dots, \xi_N$ in $H_{dR}^*(X_y/y)$ form a basis over $\kappa(y)$. Then the map*

$$\bigoplus_{i=1}^N \mathcal{O}_Y[-n_i] \longrightarrow Rf_* \Omega_{X/Y}^\bullet$$

associated to $\xi_1, \dots, \xi_N$ is an isomorphism.

Proof. By Lemma 3.5 $Rf_* \Omega_{X/Y}^\bullet$ is a perfect object of $D(\mathcal{O}_Y)$ whose formation commutes with arbitrary base change. Thus the map of the lemma is a map $a : K \rightarrow L$ between perfect objects of $D(\mathcal{O}_Y)$ whose derived restriction to any point is an isomorphism by our assumption on fibres. Then the cone C on a is a perfect object of $D(\mathcal{O}_Y)$ (Cohomology, Lemma ??) whose derived restriction to any point is zero. It follows that C is zero by More on Algebra, Lemma ?? and a is an

isomorphism. (This also uses Derived Categories of Schemes, Lemmas ?? and ?? to translate into algebra.) $\square$

We first prove the main result of this section in the following special case.

Lemma 13.2. *Let $f : X \rightarrow Y$ be a smooth proper morphism of schemes over a base S . Assume*

- (1) *Y and S are affine, and*
- (2) *there exist integers N and $n_1, \dots, n_N \geq 0$ and $\xi_i \in H_{dR}^{n_i}(X/S)$, $1 \leq i \leq N$ such that for all points $y \in Y$ the images of $\xi_1, \dots, \xi_N$ in $H_{dR}^*(X_y/y)$ form a basis over $\kappa(y)$.*

Then the map

$$\bigoplus_{i=1}^N H_{dR}^*(Y/S) \longrightarrow H_{dR}^*(X/S), \quad (a_1, \dots, a_N) \longmapsto \sum \xi_i \cup f^* a_i$$

is an isomorphism.

Proof. Say $Y = \text{Spec}(A)$ and $S = \text{Spec}(R)$. In this case $\Omega_{A/R}^\bullet$ computes $R\Gamma(Y, \Omega_{Y/S}^\bullet)$ by Lemma 3.1. Choose a finite affine open covering $\mathcal{U} : X = \bigcup_{i \in I} U_i$. Consider the complex

$$K^\bullet = \text{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U}, \Omega_{X/S}^\bullet))$$

as in Cohomology, Section ??. Let us collect some facts about this complex most of which can be found in the reference just given:

- (1) $K^\bullet$ is a complex of R -modules whose terms are A -modules,
- (2) $K^\bullet$ represents $R\Gamma(X, \Omega_{X/S}^\bullet)$ in $D(R)$ (Cohomology of Schemes, Lemma ?? and Cohomology, Lemma ??),
- (3) there is a natural map $\Omega_{A/R}^\bullet \rightarrow K^\bullet$ of complexes of R -modules which is A -linear on terms and induces the pullback map $H_{dR}^*(Y/S) \rightarrow H_{dR}^*(X/S)$ on cohomology,
- (4) $K^\bullet$ has a multiplication denoted $\wedge$ which turns it into a differential graded R -algebra,
- (5) the multiplication on $K^\bullet$ induces the cup product on $H_{dR}^*(X/S)$ (Cohomology, Section ??),
- (6) the filtration F on $\Omega_{X/S}^*$ induces a filtration

$$K^\bullet = F^0 K^\bullet \supset F^1 K^\bullet \supset F^2 K^\bullet \supset \dots$$

by subcomplexes on $K^\bullet$ such that

- (a) $F^k K^n \subset K^n$ is an A -submodule,
- (b) $F^k K^\bullet \wedge F^l K^\bullet \subset F^{k+l} K^\bullet$,
- (c) $\text{gr}^k K^\bullet$ is a complex of A -modules,
- (d) $\text{gr}^0 K^\bullet = \text{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U}, \Omega_{X/Y}^\bullet))$ and represents $R\Gamma(X, \Omega_{X/Y}^\bullet)$ in $D(A)$,
- (e) multiplication induces an isomorphism $\Omega_{A/R}^k[-k] \otimes_A \text{gr}^0 K^\bullet \rightarrow \text{gr}^k K^\bullet$

We omit the detailed proofs of these statements; please see discussion leading up to the construction of the spectral sequence in Lemma 12.1.

For every $i = 1, \dots, N$ we choose a cocycle $x_i \in K^{n_i}$ representing ξ_i . Next, we look at the map of complexes

$$\tilde{x} : M^\bullet = \bigoplus_{i=1, \dots, N} \Omega_{A/R}^\bullet[-n_i] \longrightarrow K^\bullet$$

which sends ω in the i th summand to $x_i \wedge \omega$. All that remains is to show that this map is a quasi-isomorphism. We endow $M^\bullet$ with the structure of a filtered complex by the rule

$$F^k M^\bullet = \bigoplus_{i=1, \dots, N} (\sigma_{\geq k} \Omega_{A/R}^\bullet)[-n_i]$$

With this choice the map $\tilde{x}$ is a morphism of filtered complexes. Observe that $\mathrm{gr}^0 M^\bullet = \bigoplus A[-n_i]$ and multiplication induces an isomorphism $\Omega_{A/R}^k[-k] \otimes_A \mathrm{gr}^0 M^\bullet \rightarrow \mathrm{gr}^k M^\bullet$. By construction and Lemma 13.1 we see that

$$\mathrm{gr}^0 \tilde{x} : \mathrm{gr}^0 M^\bullet \longrightarrow \mathrm{gr}^0 K^\bullet$$

is an isomorphism in $D(A)$. It follows that for all $k \geq 0$ we obtain isomorphisms

$$\mathrm{gr}^k \tilde{x} : \mathrm{gr}^k M^\bullet = \Omega_{A/R}^k[-k] \otimes_A \mathrm{gr}^0 M^\bullet \longrightarrow \Omega_{A/R}^k[-k] \otimes_A \mathrm{gr}^0 K^\bullet = \mathrm{gr}^k K^\bullet$$

in $D(A)$. Namely, the complex $\mathrm{gr}^0 K^\bullet = \mathrm{Tot}(\check{C}^\bullet(\mathcal{U}, \Omega_{X/Y}^\bullet))$ is K-flat as a complex of A -modules by Derived Categories of Schemes, Lemma ?? . Hence the tensor product on the right hand side is the derived tensor product as is true by inspection on the left hand side. Finally, taking the derived tensor product $\Omega_{A/R}^k[-k] \otimes_A^\mathbf{L} -$ is a functor on $D(A)$ and therefore sends isomorphisms to isomorphisms. Arguing by induction on k we deduce that

$$\tilde{x} : M^\bullet / F^k M^\bullet \rightarrow K^\bullet / F^k K^\bullet$$

is an isomorphism in $D(R)$ since we have the short exact sequences

$$0 \rightarrow F^k M^\bullet / F^{k+1} M^\bullet \rightarrow M^\bullet / F^{k+1} M^\bullet \rightarrow \mathrm{gr}^k M^\bullet \rightarrow 0$$

and similarly for $K^\bullet$. This proves that $\tilde{x}$ is a quasi-isomorphism as the filtrations are finite in any given degree. $\square$

Proposition 13.3. *Let $f : X \rightarrow Y$ be a smooth proper morphism of schemes over a base S . Let N and $n_1, \dots, n_N \geq 0$ be integers and let $\xi_i \in H_{dR}^{n_i}(X/S)$, $1 \leq i \leq N$. Assume for all points $y \in Y$ the images of $\xi_1, \dots, \xi_N$ in $H_{dR}^*(X_y/y)$ form a basis over $\kappa(y)$. The map*

$$\tilde{\xi} = \bigoplus \tilde{\xi}_i[-n_i] : \bigoplus \Omega_{Y/S}^\bullet[-n_i] \longrightarrow Rf_* \Omega_{X/S}^\bullet$$

(see proof) is an isomorphism in $D(Y, (Y \rightarrow S)^{-1} \mathcal{O}_S)$ and correspondingly the map

$$\bigoplus_{i=1}^N H_{dR}^*(Y/S) \longrightarrow H_{dR}^*(X/S), \quad (a_1, \dots, a_N) \longmapsto \sum \xi_i \cup f^* a_i$$

is an isomorphism.

Proof. Denote $p : X \rightarrow S$ and $q : Y \rightarrow S$ be the structure morphisms. Let $\xi'_i : \Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^\bullet[n_i]$ be the map of Remark 4.3 corresponding to ξ_i . Denote

$$\tilde{\xi}_i : \Omega_{Y/S}^\bullet \rightarrow Rf_* \Omega_{X/S}^\bullet[n_i]$$

the composition of ξ'_i with the canonical map $\Omega_{Y/S}^\bullet \rightarrow Rf_* \Omega_{X/S}^\bullet$. Using

$$R\Gamma(Y, Rf_* \Omega_{X/S}^\bullet) = R\Gamma(X, \Omega_{X/S}^\bullet)$$

on cohomology $\tilde{\xi}_i$ is the map $\eta \mapsto \xi_i \cup f^* \eta$ from $H_{dR}^m(Y/S)$ to $H_{dR}^{m+n}(X/S)$. Further, since the formation of ξ'_i commutes with restrictions to opens, so does the formation of $\tilde{\xi}_i$ commute with restriction to opens.

Thus we can consider the map

$$\tilde{\xi} = \bigoplus \tilde{\xi}_i[-n_i] : \bigoplus \Omega_{Y/S}^\bullet[-n_i] \longrightarrow Rf_* \Omega_{X/S}^\bullet$$

To prove the lemma it suffices to show that this is an isomorphism in $D(Y, q^{-1}\mathcal{O}_S)$. If we could show $\tilde{\xi}$ comes from a map of filtered complexes (with suitable filtrations), then we could appeal to the spectral sequence of Lemma 12.1 to finish the proof. This takes more work than is necessary and instead our approach will be to reduce to the affine case (whose proof does in some sense use the spectral sequence).

Indeed, if $Y' \subset Y$ is any open with inverse image $X' \subset X$, then $\tilde{\xi}|_{X'}$ induces the map

$$\bigoplus_{i=1}^N H_{dR}^*(Y'/S) \longrightarrow H_{dR}^*(X'/S), \quad (a_1, \dots, a_N) \mapsto \sum \xi_i|_{X'} \cup f^* a_i$$

on cohomology over Y' , see discussion above. Thus it suffices to find a basis for the topology on Y such that the proposition holds for the members of the basis (in particular we can forget about the map $\tilde{\xi}$ when we do this). This reduces us to the case where Y and S are affine which is handled by Lemma 13.2 and the proof is complete. $\square$

14. Projective space bundle formula

The title says it all.

Proposition 14.1. *Let $X \rightarrow S$ be a morphism of schemes. Let $\mathcal{E}$ be a locally free $\mathcal{O}_X$ -module of constant rank r . Consider the morphism $p : P = \mathbf{P}(\mathcal{E}) \rightarrow X$. Then the map*

$$\bigoplus_{i=0, \dots, r-1} H_{dR}^*(X/S) \longrightarrow H_{dR}^*(P/S)$$

given by the rule

$$(a_0, \dots, a_{r-1}) \mapsto \sum_{i=0, \dots, r-1} c_1^{dR}(\mathcal{O}_P(1))^i \cup p^*(a_i)$$

is an isomorphism.

Proof. Choose an affine open $\text{Spec}(A) \subset X$ such that $\mathcal{E}$ restricts to the trivial locally free module $\mathcal{O}_{\text{Spec}(A)}^{\oplus r}$. Then $P \times_X \text{Spec}(A) = \mathbf{P}_A^{r-1}$. Thus we see that p is proper and smooth, see Section 11. Moreover, the classes $c_1^{dR}(\mathcal{O}_P(1))^i$, $i = 0, 1, \dots, r-1$ restricted to a fibre $X_y = \mathbf{P}_y^{r-1}$ freely generate the de Rham cohomology $H_{dR}^*(X_y/y)$ over $\kappa(y)$, see Lemma 11.4. Thus we've verified the conditions of Proposition 13.3 and we win. $\square$

Remark 14.2. In the situation of Proposition 14.1 we get moreover that the map

$$\tilde{\xi} : \bigoplus_{t=0, \dots, r-1} \Omega_{X/S}^\bullet[-2t] \longrightarrow Rp_* \Omega_{P/S}^\bullet$$

is an isomorphism in $D(X, (X \rightarrow S)^{-1}\mathcal{O}_X)$ as follows immediately from the application of Proposition 13.3. Note that the arrow for $t = 0$ is simply the canonical map $c_{P/X} : \Omega_{X/S}^\bullet \rightarrow Rp_* \Omega_{P/S}^\bullet$ of Section 2. In fact, we can pin down this map further in this particular case. Namely, consider the canonical map

$$\xi' : \Omega_{P/S}^\bullet \rightarrow \Omega_{P/S}^\bullet[2]$$

of Remark 4.3 corresponding to $c_1^{dR}(\mathcal{O}_P(1))$. Then

$$\xi'[2(t-1)] \circ \dots \circ \xi'[2] \circ \xi' : \Omega_{P/S}^\bullet \rightarrow \Omega_{P/S}^\bullet[2t]$$

is the map of Remark 4.3 corresponding to $c_1^{dR}(\mathcal{O}_P(1))^t$. Tracing through the choices made in the proof of Proposition 13.3 we find the value

$$\tilde{\xi}|_{\Omega_{X/S}^\bullet[-2t]} = Rp_*\xi'[-2] \circ \dots \circ Rp_*\xi'[-2(t-1)] \circ Rp_*\xi'[-2t] \circ c_{P/X}[-2t]$$

for the restriction of our isomorphism to the summand $\Omega_{X/S}^\bullet[-2t]$. This has the following simple consequence we will use below: let

$$M = \bigoplus_{t=1, \dots, r-1} \Omega_{X/S}^\bullet[-2t] \quad \text{and} \quad K = \bigoplus_{t=0, \dots, r-2} \Omega_{X/S}^\bullet[-2t]$$

viewed as subcomplexes of the source of the arrow $\tilde{\xi}$. It follows formally from the discussion above that

$$c_{P/X} \oplus \tilde{\xi}|_M : \Omega_{X/S}^\bullet \oplus M \longrightarrow Rp_*\Omega_{P/S}^\bullet$$

is an isomorphism and that the diagram

$$\begin{array}{ccc} K & \xrightarrow{\text{id}} & M[2] \\ \tilde{\xi}|_K \downarrow & & \downarrow (\tilde{\xi}|_M)[2] \\ Rp_*\Omega_{P/S}^\bullet & \xrightarrow{Rp_*\xi'} & Rp_*\Omega_{P/S}^\bullet[2] \end{array}$$

commutes where $\text{id} : K \rightarrow M[2]$ identifies the summand corresponding to t in the decomposition of K to the summand corresponding to $t+1$ in the decomposition of M .

15. Log poles along a divisor

Let $X \rightarrow S$ be a morphism of schemes. Let $Y \subset X$ be an effective Cartier divisor. If X étale locally along Y looks like $Y \times \mathbf{A}^1$, then there is a canonical short exact sequence of complexes

$$0 \rightarrow \Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^\bullet(\log Y) \rightarrow \Omega_{Y/S}^\bullet[-1] \rightarrow 0$$

having many good properties we will discuss in this section. There is a variant of this construction where one starts with a normal crossings divisor (Étale Morphisms, Definition ??) which we will discuss elsewhere (insert future reference here).

Definition 15.1. Let $X \rightarrow S$ be a morphism of schemes. Let $Y \subset X$ be an effective Cartier divisor. We say the *de Rham complex of log poles is defined for $Y \subset X$ over S* if for all $y \in Y$ and local equation $f \in \mathcal{O}_{X,y}$ of Y we have

- (1) $\mathcal{O}_{X,y} \rightarrow \Omega_{X/S,y}$, $g \mapsto gdf$ is a split injection, and
- (2) $\Omega_{X/S,y}^p$ is f -torsion free for all p .

An easy local calculation shows that it suffices for every $y \in Y$ to find one local equation f for which conditions (1) and (2) hold.

Lemma 15.2. *Let $X \rightarrow S$ be a morphism of schemes. Let $Y \subset X$ be an effective Cartier divisor. Assume the de Rham complex of log poles is defined for $Y \subset X$ over S . There is a canonical short exact sequence of complexes*

$$0 \rightarrow \Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^\bullet(\log Y) \rightarrow \Omega_{Y/S}^\bullet[-1] \rightarrow 0$$

Proof. Our assumption is that for every $y \in Y$ and local equation $f \in \mathcal{O}_{X,y}$ of Y we have

$$\Omega_{X/S,y} = \mathcal{O}_{X,y} df \oplus M \quad \text{and} \quad \Omega_{X/S,y}^p = \wedge^{p-1}(M) df \oplus \wedge^p(M)$$

for some module M with f -torsion free exterior powers $\wedge^p(M)$. It follows that

$$\Omega_{Y/S,y}^p = \wedge^p(M/fM) = \wedge^p(M)/f \wedge^p(M)$$

Below we will tacitly use these facts. In particular the sheaves $\Omega_{X/S}^p$ have no nonzero local sections supported on Y and we have a canonical inclusion

$$\Omega_{X/S}^p \subset \Omega_{X/S}^p(Y)$$

see More on Flatness, Section ?? . Let $U = \text{Spec}(A)$ be an affine open subscheme such that $Y \cap U = V(f)$ for some nonzerodivisor $f \in A$. Let us consider the $\mathcal{O}_U$ -submodule of $\Omega_{X/S}^p(Y)|_U$ generated by $\Omega_{X/S}^p|_U$ and $d \log(f) \wedge \Omega_{X/S}^{p-1}$ where $d \log(f) = f^{-1}df$. This is independent of the choice of f as another generator of the ideal of Y on U is equal to uf for a unit $u \in A$ and we get

$$d \log(uf) - d \log(f) = d \log(u) = u^{-1}du$$

which is a section of $\Omega_{X/S}$ over U . These local sheaves glue to give a quasi-coherent submodule

$$\Omega_{X/S}^p \subset \Omega_{X/S}^p(\log Y) \subset \Omega_{X/S}^p(Y)$$

Let us agree to think of $\Omega_{Y/S}^p$ as a quasi-coherent $\mathcal{O}_X$ -module. There is a unique surjective $\mathcal{O}_X$ -linear map

$$\text{Res} : \Omega_{X/S}^p(\log Y) \rightarrow \Omega_{Y/S}^{p-1}$$

defined by the rule

$$\text{Res}(\eta' + d \log(f) \wedge \eta) = \eta|_{Y \cap U}$$

for all opens U as above and all $\eta' \in \Omega_{X/S}^p(U)$ and $\eta \in \Omega_{X/S}^{p-1}(U)$. If a form η over U restricts to zero on $Y \cap U$, then $\eta = df \wedge \eta' + f\eta''$ for some forms η' and η'' over U . We conclude that we have a short exact sequence

$$0 \rightarrow \Omega_{X/S}^p \rightarrow \Omega_{X/S}^p(\log Y) \rightarrow \Omega_{Y/S}^{p-1} \rightarrow 0$$

for all p . We still have to define the differentials $\Omega_{X/S}^p(\log Y) \rightarrow \Omega_{X/S}^{p+1}(\log Y)$. On the subsheaf $\Omega_{X/S}^p$ we use the differential of the de Rham complex of X over S . Finally, we define $d(d \log(f) \wedge \eta) = -d \log(f) \wedge d\eta$. The sign is forced on us by the Leibniz rule (on $\Omega_{X/S}^\bullet$) and it is compatible with the differential on $\Omega_{Y/S}^\bullet[-1]$ which is after all $-d_{Y/S}$ by our sign convention in Homology, Definition ?? . In this way we obtain a short exact sequence of complexes as stated in the lemma. $\square$

Definition 15.3. Let $X \rightarrow S$ be a morphism of schemes. Let $Y \subset X$ be an effective Cartier divisor. Assume the de Rham complex of log poles is defined for $Y \subset X$ over S . Then the complex

$$\Omega_{X/S}^\bullet(\log Y)$$

constructed in Lemma 15.2 is the *de Rham complex of log poles for $Y \subset X$ over S* .

This complex has many good properties.

Lemma 15.4. *Let $p : X \rightarrow S$ be a morphism of schemes. Let $Y \subset X$ be an effective Cartier divisor. Assume the de Rham complex of log poles is defined for $Y \subset X$ over S .*

- (1) *The maps $\wedge : \Omega_{X/S}^p \times \Omega_{X/S}^q \rightarrow \Omega_{X/S}^{p+q}$ extend uniquely to $\mathcal{O}_X$ -bilinear maps*

$$\wedge : \Omega_{X/S}^p(\log Y) \times \Omega_{X/S}^q(\log Y) \rightarrow \Omega_{X/S}^{p+q}(\log Y)$$

satisfying the Leibniz rule $d(\omega \wedge \eta) = d(\omega) \wedge \eta + (-1)^{\deg(\omega)} \omega \wedge d(\eta)$,

- (2) *with multiplication as in (1) the map $\Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^\bullet(\log Y)$ is a homomorphism of differential graded $\mathcal{O}_S$ -algebras,*
 (3) *via the maps in (1) we have $\Omega_{X/S}^p(\log Y) = \wedge^p(\Omega_{X/S}^1(\log Y))$, and*
 (4) *the map $\text{Res} : \Omega_{X/S}^\bullet(\log Y) \rightarrow \Omega_{Y/S}^\bullet[-1]$ satisfies*

$$\text{Res}(\omega \wedge \eta) = \text{Res}(\omega) \wedge \eta|_Y$$

for ω a local section of $\Omega_{X/S}^p(\log Y)$ and η a local section of $\Omega_{X/S}^q$.

Proof. This follows by direct calculation from the local construction of the complex in the proof of Lemma 15.2. Details omitted. $\square$

Consider a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\quad} & X \\ \downarrow & f & \downarrow \\ S' & \xrightarrow{\quad} & S \end{array}$$

of schemes. Let $Y \subset X$ be an effective Cartier divisor whose pullback $Y' = f^*Y$ is defined (Divisors, Definition ??). Assume the de Rham complex of log poles is defined for $Y \subset X$ over S and the de Rham complex of log poles is defined for $Y' \subset X'$ over S' . In this case we obtain a map of short exact sequences of complexes

$$\begin{array}{ccccccc} 0 & \longrightarrow & f^{-1}\Omega_{X/S}^\bullet & \longrightarrow & f^{-1}\Omega_{X/S}^\bullet(\log Y) & \longrightarrow & f^{-1}\Omega_{Y/S}^\bullet[-1] \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \Omega_{X'/S'}^\bullet & \longrightarrow & \Omega_{X'/S'}^\bullet(\log Y') & \longrightarrow & \Omega_{Y'/S'}^\bullet[-1] \longrightarrow 0 \end{array}$$

Linearizing, for every p we obtain a linear map $f^*\Omega_{X/S}^p(\log Y) \rightarrow \Omega_{X'/S'}^p(\log Y')$.

Lemma 15.5. *Let $f : X \rightarrow S$ be a morphism of schemes. Let $Y \subset X$ be an effective Cartier divisor. Assume the de Rham complex of log poles is defined for $Y \subset X$ over S . Denote*

$$\delta : \Omega_{Y/S}^\bullet \rightarrow \Omega_{X/S}^\bullet[2]$$

in $D(X, f^{-1}\mathcal{O}_S)$ the “boundary” map coming from the short exact sequence in Lemma 15.2. Denote

$$\xi' : \Omega_{X/S}^\bullet \rightarrow \Omega_{X/S}^\bullet[2]$$

in $D(X, f^{-1}\mathcal{O}_S)$ the map of Remark 4.3 corresponding to $\xi = c_1^{dR}(\mathcal{O}_X(-Y))$. Denote

$$\zeta' : \Omega_{Y/S}^\bullet \rightarrow \Omega_{Y/S}^\bullet[2]$$

in $D(Y, f|_Y^{-1}\mathcal{O}_S)$ the map of Remark 4.3 corresponding to $\zeta = c_1^{dR}(\mathcal{O}_X(-Y)|_Y)$. Then the diagram

$$\begin{array}{ccc} \Omega_{X/S}^\bullet & \longrightarrow & \Omega_{Y/S}^\bullet \\ \xi' \downarrow & \nearrow \delta & \downarrow \zeta' \\ \Omega_{X/S}^\bullet[2] & \longrightarrow & \Omega_{Y/S}^\bullet[2] \end{array}$$

is commutative in $D(X, f^{-1}\mathcal{O}_S)$.

Proof. More precisely, we define δ as the boundary map corresponding to the shifted short exact sequence

$$0 \rightarrow \Omega_{X/S}^\bullet[1] \rightarrow \Omega_{X/S}^\bullet(\log Y)[1] \rightarrow \Omega_{Y/S}^\bullet \rightarrow 0$$

It suffices to prove each triangle commutes. Set $\mathcal{L} = \mathcal{O}_X(-Y)$. Denote $\pi : L \rightarrow X$ the line bundle with $\pi_*\mathcal{O}_L = \bigoplus_{n \geq 0} \mathcal{L}^{\otimes n}$.

Commutativity of the upper left triangle. By Lemma 10.3 the map ξ' is the boundary map of the triangle given in Lemma 10.2. By functoriality it suffices to prove there exists a morphism of short exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Omega_{X/S}^\bullet[1] & \longrightarrow & \Omega_{L^*/S,0}^\bullet[1] & \longrightarrow & \Omega_{X/S}^\bullet \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \Omega_{X/S}^\bullet[1] & \longrightarrow & \Omega_{X/S}^\bullet(\log Y)[1] & \longrightarrow & \Omega_{Y/S}^\bullet \longrightarrow 0 \end{array}$$

where the left and right vertical arrows are the obvious ones. We can define the middle vertical arrow by the rule

$$\omega' + d \log(s) \wedge \omega \mapsto \omega' + d \log(f) \wedge \omega$$

where ω', ω are local sections of $\Omega_{X/S}^\bullet$ and where s is a local generator of $\mathcal{L}$ and $f \in \mathcal{O}_X(-Y)$ is the corresponding section of the ideal sheaf of Y in X . Since the constructions of the maps in Lemmas 10.2 and 15.2 match exactly, this works.

Commutativity of the lower right triangle. Denote $\bar{L}$ the restriction of L to Y . By Lemma 10.3 the map ζ' is the boundary map of the triangle given in Lemma 10.2 using the line bundle $\bar{L}$ on Y . By functoriality it suffices to prove there exists a morphism of short exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Omega_{X/S}^\bullet[1] & \longrightarrow & \Omega_{X/S}^\bullet(\log Y)[1] & \longrightarrow & \Omega_{Y/S}^\bullet \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \Omega_{Y/S}^\bullet[1] & \longrightarrow & \Omega_{L^*/S,0}^\bullet[1] & \longrightarrow & \Omega_{Y/S}^\bullet \longrightarrow 0 \end{array}$$

where the left and right vertical arrows are the obvious ones. We can define the middle vertical arrow by the rule

$$\omega' + d \log(f) \wedge \omega \mapsto \omega'|_Y + d \log(\bar{s}) \wedge \omega|_Y$$

where ω', ω are local sections of $\Omega_{X/S}^\bullet$ and where f is a local generator of $\mathcal{O}_X(-Y)$ viewed as a function on X and where $\bar{s}$ is $f|_Y$ viewed as a section of $\mathcal{L}|_Y = \mathcal{O}_X(-Y)|_Y$. Since the constructions of the maps in Lemmas 10.2 and 15.2 match exactly, this works. $\square$

Lemma 15.6. *Let $X \rightarrow S$ be a morphism of schemes. Let $Y \subset X$ be an effective Cartier divisor. Assume the de Rham complex of log poles is defined for $Y \subset X$ over S . Let $b \in H_{dR}^m(X/S)$ be a de Rham cohomology class whose restriction to Y is zero. Then $c_1^{dR}(\mathcal{O}_X(Y)) \cup b = 0$ in $H_{dR}^{m+2}(X/S)$.*

Proof. This follows immediately from Lemma 15.5. Namely, we have

$$c_1^{dR}(\mathcal{O}_X(Y)) \cup b = -c_1^{dR}(\mathcal{O}_X(-Y)) \cup b = -\xi'(b) = -\delta(b|_Y) = 0$$

as desired. For the second equality, see Remark 4.3. $\square$

Lemma 15.7. *Let $X \rightarrow T \rightarrow S$ be morphisms of schemes. Let $Y \subset X$ be an effective Cartier divisor. If both $X \rightarrow T$ and $Y \rightarrow T$ are smooth, then the de Rham complex of log poles is defined for $Y \subset X$ over S .*

Proof. Let $y \in Y$ be a point. By More on Morphisms, Lemma ?? there exists an integer $0 \geq m$ and a commutative diagram

$$\begin{array}{ccccc} Y & \longleftarrow & V & \longrightarrow & \mathbf{A}_T^m \\ \downarrow & & \downarrow & & \downarrow (a_1, \dots, a_m) \mapsto (a_1, \dots, a_m, 0) \\ X & \longleftarrow & U & \xrightarrow{\pi} & \mathbf{A}_T^{m+1} \end{array}$$

where $U \subset X$ is open, $V = Y \cap U$, π is étale, $V = \pi^{-1}(\mathbf{A}_T^m)$, and $y \in V$. Denote $z \in \mathbf{A}_T^m$ the image of y . Then we have

$$\Omega_{X/S, y}^p = \Omega_{\mathbf{A}_T^{m+1}/S, z}^p \otimes_{\mathcal{O}_{\mathbf{A}_T^{m+1}, z}} \mathcal{O}_{X, x}$$

by Lemma 2.2. Denote $x_1, \dots, x_{m+1}$ the coordinate functions on $\mathbf{A}_T^{m+1}$. Since the conditions (1) and (2) in Definition 15.1 do not depend on the choice of the local coordinate, it suffices to check the conditions (1) and (2) when f is the image of x_{m+1} by the flat local ring homomorphism $\mathcal{O}_{\mathbf{A}_T^{m+1}, z} \rightarrow \mathcal{O}_{X, x}$. In this way we see that it suffices to check conditions (1) and (2) for $\mathbf{A}_T^m \subset \mathbf{A}_T^{m+1}$ and the point z . To prove this case we may assume $S = \text{Spec}(A)$ and $T = \text{Spec}(B)$ are affine. Let $A \rightarrow B$ be the ring map corresponding to the morphism $T \rightarrow S$ and set $P = B[x_1, \dots, x_{m+1}]$ so that $\mathbf{A}_T^{m+1} = \text{Spec}(P)$. We have

$$\Omega_{P/A} = \Omega_{B/A} \otimes_B P \oplus \bigoplus_{j=1, \dots, m} P dx_j \oplus P dx_{m+1}$$

Hence the map $P \rightarrow \Omega_{P/A}$, $g \mapsto g dx_{m+1}$ is a split injection and x_{m+1} is a nonzerodivisor on $\Omega_{P/A}^p$ for all $p \geq 0$. Localizing at the prime ideal corresponding to z finishes the proof. $\square$

Remark 15.8. Let S be a locally Noetherian scheme. Let X be locally of finite type over S . Let $Y \subset X$ be an effective Cartier divisor. If the map

$$\mathcal{O}_{X, y}^\wedge \longrightarrow \mathcal{O}_{Y, y}^\wedge$$

has a section for all $y \in Y$, then the de Rham complex of log poles is defined for $Y \subset X$ over S . If we ever need this result we will formulate a precise statement and add a proof here.

Remark 15.9. Let S be a locally Noetherian scheme. Let X be locally of finite type over S . Let $Y \subset X$ be an effective Cartier divisor. If for every $y \in Y$ we can find a diagram of schemes over S

$$X \xleftarrow{\varphi} U \xrightarrow{\psi} V$$

with φ étale and $\psi|_{\varphi^{-1}(Y)} : \varphi^{-1}(Y) \rightarrow V$ étale, then the de Rham complex of log poles is defined for $Y \subset X$ over S . A special case is when the pair (X, Y) étale locally looks like $(V \times \mathbf{A}^1, V \times \{0\})$. If we ever need this result we will formulate a precise statement and add a proof here.

16. Calculations

In this section we calculate some Hodge and de Rham cohomology groups for a standard blowing up.

We fix a ring R and we set $S = \text{Spec}(R)$. Fix integers $0 \leq m$ and $1 \leq n$. Consider the closed immersion

$$Z = \mathbf{A}_S^m \longrightarrow \mathbf{A}_S^{m+n} = X, \quad (a_1, \dots, a_m) \mapsto (a_1, \dots, a_m, 0, \dots, 0).$$

We are going to consider the blowing up L of X along the closed subscheme Z . Write

$$X = \mathbf{A}_S^{m+n} = \text{Spec}(A) \quad \text{with} \quad A = R[x_1, \dots, x_m, y_1, \dots, y_n]$$

We will consider $A = R[x_1, \dots, x_m, y_1, \dots, y_n]$ as a graded R -algebra by setting $\deg(x_i) = 0$ and $\deg(y_j) = 1$. With this grading we have

$$P = \text{Proj}(A) = \mathbf{A}_S^m \times_S \mathbf{P}_S^{n-1} = Z \times_S \mathbf{P}_S^{n-1} = \mathbf{P}_Z^{n-1}$$

Observe that the ideal cutting out Z in X is the ideal A_+ . Hence L is the Proj of the Rees algebra

$$A \oplus A_+ \oplus (A_+)^2 \oplus \dots = \bigoplus_{d \geq 0} A_{\geq d}$$

Hence L is an example of the phenomenon studied in more generality in More on Morphisms, Section ??; we will use the observations we made there without further mention. In particular, we have a commutative diagram

$$\begin{array}{ccccc} P & \xrightarrow{0} & L & \xrightarrow{\pi} & P \\ p \downarrow & & \downarrow b & & \downarrow p \\ Z & \xrightarrow{i} & X & \longrightarrow & Z \end{array}$$

such that $\pi : L \rightarrow P$ is a line bundle over $P = Z \times_S \mathbf{P}_S^{n-1}$ with zero section 0 whose image $E = 0(P) \subset L$ is the exceptional divisor of the blowup b .

Lemma 16.1. *For $a \geq 0$ we have*

- (1) *the map $\Omega_{X/S}^a \rightarrow b_* \Omega_{L/S}^a$ is an isomorphism,*
- (2) *the map $\Omega_{Z/S}^a \rightarrow p_* \Omega_{P/S}^a$ is an isomorphism, and*
- (3) *the map $Rb_* \Omega_{L/S}^a \rightarrow i_* R p_* \Omega_{P/S}^a$ is an isomorphism on cohomology sheaves in degree ≥ 1 .*

Proof. Let us first prove part (2). Since $P = Z \times_S \mathbf{P}_S^{n-1}$ we see that

$$\Omega_{P/S}^a = \bigoplus_{a=r+s} \text{pr}_1^* \Omega_{Z/S}^r \otimes \text{pr}_2^* \Omega_{\mathbf{P}_S^{n-1}/S}^s$$

Recalling that $p = \text{pr}_1$ by the projection formula (Cohomology, Lemma ??) we obtain

$$p_*\Omega_{P/S}^a = \bigoplus_{a=r+s} \Omega_{Z/S}^r \otimes \text{pr}_{1,*}\text{pr}_2^*\Omega_{\mathbf{P}_S^{n-1}/S}^s$$

By the calculations in Section 11 and in particular in the proof of Lemma 11.3 we have $\text{pr}_{1,*}\text{pr}_2^*\Omega_{\mathbf{P}_S^{n-1}/S}^s = 0$ except if $s = 0$ in which case we get $\text{pr}_{1,*}\mathcal{O}_P = \mathcal{O}_Z$. This proves (2).

By the material in Section 10 and in particular Lemma 10.4 we have $\pi_*\Omega_{L/S}^a = \Omega_{P/S}^a \oplus \bigoplus_{k \geq 1} \Omega_{L/S,k}^a$. Since the composition $\pi \circ 0$ in the diagram above is the identity morphism on P to prove part (3) it suffices to show that $\Omega_{L/S,k}^a$ has vanishing higher cohomology for $k > 0$. By Lemmas 10.2 and 10.4 there are short exact sequences

$$0 \rightarrow \Omega_{P/S}^a \otimes \mathcal{O}_P(k) \rightarrow \Omega_{L/S,k}^a \rightarrow \Omega_{P/S}^{a-1} \otimes \mathcal{O}_P(k) \rightarrow 0$$

where $\Omega_{P/S}^{a-1} = 0$ if $a = 0$. Since $P = Z \times_S \mathbf{P}_S^{n-1}$ we have

$$\Omega_{P/S}^a = \bigoplus_{i+j=a} \Omega_{Z/S}^i \boxtimes \Omega_{\mathbf{P}_S^{n-1}/S}^j$$

by Lemma 8.1. Since $\Omega_{Z/S}^i$ is free of finite rank we see that it suffices to show that the higher cohomology of $\mathcal{O}_Z \boxtimes \Omega_{\mathbf{P}_S^{n-1}/S}^j(k)$ is zero for $k > 0$. This follows from Lemma 11.2 applied to $P = Z \times_S \mathbf{P}_S^{n-1} = \mathbf{P}_Z^{n-1}$ and the proof of (3) is complete.

We still have to prove (1). If $n = 1$, then we are blowing up an effective Cartier divisor and b is an isomorphism and we have (1). If $n > 1$, then the composition

$$\Gamma(X, \Omega_{X/S}^a) \rightarrow \Gamma(L, \Omega_{L/S}^a) \rightarrow \Gamma(L \setminus E, \Omega_{L/S}^a) = \Gamma(X \setminus Z, \Omega_{X/S}^a)$$

is an isomorphism as $\Omega_{X/S}^a$ is finite free (small detail omitted). Thus the only way (1) can fail is if there are nonzero elements of $\Gamma(L, \Omega_{L/S}^a)$ which vanish outside of $E = 0(P)$. Since L is a line bundle over P with zero section $0 : P \rightarrow L$, it suffices to show that on a line bundle there are no nonzero sections of a sheaf of differentials which vanish identically outside the zero section. The reader sees this is true either (preferably) by a local calculation or by using that $\Omega_{L/S,k} \subset \Omega_{L^*/S,k}$ (see references above). $\square$

We suggest the reader skip to the next section at this point.

Lemma 16.2. *For $a \geq 0$ there are canonical maps*

$$b^*\Omega_{X/S}^a \longrightarrow \Omega_{L/S}^a \longrightarrow b^*\Omega_{X/S}^a \otimes_{\mathcal{O}_L} \mathcal{O}_L((n-1)E)$$

whose composition is induced by the inclusion $\mathcal{O}_L \subset \mathcal{O}_L((n-1)E)$.

Proof. The first arrow in the displayed formula is discussed in Section 2. To get the second arrow we have to show that if we view a local section of $\Omega_{L/S}^a$ as a “meromorphic section” of $b^*\Omega_{X/S}^a$, then it has a pole of order at most $n-1$ along E . To see this we work on affine local charts on L . Namely, recall that L is covered by the spectra of the affine blowup algebras $A[\frac{I}{y_i}]$ where $I = A_+$ is the ideal generated by $y_1, \dots, y_n$. See Algebra, Section ?? and Divisors, Lemma ??. By symmetry it is enough to work on the chart corresponding to $i = 1$. Then

$$A[\frac{I}{y_1}] = R[x_1, \dots, x_m, y_1, t_2, \dots, t_n]$$

where $t_i = y_i/y_1$, see More on Algebra, Lemma ???. Thus the module $\Omega_{L/S}^1$ is over the corresponding affine open freely generated by $dx_1, \dots, dx_m, dy_1$, and $dt_1, \dots, dt_n$. Of course, the first $m+1$ of these generators come from $b^*\Omega_{X/S}^1$ and for the remaining $n-1$ we have

$$dt_j = d\frac{y_j}{y_1} = \frac{1}{y_1}dy_j - \frac{y_j}{y_1^2}dy_1 = \frac{dy_j - t_j dy_1}{y_1}$$

which has a pole of order 1 along E since E is cut out by y_1 on this chart. Since the wedges of a of these elements give a basis of $\Omega_{L/S}^a$ over this chart, and since there are at most $n-1$ of the dt_j involved this finishes the proof. $\square$

Lemma 16.3. *Let $E = 0(P)$ be the exceptional divisor of the blowing up b . For any locally free $\mathcal{O}_X$ -module $\mathcal{E}$ and $0 \leq i \leq n-1$ the map*

$$\mathcal{E} \longrightarrow Rb_*(b^*\mathcal{E} \otimes_{\mathcal{O}_L} \mathcal{O}_L(iE))$$

is an isomorphism in $D(\mathcal{O}_X)$.

Proof. By the projection formula it is enough to show this for $\mathcal{E} = \mathcal{O}_X$, see Cohomology, Lemma ???. Since X is affine it suffices to show that the maps

$$H^0(X, \mathcal{O}_X) \rightarrow H^0(L, \mathcal{O}_L) \rightarrow H^0(L, \mathcal{O}_L(iE))$$

are isomorphisms and that $H^j(X, \mathcal{O}_L(iE)) = 0$ for $j > 0$ and $0 \leq i \leq n-1$, see Cohomology of Schemes, Lemma ???. Since π is affine, we can compute global sections and cohomology after taking π_* , see Cohomology of Schemes, Lemma ???. If $n = 1$, then $L \rightarrow X$ is an isomorphism and $i = 0$ hence the first statement holds. If $n > 1$, then we consider the composition

$$H^0(X, \mathcal{O}_X) \rightarrow H^0(L, \mathcal{O}_L) \rightarrow H^0(L, \mathcal{O}_L(iE)) \rightarrow H^0(L \setminus E, \mathcal{O}_L) = H^0(X \setminus Z, \mathcal{O}_X)$$

Since $H^0(X \setminus Z, \mathcal{O}_X) = H^0(X, \mathcal{O}_X)$ in this case as Z has codimension $n \geq 2$ in X (details omitted) we conclude the first statement holds. For the second, recall that $\mathcal{O}_L(E) = \mathcal{O}_L(-1)$, see Divisors, Lemma ???. Hence we have

$$\pi_*\mathcal{O}_L(iE) = \pi_*\mathcal{O}_L(-i) = \bigoplus_{k \geq -i} \mathcal{O}_P(k)$$

as discussed in More on Morphisms, Section ???. Thus we conclude by the vanishing of the cohomology of twists of the structure sheaf on $P = \mathbf{P}_Z^{n-1}$ shown in Cohomology of Schemes, Lemma ???. $\square$

17. Blowing up and de Rham cohomology

Fix a base scheme S , a smooth morphism $X \rightarrow S$, and a closed subscheme $Z \subset X$ which is also smooth over S . Denote $b : X' \rightarrow X$ the blowing up of X along Z . Denote $E \subset X'$ the exceptional divisor. Picture

$$(17.0.1) \quad \begin{array}{ccc} E & \xrightarrow{j} & X' \\ p \downarrow & & \downarrow b \\ Z & \xrightarrow{i} & X \end{array}$$

Our goal in this section is to prove that the map $b^* : H_{dR}^*(X/S) \rightarrow H_{dR}^*(X'/S)$ is injective (although a lot more can be said).

Lemma 17.1. *With notation as in More on Morphisms, Lemma ?? for $a \geq 0$ we have*

- (1) the map $\Omega_{X/S}^a \rightarrow b_*\Omega_{X'/S}^a$ is an isomorphism,
- (2) the map $\Omega_{Z/S}^a \rightarrow p_*\Omega_{E/S}^a$ is an isomorphism,
- (3) the map $Rb_*\Omega_{X'/S}^a \rightarrow i_*Rp_*\Omega_{E/S}^a$ is an isomorphism on cohomology sheaves in degree ≥ 1 .

Proof. Let $\epsilon : X_1 \rightarrow X$ be a surjective étale morphism. Denote $i_1 : Z_1 \rightarrow X_1$, $b_1 : X'_1 \rightarrow X_1$, $E_1 \subset X'_1$, and $p_1 : E_1 \rightarrow Z_1$ the base changes of the objects considered in More on Morphisms, Lemma ???. Observe that i_1 is a closed immersion of schemes smooth over S and that b_1 is the blowing up with center Z_1 by Divisors, Lemma ??. Suppose that we can prove (1), (2), and (3) for the morphisms b_1 , p_1 , and i_1 . Then by Lemma 2.2 we obtain that the pullback by ϵ of the maps in (1), (2), and (3) are isomorphisms. As ϵ is a surjective flat morphism we conclude. Thus working étale locally, by More on Morphisms, Lemma ??, we may assume we are in the situation discussed in Section 16. In this case the lemma is the same as Lemma 16.1. $\square$

Lemma 17.2. *With notation as in More on Morphisms, Lemma ?? and denoting $f : X \rightarrow S$ the structure morphism there is a canonical distinguished triangle*

$$\Omega_{X/S}^\bullet \rightarrow Rb_*(\Omega_{X'/S}^\bullet) \oplus i_*\Omega_{Z/S}^\bullet \rightarrow i_*Rp_*(\Omega_{E/S}^\bullet) \rightarrow \Omega_{X/S}^\bullet[1]$$

in $D(X, f^{-1}\mathcal{O}_S)$ where the four maps

$$\begin{array}{ccc} \Omega_{X/S}^\bullet & \rightarrow & Rb_*(\Omega_{X'/S}^\bullet), \\ \Omega_{X/S}^\bullet & \rightarrow & i_*\Omega_{Z/S}^\bullet, \\ Rb_*(\Omega_{X'/S}^\bullet) & \rightarrow & i_*Rp_*(\Omega_{E/S}^\bullet), \\ i_*\Omega_{Z/S}^\bullet & \rightarrow & i_*Rp_*(\Omega_{E/S}^\bullet) \end{array}$$

are the canonical ones (Section 2), except with sign reversed for one of them.

Proof. Choose a distinguished triangle

$$C \rightarrow Rb_*\Omega_{X'/S}^\bullet \oplus i_*\Omega_{Z/S}^\bullet \rightarrow i_*Rp_*\Omega_{E/S}^\bullet \rightarrow C[1]$$

in $D(X, f^{-1}\mathcal{O}_S)$. It suffices to show that $\Omega_{X/S}^\bullet$ is isomorphic to C in a manner compatible with the canonical maps. By the axioms of triangulated categories there exists a map of distinguished triangles

$$\begin{array}{ccccccc} C' & \longrightarrow & b_*\Omega_{X'/S}^\bullet \oplus i_*\Omega_{Z/S}^\bullet & \longrightarrow & i_*p_*\Omega_{E/S}^\bullet & \longrightarrow & C'[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ C & \longrightarrow & Rb_*\Omega_{X'/S}^\bullet \oplus i_*\Omega_{Z/S}^\bullet & \longrightarrow & i_*Rp_*\Omega_{E/S}^\bullet & \longrightarrow & C[1] \end{array}$$

By Lemma 17.1 part (3) and Derived Categories, Proposition ?? we conclude that $C' \rightarrow C$ is an isomorphism. By Lemma 17.1 part (2) the map $i_*\Omega_{Z/S}^\bullet \rightarrow i_*p_*\Omega_{E/S}^\bullet$ is an isomorphism. Thus $C' = b_*\Omega_{X'/S}^\bullet$ in the derived category. Finally we use Lemma 17.1 part (1) tells us this is equal to $\Omega_{X/S}^\bullet$. We omit the verification this is compatible with the canonical maps. $\square$

Proposition 17.3. *With notation as in More on Morphisms, Lemma ?? the map $\Omega_{X/S}^\bullet \rightarrow Rb_*\Omega_{X'/S}^\bullet$ has a splitting in $D(X, (X \rightarrow S)^{-1}\mathcal{O}_S)$.*

Proof. Consider the triangle constructed in Lemma 17.2. We claim that the map

$$Rb_*(\Omega_{X'/S}^\bullet) \oplus i_*\Omega_{Z/S}^\bullet \rightarrow i_*Rp_*(\Omega_{E/S}^\bullet)$$

has a splitting whose image contains the summand $i_*\Omega_{Z/S}^\bullet$. By Derived Categories, Lemma ?? this will show that the first arrow of the triangle has a splitting which vanishes on the summand $i_*\Omega_{Z/S}^\bullet$ which proves the lemma. We will prove the claim by decomposing $Rp_*\Omega_{E/S}^\bullet$ into a direct sum where the first piece corresponds to $\Omega_{Z/S}^\bullet$ and the second piece can be lifted through $Rb_*\Omega_{X'/S}^\bullet$.

Proof of the claim. We may decompose X into open and closed subschemes having fixed relative dimension to S , see Morphisms, Lemma ?. Since the derived category $D(X, f^{-1}\mathcal{O}_S)$ correspondingly decomposes as a product of categories, we may assume X has fixed relative dimension N over S . We may decompose $Z = \coprod Z_m$ into open and closed subschemes of relative dimension $m \geq 0$ over S . The restriction $i_m : Z_m \rightarrow X$ of i to Z_m is a regular immersion of codimension $N - m$, see Divisors, Lemma ?. Let $E = \coprod E_m$ be the corresponding decomposition, i.e., we set $E_m = p^{-1}(Z_m)$. If $p_m : E_m \rightarrow Z_m$ denotes the restriction of p to E_m , then we have a canonical isomorphism

$$\tilde{\xi}_m : \bigoplus_{t=0, \dots, N-m-1} \Omega_{Z_m/S}^\bullet[-2t] \longrightarrow Rp_{m,*}\Omega_{E_m/S}^\bullet$$

in $D(Z_m, (Z_m \rightarrow S)^{-1}\mathcal{O}_S)$ where in degree 0 we have the canonical map $\Omega_{Z_m/S}^\bullet \rightarrow Rp_{m,*}\Omega_{E_m/S}^\bullet$. See Remark 14.2. Thus we have an isomorphism

$$\tilde{\xi} : \bigoplus_m \bigoplus_{t=0, \dots, N-m-1} \Omega_{Z_m/S}^\bullet[-2t] \longrightarrow Rp_*(\Omega_{E/S}^\bullet)$$

in $D(Z, (Z \rightarrow S)^{-1}\mathcal{O}_S)$ whose restriction to the summand $\Omega_{Z/S}^\bullet = \bigoplus \Omega_{Z_m/S}^\bullet$ of the source is the canonical map $\Omega_{Z/S}^\bullet \rightarrow Rp_*(\Omega_{E/S}^\bullet)$. Consider the subcomplexes M_m and K_m of the complex $\bigoplus_{t=0, \dots, N-m-1} \Omega_{Z_m/S}^\bullet[-2t]$ introduced in Remark 14.2. We set

$$M = \bigoplus M_m \quad \text{and} \quad K = \bigoplus K_m$$

We have $M = K[-2]$ and by construction the map

$$c_{E/Z} \oplus \tilde{\xi}|_M : \Omega_{Z/S}^\bullet \oplus M \longrightarrow Rp_*(\Omega_{E/S}^\bullet)$$

is an isomorphism (see remark referenced above).

Consider the map

$$\delta : \Omega_{E/S}^\bullet[-2] \longrightarrow \Omega_{X'/S}^\bullet$$

in $D(X', (X' \rightarrow S)^{-1}\mathcal{O}_S)$ of Lemma 15.5 with the property that the composition

$$\Omega_{E/S}^\bullet[-2] \longrightarrow \Omega_{X'/S}^\bullet \longrightarrow \Omega_{E/S}^\bullet$$

is the map θ' of Remark 4.3 for $c_1^{dR}(\mathcal{O}_{X'}(-E))|_E = c_1^{dR}(\mathcal{O}_E(1))$. The final assertion of Remark 14.2 tells us that the diagram

$$\begin{array}{ccc} K[-2] & \xrightarrow{\text{id}} & M \\ (\tilde{\xi}|_K)[-2] \downarrow & & \downarrow \tilde{x}|_M \\ Rp_*\Omega_{E/S}^\bullet[-2] & \xrightarrow{Rp_*\theta'} & Rp_*\Omega_{E/S}^\bullet \end{array}$$

commutes. Thus we see that we can obtain the desired splitting of the claim as the map

$$\begin{aligned}
Rp_*(\Omega_{E/S}^\bullet) &\xrightarrow{(c_{E/Z} \oplus \tilde{\xi}|_M)^{-1}} \Omega_{Z/S}^\bullet \oplus M \\
&\xrightarrow{\text{id} \oplus \text{id}^{-1}} \Omega_{Z/S}^\bullet \oplus K[-2] \\
&\xrightarrow{\text{id} \oplus (\tilde{\xi}|_K)[-2]} \Omega_{Z/S}^\bullet \oplus Rp_*\Omega_{E/S}^\bullet[-2] \\
&\xrightarrow{\text{id} \oplus Rb_*\delta} \Omega_{Z/S}^\bullet \oplus Rb_*\Omega_{X'/S}^\bullet
\end{aligned}$$

The relationship between θ' and δ stated above together with the commutative diagram involving θ' , $\tilde{\xi}|_K$, and $\tilde{\xi}|_M$ above are exactly what's needed to show that this is a section to the canonical map $\Omega_{Z/S}^\bullet \oplus Rb_*(\Omega_{X'/S}^\bullet) \rightarrow Rp_*(\Omega_{E/S}^\bullet)$ and the proof of the claim is complete. $\square$

Lemma 17.5 shows that producing the splitting on Hodge cohomology is a good deal easier than the result of Proposition 17.3. We urge the reader to skip ahead to the next section.

Lemma 17.4. *Let $i : Z \rightarrow X$ be a closed immersion of schemes which is regular of codimension c . Then $\text{Ext}_{\mathcal{O}_X}^q(i_*\mathcal{F}, \mathcal{E}) = 0$ for $q < c$ for $\mathcal{E}$ locally free on X and $\mathcal{F}$ any $\mathcal{O}_Z$ -module.*

Proof. By the local to global spectral sequence of Ext it suffices to prove this affine locally on X . See Cohomology, Section ???. Thus we may assume $X = \text{Spec}(A)$ and there exists a regular sequence $f_1, \dots, f_c$ in A such that $Z = \text{Spec}(A/(f_1, \dots, f_c))$. We may assume $c \geq 1$. Then we see that $f_1 : \mathcal{E} \rightarrow \mathcal{E}$ is injective. Since $i_*\mathcal{F}$ is annihilated by f_1 this shows that the lemma holds for $i = 0$ and that we have a surjection

$$\text{Ext}_{\mathcal{O}_X}^{q-1}(i_*\mathcal{F}, \mathcal{E}/f_1\mathcal{E}) \longrightarrow \text{Ext}_{\mathcal{O}_X}^q(i_*\mathcal{F}, \mathcal{E})$$

Thus it suffices to show that the source of this arrow is zero. Next we repeat this argument: if $c \geq 2$ the map $f_2 : \mathcal{E}/f_1\mathcal{E} \rightarrow \mathcal{E}/f_1\mathcal{E}$ is injective. Since $i_*\mathcal{F}$ is annihilated by f_2 this shows that the lemma holds for $q = 1$ and that we have a surjection

$$\text{Ext}_{\mathcal{O}_X}^{q-2}(i_*\mathcal{F}, \mathcal{E}/f_1\mathcal{E} + f_2\mathcal{E}) \longrightarrow \text{Ext}_{\mathcal{O}_X}^{q-1}(i_*\mathcal{F}, \mathcal{E}/f_1\mathcal{E})$$

Continuing in this fashion the lemma is proved. $\square$

Lemma 17.5. *With notation as in More on Morphisms, Lemma ?? for $a \geq 0$ there is a unique arrow $Rb_*\Omega_{X'/S}^a \rightarrow \Omega_{X/S}^a$ in $D(\mathcal{O}_X)$ whose composition with $\Omega_{X/S}^a \rightarrow Rb_*\Omega_{X'/S}^a$ is the identity on $\Omega_{X/S}^a$.*

Proof. We may decompose X into open and closed subschemes having fixed relative dimension to S , see Morphisms, Lemma ???. Since the derived category $D(X, f^{-1}\mathcal{O}_S)$ correspondingly decomposes as a product of categories, we may assume X has fixed relative dimension N over S . We may decompose $Z = \coprod Z_m$ into open and closed subschemes of relative dimension $m \geq 0$ over S . The restriction $i_m : Z_m \rightarrow X$ of i to Z_m is a regular immersion of codimension $N - m$, see Divisors, Lemma ???. Let $E = \coprod E_m$ be the corresponding decomposition, i.e., we set $E_m = p^{-1}(Z_m)$. We claim that there are natural maps

$$b^*\Omega_{X/S}^a \rightarrow \Omega_{X'/S}^a \rightarrow b^*\Omega_{X/S}^a \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{X'}(\sum (N - m - 1)E_m)$$

whose composition is induced by the inclusion $\mathcal{O}_{X'} \rightarrow \mathcal{O}_{X'}(\sum(N-m-1)E_m)$. Namely, in order to prove this, it suffices to show that the cokernel of the first arrow is locally on X' annihilated by a local equation of the effective Cartier divisor $\sum(N-m-1)E_m$. To see this in turn we can work étale locally on X as in the proof of Lemma 17.1 and apply Lemma 16.2. Computing étale locally using Lemma 16.3 we see that the induced composition

$$\Omega_{X/S}^a \rightarrow Rb_*\Omega_{X'/S}^a \rightarrow Rb_*\left(b^*\Omega_{X/S}^a \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{X'}(\sum(N-m-1)E_m)\right)$$

is an isomorphism in $D(\mathcal{O}_X)$ which is how we obtain the existence of the map in the lemma.

For uniqueness, it suffices to show that there are no nonzero maps from $\tau_{\geq 1}Rb_*\Omega_{X'/S}$ to $\Omega_{X/S}^a$ in $D(\mathcal{O}_X)$. For this it suffices in turn to show that there are no nonzero maps from $R^qb_*\Omega_{X'/S}[-q]$ to $\Omega_{X/S}^a$ in $D(\mathcal{O}_X)$ for $q \geq 1$ (details omitted). By Lemma 17.1 we see that $R^qb_*\Omega_{X'/S} \cong i_*R^qp_*\Omega_{E/S}^a$ is the pushforward of a module on $Z = \coprod Z_m$. Moreover, observe that the restriction of $R^qp_*\Omega_{E/S}^a$ to Z_m is nonzero only for $q < N-m$. Namely, the fibres of $E_m \rightarrow Z_m$ have dimension $N-m-1$ and we can apply Limits, Lemma ???. Thus the desired vanishing follows from Lemma 17.4. $\square$

18. Comparing sheaves of differential forms

The goal of this section is to construct for every locally quasi-finite syntomic morphism of schemes $f : Y \rightarrow X$ maps

$$c_{Y/X}^p : \Omega_{Y/\mathbf{Z}}^p \longrightarrow f^*\Omega_{X/\mathbf{Z}}^p \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$$

for all $p \geq 0$ satisfying the following properties

- (1) $c_{Y/X}^0(1) = \delta(NL_{Y/X})$ see Discriminants, Section ??,
- (2) $c_{Y/X}^{q+p}(\omega \wedge \eta) = \omega \wedge c_{Y/X}^p(\eta)$ for local sections ω of $f^*\Omega_{X/\mathbf{Z}}^q$ and η of $\Omega_{Y/\mathbf{Z}}^p$,
- (3) the formation of $c_{Y/X}^p$ is compatible with restriction to opens and with base change (as formulated in Lemma 18.4).

Note that these conditions imply that the maps $c_{Y/X}^p$ are $\mathcal{O}_Y$ -linear and that the composition with $f^*\Omega_{X/\mathbf{Z}}^p \rightarrow \Omega_{Y/\mathbf{Z}}^p$ is multiplication by $\delta(NL_{Y/X})$. We will in fact show that there exists a unique collection of operators $c_{Y/X}^p$ with properties (1), (2), (3). This result will be applied in Section 19 to the construction of the trace map on the de Rham complexes if f is finite.

Lemma 18.1. *Let R be a ring and consider a commutative diagram*

$$\begin{array}{ccccccc} 0 & \longrightarrow & K^0 & \longrightarrow & L^0 & \longrightarrow & M^0 \longrightarrow 0 \\ & & & & \uparrow \partial & & \uparrow \\ & & & & L^{-1} & \xlongequal{\quad} & M^{-1} \end{array}$$

of R -modules with exact top row and M^0 and M^{-1} finite free of the same rank. For $p \geq 0$ there are canonical maps

$$c^p : \wedge^p(H^0(L^\bullet)) \longrightarrow \wedge^p(K^0) \otimes_R \det(M^\bullet)$$

with the following properties:

- (1) $c^0(1) = \delta(M^\bullet)$, and

(2) $c^{q+p}(\omega \wedge \eta) = \omega \wedge c^p(\eta)$ for $\omega \in \wedge^q(K^0)$ and $\eta \in \wedge^p(H^0(L^\bullet))$.

In particular, the composition of c^p with the map $\wedge^p(K^0) \rightarrow \wedge^p(H^0(L^\bullet))$ is multiplication with $\delta(M^\bullet)$.

Proof. Say M^0 and M^{-1} are free of rank n . For every $p \geq 0$ there is a unique surjection

$$\pi_p : \wedge^{p+n}(L^0) \longrightarrow \wedge^p(K^0) \otimes \wedge^n(M^0)$$

with the following two properties; (i) we have $\pi_p(k_1 \wedge \dots \wedge k_p \wedge l_1 \wedge \dots \wedge l_n) = k_1 \wedge \dots \wedge k_p \otimes m_1 \wedge \dots \wedge m_n$ if $k_1, \dots, k_p \in K^0$ and $l_1, \dots, l_n \in L^0$ map to $m_1, \dots, m_n \in M^0$ and (ii) the kernel of π_p is the submodule generated by wedges $l_1 \wedge \dots \wedge l_{p+n}$ such that $> p$ of the l_j are in K^0 . Pick a basis $e_1, \dots, e_n$ for $L^{-1} = M^{-1}$. Then we define our map by the rule

$$\eta \mapsto \pi_p(\tilde{\eta} \wedge \partial(e_1) \wedge \dots \wedge \partial(e_n)) \otimes (e_1 \wedge \dots \wedge e_n)^{\otimes -1}$$

where $\eta \in \wedge^p(H^0(L^\bullet))$ and $\tilde{\eta} \in \wedge^p(L^0)$ is a lift of η . The expression on the right hand side makes sense because $\det(M^\bullet) = \wedge^n(M^0) \otimes \wedge^n(M^{-1})^{\otimes -1}$. The expression on the right is independent of the choice of $\tilde{\eta}$ since $\partial(\xi) \wedge \partial(e_1) \wedge \dots \wedge \partial(e_n) = 0$ for any $\xi \in L^{-1}$. A straightforward calculation shows that the map is independent of the choice of the basis of $L^{-1} = M^{-1}$. It is immediate from the definition of $\delta(M^\bullet)$ in More on Algebra, Section ?? that $c^0(1) = \delta(M^\bullet)$. Finally, the rule in statement (2) can be directly verified from the definition. $\square$

Lemma 18.2. Let $R_1 \rightarrow R_2$ be a ring homomorphism. For $i = 1, 2$ consider commutative diagrams

$$\begin{array}{ccccccc} 0 & \longrightarrow & K_i^0 & \longrightarrow & L_i^0 & \longrightarrow & M_i^0 \longrightarrow 0 \\ & & & & \uparrow \partial_i & & \uparrow \\ & & & & L_i^{-1} & = & M_i^{-1} \end{array}$$

of R_i -modules as in Lemma 18.1. Assume we have maps $K_1^0 \rightarrow K_2^0$, $L_1^j \rightarrow L_2^j$, $M_1^j \rightarrow M_2^j$ compatible with the given ring map $R_1 \rightarrow R_2$ and compatible with the maps in the displayed diagrams. If the maps $M_1^j \otimes_{R_1} R_2 \rightarrow M_2^j$ are isomorphisms, then the diagrams

$$\begin{array}{ccc} \wedge^p(H^0(L_1^\bullet)) & \xrightarrow{c^p} & \wedge^p(K_1^0) \otimes_{R_1} \det(M_1^\bullet) \\ \downarrow & & \downarrow \\ \wedge^p(H^0(L_2^\bullet)) & \xrightarrow{c^p} & \wedge^p(K_2^0) \otimes_{R_2} \det(M_2^\bullet) \end{array}$$

commute.

Proof. This follows from the explicit description of the map given in the proof of Lemma 18.1. Note that we need the assumption that $M_1^j \otimes_{R_1} R_2 \rightarrow M_2^j$ are isomorphisms to see that the basis we use for M_1^{-1} maps to a basis for M_2^{-1} (and hence in checking the compatibility in both constructions we can use the “same” basis). $\square$

Remark 18.3. Let A be a ring. Let $P = A[x_1, \dots, x_n]$. Let $f_1, \dots, f_n \in P$ and set $B = P/(f_1, \dots, f_n)$. Assume $A \rightarrow B$ is quasi-finite. Then B is a relative global complete intersection over A (Algebra, Definition ??) and $(f_1, \dots, f_n)/(f_1, \dots, f_n)^2$

is free with generators the classes $\bar{f}_i$ by Algebra, Lemma ?? . Consider the following diagram

$$\begin{array}{ccccc} \Omega_{A/\mathbf{Z}} \otimes_A B & \longrightarrow & \Omega_{P/\mathbf{Z}} \otimes_P B & \longrightarrow & \Omega_{P/A} \otimes_P B \\ & & \uparrow & & \uparrow \\ & & (f_1, \dots, f_n)/(f_1, \dots, f_n)^2 & = & (f_1, \dots, f_n)/(f_1, \dots, f_n)^2 \end{array}$$

The right column represents $NL_{B/A}$ in $D(B)$ hence has cohomology $\Omega_{B/A}$ in degree 0. The top row is the split short exact sequence $0 \rightarrow \Omega_{A/\mathbf{Z}} \otimes_A B \rightarrow \Omega_{P/\mathbf{Z}} \otimes_P B \rightarrow \Omega_{P/A} \otimes_P B \rightarrow 0$. The middle column has cohomology $\Omega_{B/\mathbf{Z}}$ in degree 0 by Algebra, Lemma ?? . Thus by Lemma 18.1 we obtain canonical B -module maps

$$\Omega_{B/\mathbf{Z}}^p \longrightarrow \Omega_{A/\mathbf{Z}}^p \otimes_A \det(NL_{B/A})$$

for $p \geq 0$ satisfying properties (1) and (2) and in particular satisfy the property that the composition with $\Omega_{A/\mathbf{Z}}^p \rightarrow \Omega_{B/\mathbf{Z}}^p$ is multiplication by $\delta(NL_{B/A})$.

Lemma 18.4. *Consider a commutative diagram*

$$\begin{array}{ccc} Y' & \xrightarrow{\quad} & Y \\ f' \downarrow & & \downarrow f \\ X' & \xrightarrow{\quad a \quad} & X \end{array}$$

of schemes which induces an isomorphism of Y' with an open subscheme of $X' \times_X Y$. Assume f is locally quasi-finite and syntomic and assume given maps $c_{Y/X}^p : \Omega_{Y/\mathbf{Z}}^p \rightarrow f^* \Omega_{X/\mathbf{Z}}^p \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$ for $p \geq 0$ satisfying (1) and (2). Then there is at most one collection of maps $c_{Y'/X'}^p : \Omega_{Y'/\mathbf{Z}}^p \rightarrow (f')^* \Omega_{X'/\mathbf{Z}}^p \otimes_{\mathcal{O}_{Y'}} \det(NL_{Y'/X'})$ for $p \geq 0$ satisfying (1) and (2) such that the diagrams

$$\begin{array}{ccc} b^* \Omega_{Y/\mathbf{Z}}^p & \xrightarrow{b^* c_{Y/X}^p} & b^* (f^* \Omega_{X/\mathbf{Z}}^p \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})) = (f')^* a^* \Omega_{X/\mathbf{Z}}^p \otimes_{\mathcal{O}_Y} b^* \det(NL_{Y/X}) \\ \downarrow & & \downarrow \\ \Omega_{Y'/\mathbf{Z}}^p & \xrightarrow{c_{Y'/X'}^p} & (f')^* \Omega_{X'/\mathbf{Z}}^p \otimes_{\mathcal{O}_{Y'}} \det(NL_{Y'/X'}) \end{array}$$

commute for all $p \geq 0$. Here the vertical arrows use the maps $b^* \Omega_{Y/\mathbf{Z}}^p \rightarrow \Omega_{Y'/\mathbf{Z}}^p$ and $a^* \Omega_{X/\mathbf{Z}}^p \rightarrow \Omega_{X'/\mathbf{Z}}^p$ of Section 2 as well as the identification $b^* \det(NL_{Y/X}) = \det(NL_{Y'/X'})$ of Discriminants, Section ?? .

Proof. The map

$$(f')^* \Omega_{X'/\mathbf{Z}} \oplus b^* \Omega_{Y/\mathbf{Z}} \longrightarrow \Omega_{Y'/\mathbf{Z}}$$

is surjective because Y' is an open subscheme of the fibre product; the corresponding algebra statement is that $\Omega_{B \otimes_A C/\mathbf{Z}}$ is generated as a module by the images of $\Omega_{B/\mathbf{Z}}$ and $\Omega_{C/\mathbf{Z}}$. Thus for all p the map

$$\bigoplus_{i=0, \dots, p} (f')^* \Omega_{X'/\mathbf{Z}}^i \otimes b^* \Omega_{Y/\mathbf{Z}}^{p-i} \longrightarrow \Omega_{Y'/\mathbf{Z}}^p$$

is surjective. Conditions (1) and (2) combined with the commutativity of the diagram in the lemma, implies that the map $c_{Y'/X'}^p$ is determined (but existence is not immediate). $\square$

Lemma 18.5. *Consider a commutative diagram*

$$\begin{array}{ccc} Y' & \xrightarrow{b} & Y \\ f' \downarrow & & \downarrow f \\ X' & \xrightarrow{a} & X \end{array}$$

of schemes which induces an isomorphism of Y' with an open subscheme of $X' \times_X Y$. Assume f is locally quasi-finite and syntomic. Then for every $y' \in Y'$ we can find opens $V' \subset Y'$, $V \subset Y$, $U' \subset X'$, $U \subset X$ with $y' \in V'$, with $f'(V') \subset U'$, $b(V') \subset V$, $a(U') \subset U$, $f(V) \subset U$, and such that for $p \geq 0$ there are maps $c_{V/U}^p$ and $c_{V'/U'}^p$ satisfying (1) and (2) which are compatible with the diagram

$$\begin{array}{ccc} V' & \xrightarrow{b} & V \\ \downarrow & & \downarrow \\ U' & \xrightarrow{a} & U \end{array}$$

in the sense explained in Lemma 18.4.

Proof. It is clear that we may replace Y' by $X' \times_X Y$ in order to prove this. Pick affine opens $V \subset Y$ and $U \subset X$ with V containing the image of y' as in Discriminants, Lemma ?? part (5). Pick an affine open $U' \subset X'$ containing the image of y' and mapping into U . After replacing X, Y, X', Y' by $U, V, U', U' \times_U V$ we may assume we are in the situation described in the next paragraph.

Assume that $X' = \text{Spec}(A')$, $X = \text{Spec}(A)$, $Y = \text{Spec}(B)$, and $Y' = \text{Spec}(A' \otimes_A B)$ with $B = A[x_1, \dots, x_n]/(f_1, \dots, f_n)$ a relative global complete intersection over A . Then also $B' = A'[x_1, \dots, x_n]/(f'_1, \dots, f'_n)$ is a relative global complete intersection over A' where f'_i is the image of f_i in $A'[x_1, \dots, x_n]$, see Algebra, Lemma ??. The construction in Remark 18.3 provides us with the maps $c_{V/U}^p$ and $c_{V'/U'}^p$. These maps are compatible by Lemma 18.2 and the compatibility of the presentations of B and B' over A and A' . Some details omitted. $\square$

Lemma 18.6. *There exists a unique rule that to every locally quasi-finite syntomic morphism of schemes $f : Y \rightarrow X$ assigns $\mathcal{O}_Y$ -module maps*

$$c_{Y/X}^p : \Omega_{Y/\mathbf{Z}}^p \longrightarrow f^* \Omega_{X/\mathbf{Z}}^p \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$$

for $p \geq 0$ satisfying (1), (2), and (3). In particular, the composition of $c_{Y/X}^p$ with $f^ \Omega_{X/\mathbf{Z}}^p \rightarrow \Omega_{Y/\mathbf{Z}}^p$ is multiplication by $\delta(NL_{Y/X})$.*

Proof. This proof is very similar to the proof of Discriminants, Proposition ?? and we suggest the reader look at that proof first.

Let us reformulate the statement. Consider the category $\mathcal{C}$ whose objects, denoted Y/X , are locally quasi-finite syntomic morphism $f : Y \rightarrow X$ of schemes and whose morphisms $b/a : Y'/X' \rightarrow Y/X$ are commutative diagrams

$$\begin{array}{ccc} Y' & \xrightarrow{b} & Y \\ f' \downarrow & & \downarrow f \\ X' & \xrightarrow{a} & X \end{array}$$

which induce an isomorphism of Y' with an open subscheme of $X' \times_X Y$. The lemma means that for every object Y/X of $\mathcal{C}$ we have maps $c_{Y/X}^p$, $p \geq 0$ with properties (1) and (2), and for every morphism $b/a : Y'/X' \rightarrow Y/X$ of $\mathcal{C}$ we have that $c_{Y/X}^p$ and $c_{Y'/X'}^p$ are compatible as in Lemma 18.4.

Given Y/X in $\mathcal{C}$ and $y \in Y$ we can find an affine open $V \subset Y$ and $U \subset X$ with $f(V) \subset U$ such that there exists some maps

$$\Omega_{Y/\mathbf{Z}}^p|_V \longrightarrow \left(f^* \Omega_{X/\mathbf{Z}}^p \otimes_{\mathcal{O}_Y} \det(NL_{Y/X}) \right)|_V$$

with properties (1) and (2). This follows from picking affine opens as in Discriminants, Lemma ?? part (5) and then using the construction in Remark 18.3.

Note that the étale locus of f is exactly the set of points where $\delta(NL_{Y/X})$ does not vanish, see discussion in Discriminants, Section ?. In particular $f^* \Omega_{X/\mathbf{Z}}^p \rightarrow \Omega_{Y/\mathbf{Z}}^p$ becomes an isomorphism after inverting $\delta(NL_{Y/X})$. We conclude that if $\Omega_{X/\mathbf{Z}}^p$ is finite locally free and the annihilator of $\delta(NL_{Y/X})$ in $\mathcal{O}_Y$ is zero, then the local maps constructed in the previous paragraph are unique and automatically glue!

Let $\mathcal{C}_{nice} \subset \mathcal{C}$ denote the full subcategory of Y/X such that

- (1) $\Omega_{X/\mathbf{Z}}$ is locally free, and
- (2) the annihilator of $\delta(NL_{Y/X})$ in $\mathcal{O}_Y$ is zero.

By the remarks in the previous paragraph, we see that for any object Y/X of $\mathcal{C}_{nice}$ we have unique maps $c_{Y/X}^p$, $p \geq 0$ satisfying conditions (1) and (2). If $b/a : Y'/X' \rightarrow Y/X$ is a morphism of $\mathcal{C}_{nice}$, then the maps $c_{Y/X}^p$ and $c_{Y'/X'}^p$ are compatible as in Lemma 18.4: namely, locally there do exist compatible maps by Lemma 18.5 and the uniqueness just mentioned shows these maps agree with $c_{Y/X}^p$ and $c_{Y'/X'}^p$. In other words, we have solved the problem on the full subcategory $\mathcal{C}_{nice}$. For Y/X in $\mathcal{C}_{nice}$ we continue to denote $c_{Y/X}^p$ the solution we've just found.

In fact, more generally, suppose we have a morphism $b/a : Y'/X' \rightarrow Y/X$ of $\mathcal{C}$ such that Y/X is an object of $\mathcal{C}_{nice}$. The same argument, but this time using the uniqueness in Lemma 18.4, tells us there are maps $c_{Y'/X'}^p$, $p \geq 0$ satisfying conditions (1) and (2) compatible with the already constructed maps $c_{Y/X}^p$. Let us call this the base change $(b/a)^* c_{Y/X}^p$.

Consider morphisms

$$Y_1/X_1 \xleftarrow{b_1/a_1} Y/X \xrightarrow{b_2/a_2} Y_2/X_2$$

in $\mathcal{C}$ such that Y_1/X_1 and Y_2/X_2 are objects of $\mathcal{C}_{nice}$. **Claim.** The two base changes $(b_i/a_i)^* c_{Y_i/X_i}^p$, $i = 1, 2$ are equal. We will first show that the claim implies the lemma and then we will prove the claim.

Let $d, n \geq 1$ and consider the locally quasi-finite syntomic morphism $Y_{n,d} \rightarrow X_{n,d}$ constructed in Discriminants, Example ?. Then $Y_{n,d}$ and $X_{n,d}$ are irreducible schemes of finite type and smooth over $\mathbf{Z}$. Namely, $X_{n,d}$ is a spectrum of a polynomial ring over $\mathbf{Z}$ and $Y_{n,d}$ is an open subscheme of such. The morphism $Y_{n,d} \rightarrow X_{n,d}$ is locally quasi-finite syntomic and étale over a dense open, see Discriminants, Lemma ?. Thus $\delta(NL_{Y_{n,d}/X_{n,d}})$ is nonzero: for example we have the local description of $\delta(NL_{Y/X})$ in Discriminants, Remark ?? and we have the local description of étale morphisms in Morphisms, Lemma ?? part (8). Now a nonzero section of

an invertible module over an irreducible regular scheme has vanishing annihilator. Thus $Y_{n,d}/X_{n,d}$ is an object of $\mathcal{C}_{nice}$.

Let Y/X be an arbitrary object of $\mathcal{C}$. Let $y \in Y$. By Discriminants, Lemma ?? we can find $n, d \geq 1$ and morphisms

$$Y/X \leftarrow V/U \xrightarrow{b/a} Y_{n,d}/X_{n,d}$$

of $\mathcal{C}$ such that $V \subset Y$ and $U \subset X$ are open. Then we have the base change $c_{V/U}^p = (b/a)^* c_{Y_{n,d}/X_{n,d}}^p$. The claim guarantees these locally constructed maps $c_{V/U}^p$ glue! Thus we get a well defined global maps $c_{Y/X}^p$ with properties (1) and (2). Finally, let $b/a : Y'/X' \rightarrow Y/X$ be an arbitray morphism of $\mathcal{C}$. We have the maps $c_{Y'/X'}^p$, $p \geq 0$ and $c_{Y/X}^p$, $p \geq 0$ constructed in this paragraph. To check they are compatible as in Lemma 18.4 we may work locally on Y'/X' and Y/X . Thus we may assume that there exists a morphism $Y/X \rightarrow Y_{n,d}/X_{n,d}$. By construction, both the family $c_{Y'/X'}^p$ and the family $c_{X/Y}^p$ are compatible with the map towards $Y_{n,d}/X_{n,d}$. It follows easily from this (and the uniqueness in Lemma 18.4) that $c_{Y'/X'}^p$ is compatible with $c_{Y/X}^p$. Thus it remains to prove the claim.

In the rest of the proof we prove the claim. We may pick a point $y \in Y$ and prove the maps agree in an open neighbourhood of y . Thus we may replace Y_1, Y_2 by open neighbourhoods of the image of y in Y_1 and Y_2 . Therefore we may assume Y, X, Y_1, X_1, Y_2, X_2 are affine, say they are the spectra of rings B, A, B_1, A_1, B_2, A_2 . Picture

$$\begin{array}{ccccc} B_1 & \longrightarrow & B & \longleftarrow & B_2 \\ \uparrow & & \uparrow & & \uparrow \\ A_1 & \longrightarrow & A & \longleftarrow & A_2 \end{array}$$

By assumption the spectrum of B is an affine open of both the spectrum of $A \otimes_{A_1} B_1$ and $A \otimes_{A_2} B_2$. Shrinking more we may assume there exist elements $g_i \in A \otimes_{A_i} B_i$ such that our maps give isomorphisms $(A \otimes_{A_i} B_i)_{g_i} = B$, see Properties, Lemma ???. Let x_α, y_β be a sufficiently large collection of variables such that we may choose surjections

$$A'_1 = A_1[x_\alpha] \rightarrow A \quad \text{and} \quad A'_2 = A_2[y_\beta] \rightarrow A$$

of A_1 and A_2 -algebras. Then we can choose lifts $h_i \in A'_i \otimes_{A_i} B_i$ of $g_i \in A \otimes_{A_i} B_i$ and we consider the diagram

$$\begin{array}{ccccc} (A'_1 \otimes_{A_1} B_1)_{h_1} & \longrightarrow & B & \longleftarrow & (A'_2 \otimes_{A_1} B_2)_{h_2} \\ \uparrow & & \uparrow & & \uparrow \\ A'_1 & \longrightarrow & A & \longleftarrow & A'_2 \end{array}$$

By construction the two squares are cocartesian. Next, we consider the ring map

$$A' = A'_1 \times_A A'_2 \longrightarrow B' = (A'_1 \otimes_{A_1} B_1)_{h_1} \times_B (A'_2 \otimes_{A_1} B_2)_{h_2}$$

By More on Algebra, Lemma ?? we have $A'_1 \otimes_{A'} B' = (A'_1 \otimes_{A_1} B_1)_{h_1}$ and $A'_2 \otimes_{A'} B' = (A'_2 \otimes_{A_2} B_2)_{h_2}$. In particular the fibres of the morphism $Y' = \text{Spec}(B') \rightarrow \text{Spec}(A') = X'$ are open subschemes of base changes of the fibres of the maps $Y_i \rightarrow X_i$, hence finite and local complete intersections (see Discriminants, Lemma ??). By More on Algebra, Lemma ?? the ring map $A' \rightarrow B'$ is flat and of finite

presentation. Thus by Discriminants, Lemma ?? part (3) we conclude that Y'/X' is an object of $\mathcal{C}$. Consider now the commutative diagram

$$\begin{array}{ccccc}
 & & Y/X & & \\
 & \swarrow^{b_1/a_1} & & \searrow^{b_2/a_2} & \\
 & Y'_1/X'_1 & & Y'_2/X'_2 & \\
 & \swarrow & & \searrow & \\
 Y_1/X_1 & & Y'/X' & & Y_2/X_2
 \end{array}$$

where Y'_i/X'_i corresponds to $A'_i \rightarrow (A'_i \otimes_{A_i} B_i)_{h_i}$. Note that Y'_i/X'_i is an object of $\mathcal{C}_{nice}$ as it obtained by taking an open subscheme of the product of an infinite dimensional affine space with Y_i/X_i ; small detail omitted. In particular, the pullback of c_{Y_i/X_i}^p via (b_i/a_i) is the same as the pullback of $c_{Y'_i/X'_i}^p$ via $Y/X \rightarrow Y'_i/X'_i$. Now we would be done if Y'/X' is an object of $\mathcal{C}_{nice}$, but this is almost never the case. Namely, then pulling back $c_{Y'/X'}^p$ around the two sides of the square, we would obtain the desired conclusion. To get around the problem that Y'/X' is not in $\mathcal{C}_{nice}$ we note the arguments above show that, after possibly shrinking all of the schemes $X, Y, X'_1, Y'_1, X'_2, Y'_2, X', Y'$ we can find some $n, d \geq 1$, and extend the diagram like so:

$$\begin{array}{ccc}
 & Y/X & \\
 & \swarrow & \searrow \\
 Y'_1/X'_1 & & Y'_2/X'_2 \\
 & \swarrow & \searrow \\
 & Y'/X' & \\
 & \downarrow & \\
 & Y_{n,d}/X_{n,d} &
 \end{array}$$

and then we can use the already given argument by pulling back from $c_{Y_{n,d}/X_{n,d}}^p$. This finishes the proof. $\square$

19. Trace maps on de Rham complexes

A reference for some of the material in this section is [Gar84]. Let S be a scheme. Let $f : Y \rightarrow X$ be a finite locally free morphism of schemes over S . Then there is a trace map $\text{Trace}_f : f_* \mathcal{O}_Y \rightarrow \mathcal{O}_X$, see Discriminants, Section ???. In this situation a trace map on de Rham complexes is a map of complexes

$$\Theta_{Y/X} : f_* \Omega_{Y/S}^\bullet \longrightarrow \Omega_{X/S}^\bullet$$

such that $\Theta_{Y/X}$ is equal to Trace_f in degree 0 and satisfies

$$\Theta_{Y/X}(\omega \wedge \eta) = \omega \wedge \Theta_{Y/X}(\eta)$$

for local sections ω of $\Omega_{X/S}^\bullet$ and η of $f_* \Omega_{Y/S}^\bullet$. It is not clear to us whether such a trace map $\Theta_{Y/X}$ exists for every finite locally free morphism $Y \rightarrow X$; please email stacks.project@gmail.com if you have a counterexample or a proof.

Example 19.1. Here is an example where we do not have a trace map on de Rham complexes. For example, consider the $\mathbf{C}$ -algebra $B = \mathbf{C}[x, y]$ with action of $G = \{\pm 1\}$ given by $x \mapsto -x$ and $y \mapsto -y$. The invariants $A = B^G$ form a normal domain of finite type over $\mathbf{C}$ generated by x^2, xy, y^2 . We claim that for the inclusion $A \subset B$ there is no reasonable trace map $\Omega_{B/\mathbf{C}} \rightarrow \Omega_{A/\mathbf{C}}$ on 1-forms. Namely, consider the element $\omega = xdy \in \Omega_{B/\mathbf{C}}$. Since ω is invariant under the action of G if a “reasonable” trace map exists, then 2ω should be in the image of $\Omega_{A/\mathbf{C}} \rightarrow \Omega_{B/\mathbf{C}}$. This is not the case: there is no way to write 2ω as a linear combination of $d(x^2)$, $d(xy)$, and $d(y^2)$ even with coefficients in B . This example contradicts the main theorem in [Zan99].

Lemma 19.2. *There exists a unique rule that to every finite syntomic morphism of schemes $f : Y \rightarrow X$ assigns $\mathcal{O}_X$ -module maps*

$$\Theta_{Y/X}^p : f_*\Omega_{Y/\mathbf{Z}}^p \longrightarrow \Omega_{X/\mathbf{Z}}^p$$

satisfying the following properties

- (1) *the composition with $\Omega_{X/\mathbf{Z}}^p \otimes_{\mathcal{O}_X} f_*\mathcal{O}_Y \rightarrow f_*\Omega_{Y/\mathbf{Z}}^p$ is equal to $\text{id} \otimes \text{Trace}_f$ where $\text{Trace}_f : f_*\mathcal{O}_Y \rightarrow \mathcal{O}_X$ is the map from Discriminants, Section ??,*
- (2) *the rule is compatible with base change.*

Proof. First, assume that X is locally Noetherian. By Lemma 18.6 we have a canonical map

$$c_{Y/X}^p : \Omega_{Y/\mathbf{Z}}^p \longrightarrow f^*\Omega_{X/\mathbf{Z}}^p \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$$

By Discriminants, Proposition ?? we have a canonical isomorphism

$$c_{Y/X} : \det(NL_{Y/X}) \rightarrow \omega_{Y/X}$$

mapping $\delta(NL_{Y/X})$ to $\tau_{Y/X}$. Combined these maps give

$$c_{Y/X}^p \otimes c_{Y/X} : \Omega_{Y/\mathbf{Z}}^p \longrightarrow f^*\Omega_{X/\mathbf{Z}}^p \otimes_{\mathcal{O}_Y} \omega_{Y/X}$$

By Discriminants, Section ?? this is the same thing as a map

$$\Theta_{Y/X}^p : f_*\Omega_{Y/\mathbf{Z}}^p \longrightarrow \Omega_{X/\mathbf{Z}}^p$$

Recall that the relationship between $c_{Y/X}^p \otimes c_{Y/X}$ and $\Theta_{Y/X}^p$ uses the evaluation map $f_*\omega_{Y/X} \rightarrow \mathcal{O}_X$ which sends $\tau_{Y/X}$ to $\text{Trace}_f(1)$, see Discriminants, Section ?. Hence property (1) holds. Property (2) holds for base changes by $X' \rightarrow X$ with X' locally Noetherian because both $c_{Y/X}^p$ and $c_{Y/X}$ are compatible with such base changes. For $f : Y \rightarrow X$ finite syntomic and X locally Noetherian, we will continue to denote $\Theta_{Y/X}^p$ the solution we’ve just found.

Uniqueness. Suppose that we have a finite syntomic morphism $f : Y \rightarrow X$ such that X is smooth over $\text{Spec}(\mathbf{Z})$ and f is étale over a dense open of X . We claim that in this case $\Theta_{Y/X}^p$ is uniquely determined by property (1). Namely, consider the maps

$$\Omega_{X/\mathbf{Z}}^p \otimes_{\mathcal{O}_X} f_*\mathcal{O}_Y \rightarrow f_*\Omega_{Y/\mathbf{Z}}^p \rightarrow \Omega_{X/\mathbf{Z}}^p$$

The sheaf $\Omega_{X/\mathbf{Z}}^p$ is torsion free (by the assumed smoothness), hence it suffices to check that the restriction of $\Theta_{Y/X}^p$ is uniquely determined over the dense open over which f is étale, i.e., we may assume f is étale. However, if f is étale, then $f^*\Omega_{X/\mathbf{Z}} = \Omega_{Y/\mathbf{Z}}$ hence the first arrow in the displayed equation is an isomorphism. Since we’ve pinned down the composition, this guarantees uniqueness.

Let $f : Y \rightarrow X$ be a finite syntomic morphism of locally Noetherian schemes. Let $x \in X$. By Discriminants, Lemma ?? we can find $d \geq 1$ and a commutative diagram

$$\begin{array}{ccccc} Y & \longleftarrow & V & \longrightarrow & V_d \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & U & \longrightarrow & U_d \end{array}$$

such that $x \in U \subset X$ is open, $V = f^{-1}(U)$ and $V = U \times_{U_d} V_d$. Thus $\Theta_{Y/X}^p|_V$ is the pullback of the map Θ_{V_d/U_d}^p . However, by the discussion on uniqueness above and Discriminants, Lemmas ?? and ?? the map Θ_{V_d/U_d}^p is uniquely determined by the requirement (1). Hence uniqueness holds.

At this point we know that we have existence and uniqueness for all finite syntomic morphisms $Y \rightarrow X$ with X locally Noetherian. We could now give an argument similar to the proof of Lemma 18.6 to extend to general X . However, instead it is possible to directly use absolute Noetherian approximation to finish the proof. Namely, to construct $\Theta_{Y/X}^p$ it suffices to do so Zariski locally on X (provided we also show the uniqueness). Hence we may assume X is affine (small detail omitted). Then we can write $X = \lim_{i \in I} X_i$ as the limit over a directed set I of Noetherian affine schemes. By Algebra, Lemma ?? we can find $0 \in I$ and a finitely presented morphism of affines $f_0 : Y_0 \rightarrow X_0$ whose base change to X is $Y \rightarrow X$. After increasing 0 we may assume $Y_0 \rightarrow X_0$ is finite and syntomic, see Algebra, Lemma ?? and ??. For $i \geq 0$ also the base change $f_i : Y_i = Y_0 \times_{X_0} X_i \rightarrow X_i$ is finite syntomic. Then

$$\Gamma(X, f_* \Omega_{Y/\mathbf{Z}}^p) = \Gamma(Y, \Omega_{Y/\mathbf{Z}}^p) = \operatorname{colim}_{i \geq 0} \Gamma(Y_i, \Omega_{Y_i/\mathbf{Z}}^p) = \operatorname{colim}_{i \geq 0} \Gamma(X_i, f_{i,*} \Omega_{Y_i/X_i}^p)$$

Hence we can (and are forced to) define $\Theta_{Y/X}^p$ as the colimit of the maps Θ_{Y_i/X_i}^p . This map is compatible with any cartesian diagram

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

with X' affine as we know this for the case of Noetherian affine schemes by the arguments given above (small detail omitted; hint: if we also write $X' = \lim_{j \in J} X'_j$ then for every $i \in I$ there is a $j \in J$ and a morphism $X'_j \rightarrow X_i$ compatible with the morphism $X' \rightarrow X$). This finishes the proof. $\square$

Proposition 19.3. *Let $f : Y \rightarrow X$ be a finite syntomic morphism of schemes. The maps $\Theta_{Y/X}^p$ of Lemma 19.2 define a map of complexes*

$$\Theta_{Y/X} : f_* \Omega_{Y/\mathbf{Z}}^\bullet \longrightarrow \Omega_{X/\mathbf{Z}}^\bullet$$

with the following properties

- (1) *in degree 0 we get $\operatorname{Trace}_f : f_* \mathcal{O}_Y \rightarrow \mathcal{O}_X$, see Discriminants, Section ??,*
- (2) *we have $\Theta_{Y/X}(\omega \wedge \eta) = \omega \wedge \Theta_{Y/X}(\eta)$ for ω in $\Omega_{X/\mathbf{Z}}^\bullet$ and η in $f_* \Omega_{Y/\mathbf{Z}}^\bullet$,*
- (3) *if f is a morphism over a base scheme S , then $\Theta_{Y/X}$ induces a map of complexes $f_* \Omega_{Y/S}^\bullet \rightarrow \Omega_{X/S}^\bullet$.*

Proof. By Discriminants, Lemma ?? for every $x \in X$ we can find $d \geq 1$ and a commutative diagram

$$\begin{array}{ccccccc} Y & \longleftarrow & V & \longrightarrow & V_d & \longrightarrow & Y_d = \text{Spec}(B_d) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & U & \longrightarrow & U_d & \longrightarrow & X_d = \text{Spec}(A_d) \end{array}$$

such that $x \in U \subset X$ is affine open, $V = f^{-1}(U)$ and $V = U \times_{U_d} V_d$. Write $U = \text{Spec}(A)$ and $V = \text{Spec}(B)$ and observe that $B = A \otimes_{A_d} B_d$ and recall that $B_d = A_d e_1 \oplus \dots \oplus A_d e_d$. Suppose we have $a_1, \dots, a_r \in A$ and $b_1, \dots, b_s \in B$. We may write $b_j = \sum a_{j,l} e_l$ with $a_{j,l} \in A$. Set $N = r + sd$ and consider the factorizations

$$\begin{array}{ccccc} V & \longrightarrow & V' = \mathbf{A}^N \times V_d & \longrightarrow & V_d \\ \downarrow & & \downarrow & & \downarrow \\ U & \longrightarrow & U' = \mathbf{A}^N \times U_d & \longrightarrow & U_d \end{array}$$

Here the horizontal lower right arrow is given by the morphism $U \rightarrow U_d$ (from the earlier diagram) and the morphism $U \rightarrow \mathbf{A}^N$ given by $a_1, \dots, a_r, a_{1,1}, \dots, a_{s,d}$. Then we see that the functions $a_1, \dots, a_r$ are in the image of $\Gamma(U', \mathcal{O}_{U'}) \rightarrow \Gamma(U, \mathcal{O}_U)$ and the functions $b_1, \dots, b_s$ are in the image of $\Gamma(V', \mathcal{O}_{V'}) \rightarrow \Gamma(V, \mathcal{O}_V)$. In this way we see that for any finite collection of elements⁴ of the groups

$$\Gamma(V, \Omega_{Y/\mathbf{Z}}^i), \quad i = 0, 1, 2, \dots \quad \text{and} \quad \Gamma(U, \Omega_{X/\mathbf{Z}}^j), \quad j = 0, 1, 2, \dots$$

we can find a factorizations $V \rightarrow V' \rightarrow V_d$ and $U \rightarrow U' \rightarrow U_d$ with $V' = \mathbf{A}^N \times V_d$ and $U' = \mathbf{A}^N \times U_d$ as above such that these sections are the pullbacks of sections from

$$\Gamma(V', \Omega_{V'/\mathbf{Z}}^i), \quad i = 0, 1, 2, \dots \quad \text{and} \quad \Gamma(U', \Omega_{U'/\mathbf{Z}}^j), \quad j = 0, 1, 2, \dots$$

The upshot of this is that to check $d \circ \Theta_{Y/X} = \Theta_{Y/X} \circ d$ it suffices to check this is true for $\Theta_{V'/U'}$. Similarly, for property (2) of the lemma.

By Discriminants, Lemmas ?? and ?? the scheme U_d is smooth and the morphism $V_d \rightarrow U_d$ is étale over a dense open of U_d . Hence the same is true for the morphism $V' \rightarrow U'$. Since $\Omega_{U'/\mathbf{Z}}$ is locally free and hence $\Omega_{U'/\mathbf{Z}}^p$ is torsion free, it suffices to check the desired relations after restricting to the open over which V' is finite étale. Then we may check the relations after a surjective étale base change. Hence we may split the finite étale cover and assume we are looking at a morphism of the form

$$\coprod_{i=1, \dots, d} W \longrightarrow W$$

with W smooth over $\mathbf{Z}$. In this case any local properties of our construction are trivial to check (provided they are true). This finishes the proof of (1) and (2).

Finally, we observe that (3) follows from (2) because $\Omega_{Y/S}$ is the quotient of $\Omega_{Y/\mathbf{Z}}$ by the submodule generated by pullbacks of local sections of $\Omega_{S/\mathbf{Z}}$. $\square$

⁴After all these elements will be finite sums of elements of the form $a_0 da_1 \wedge \dots \wedge da_i$ with $a_0, \dots, a_i \in A$ or finite sums of elements of the form $b_0 db_1 \wedge \dots \wedge db_j$ with $b_0, \dots, b_j \in B$.

Example 19.4. Let A be a ring. Let $f = x^d + \sum_{0 \leq i < d} a_{d-i} x^i \in A[x]$. Let $B = A[x]/(f)$. By Proposition 19.3 we have a morphism of complexes

$$\Theta_{B/A} : \Omega_B^\bullet \longrightarrow \Omega_A^\bullet$$

In particular, if $t \in B$ denotes the image of $x \in A[x]$ we can consider the elements

$$\Theta_{B/A}(t^i dt) \in \Omega_A^1, \quad i = 0, \dots, d-1$$

What are these elements? By the same principle as used in the proof of Proposition 19.3 it suffices to compute this in the universal case, i.e., when $A = \mathbf{Z}[a_1, \dots, a_d]$ or even when A is replaced by the fraction field $\mathbf{Q}(a_1, \dots, a_d)$. Writing symbolically

$$f = \prod_{i=1, \dots, d} (x - \alpha_i)$$

we see that over $\mathbf{Q}(\alpha_1, \dots, \alpha_d)$ the algebra B becomes split:

$$\mathbf{Q}(a_0, \dots, a_{d-1})[x]/(f) \longrightarrow \prod_{i=1, \dots, d} \mathbf{Q}(\alpha_1, \dots, \alpha_d), \quad t \longmapsto (\alpha_1, \dots, \alpha_d)$$

Thus for example

$$\Theta(dt) = \sum d\alpha_i = -da_1$$

Next, we have

$$\Theta(tdt) = \sum \alpha_i d\alpha_i = a_1 da_1 - da_2$$

Next, we have

$$\Theta(t^2 dt) = \sum \alpha_i^2 d\alpha_i = -a_1^2 da_1 + a_1 da_2 + a_2 da_1 - da_3$$

(modulo calculation error), and so on. This suggests that if $f(x) = x^d - a$ then

$$\Theta_{B/A}(t^i dt) = \begin{cases} 0 & \text{if } i = 0, \dots, d-2 \\ da & \text{if } i = d-1 \end{cases}$$

in Ω_A . This is true for in this particular case one can do the calculation for the extension $\mathbf{Q}(a)[x]/(x^d - a)$ to verify this directly.

Lemma 19.5. *Let p be a prime number. Let $X \rightarrow S$ be a smooth morphism of relative dimension d of schemes in characteristic p . The relative Frobenius $F_{X/S} : X \rightarrow X^{(p)}$ of X/S (Varieties, Definition ??) is finite syntomic and the corresponding map*

$$\Theta_{X/X^{(p)}} : F_{X/S,*} \Omega_{X/S}^\bullet \rightarrow \Omega_{X^{(p)}/S}^\bullet$$

is zero in all degrees except in degree d where it defines a surjection.

Proof. Observe that $F_{X/S}$ is a finite morphism by Varieties, Lemma ??. To prove that $F_{X/S}$ is flat, it suffices to show that the morphism $F_{X/S,s} : X_s \rightarrow X_s^{(p)}$ between fibres is flat for all $s \in S$, see More on Morphisms, Theorem ??. Flatness of $X_s \rightarrow X_s^{(p)}$ follows from Algebra, Lemma ?? (and the finiteness already shown). By More on Morphisms, Lemma ?? the morphism $F_{X/S}$ is a local complete intersection morphism. Hence $F_{X/S}$ is finite syntomic (see More on Morphisms, Lemma ??).

For every point $x \in X$ we may choose a commutative diagram

$$\begin{array}{ccc} X & \longleftarrow & U \\ \downarrow & & \downarrow \pi \\ S & \longleftarrow & \mathbf{A}_S^d \end{array}$$

where π is étale and $x \in U$ is open in X , see Morphisms, Lemma ???. Observe that $\mathbf{A}_S^d \rightarrow \mathbf{A}_S^d$, $(x_1, \dots, x_d) \mapsto (x_1^p, \dots, x_d^p)$ is the relative Frobenius for $\mathcal{A}_S^d$ over S . The commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{F_{X/S}} & U^{(p)} \\ \pi \downarrow & & \downarrow \pi^{(p)} \\ \mathbf{A}_S^d & \xrightarrow{x_i \mapsto x_i^p} & \mathbf{A}_S^d \end{array}$$

of Varieties, Lemma ?? for $\pi : U \rightarrow \mathbf{A}_S^d$ is cartesian by Étale Morphisms, Lemma ??. Since the construction of Θ is compatible with base change and since $\Omega_{U/S} = \pi^* \Omega_{\mathbf{A}_S^d/S}$ (Lemma 2.2) we conclude that it suffices to show the lemma for $\mathbf{A}_S^d$.

Let A be a ring of characteristic p . Consider the unique A -algebra homomorphism $A[y_1, \dots, y_d] \rightarrow A[x_1, \dots, x_d]$ sending y_i to x_i^p . The arguments above reduce us to computing the map

$$\Theta^i : \Omega_{A[x_1, \dots, x_d]/A}^i \rightarrow \Omega_{A[y_1, \dots, y_d]/A}^i$$

We urge the reader to do the computation in this case for themselves. As in Example 19.4 we may reduce this to computing a formula for Θ^i in the universal case

$$\mathbf{Z}[y_1, \dots, y_d] \rightarrow \mathbf{Z}[x_1, \dots, x_d], \quad y_i \mapsto x_i^p$$

In turn, we can find the formula for Θ^i by computing in the complex case, i.e., for the $\mathbf{C}$ -algebra map

$$\mathbf{C}[y_1, \dots, y_d] \rightarrow \mathbf{C}[x_1, \dots, x_d], \quad y_i \mapsto x_i^p$$

We may even invert $x_1, \dots, x_d$ and $y_1, \dots, y_d$. In this case, we have $dx_i = p^{-1}x_i^{-p+1}dy_i$. Hence we see that

$$\begin{aligned} \Theta^i(x_1^{e_1} \dots x_d^{e_d} dx_1 \wedge \dots \wedge dx_i) &= p^{-i} \Theta^i(x_1^{e_1-p+1} \dots x_i^{e_i-p+1} x_{i+1}^{e_{i+1}} \dots x_d^{e_d} dy_1 \wedge \dots \wedge dy_i) \\ &= p^{-i} \text{Trace}(x_1^{e_1-p+1} \dots x_i^{e_i-p+1} x_{i+1}^{e_{i+1}} \dots x_d^{e_d}) dy_1 \wedge \dots \wedge dy_i \end{aligned}$$

by the properties of Θ^i . An elementary computation shows that the trace in the expression above is zero unless $e_1, \dots, e_i$ are congruent to -1 modulo p and $e_{i+1}, \dots, e_d$ are divisible by p . Moreover, in this case we obtain

$$p^{d-i} y_1^{(e_1-p+1)/p} \dots y_i^{(e_i-p+1)/p} y_{i+1}^{e_{i+1}/p} \dots y_d^{e_d/p} dy_1 \wedge \dots \wedge dy_i$$

We conclude that we get zero in characteristic p unless $d = i$ and in this case we get every possible d -form. $\square$

20. Poincaré duality

In this section we prove Poincaré duality for the de Rham cohomology of a proper smooth scheme over a field. Let us first explain how this works for Hodge cohomology.

Lemma 20.1. *Let k be a field. Let X be a nonempty smooth proper scheme over k equidimensional of dimension d . There exists a k -linear map*

$$t : H^d(X, \Omega_{X/k}^d) \longrightarrow k$$

unique up to precomposing by multiplication by a unit of $H^0(X, \mathcal{O}_X)$ with the following property: for all p, q the pairing

$$H^q(X, \Omega_{X/k}^p) \times H^{d-q}(X, \Omega_{X/k}^{d-p}) \longrightarrow k, \quad (\xi, \xi') \longmapsto t(\xi \cup \xi')$$

is perfect.

Proof. By Duality for Schemes, Lemma ?? we have $\omega_X^\bullet = \Omega_{X/k}^d[d]$. Since $\Omega_{X/k}$ is locally free of rank d (Morphisms, Lemma ??) we have

$$\Omega_{X/k}^d \otimes_{\mathcal{O}_X} (\Omega_{X/k}^p)^\vee \cong \Omega_{X/k}^{d-p}$$

Thus we obtain a k -linear map $t : H^d(X, \Omega_{X/k}^d) \rightarrow k$ such that the statement is true by Duality for Schemes, Lemma ??. In particular the pairing $H^0(X, \mathcal{O}_X) \times H^d(X, \Omega_{X/k}^d) \rightarrow k$ is perfect, which implies that any k -linear map $t' : H^d(X, \Omega_{X/k}^d) \rightarrow k$ is of the form $\xi \mapsto t(g\xi)$ for some $g \in H^0(X, \mathcal{O}_X)$. Of course, in order for t' to still produce a duality between $H^0(X, \mathcal{O}_X)$ and $H^d(X, \Omega_{X/k}^d)$ we need g to be a unit. Denote $\langle -, - \rangle_{p,q}$ the pairing constructed using t and denote $\langle -, - \rangle'_{p,q}$ the pairing constructed using t' . Clearly we have

$$\langle \xi, \xi' \rangle'_{p,q} = \langle g\xi, \xi' \rangle_{p,q}$$

for $\xi \in H^q(X, \Omega_{X/k}^p)$ and $\xi' \in H^{d-q}(X, \Omega_{X/k}^{d-p})$. Since g is a unit, i.e., invertible, we see that using t' instead of t we still get perfect pairings for all p, q . $\square$

Lemma 20.2. *Let k be a field. Let X be a smooth proper scheme over k . The map*

$$d : H^0(X, \mathcal{O}_X) \rightarrow H^0(X, \Omega_{X/k}^1)$$

is zero.

Proof. Since X is smooth over k it is geometrically reduced over k , see Varieties, Lemma ??. Hence $H^0(X, \mathcal{O}_X) = \prod k_i$ is a finite product of finite separable field extensions k_i/k , see Varieties, Lemma ??. It follows that $\Omega_{H^0(X, \mathcal{O}_X)/k} = \prod \Omega_{k_i/k} = 0$ (see for example Algebra, Lemma ??). Since the map of the lemma factors as

$$H^0(X, \mathcal{O}_X) \rightarrow \Omega_{H^0(X, \mathcal{O}_X)/k} \rightarrow H^0(X, \Omega_{X/k}^1)$$

by functoriality of the de Rham complex (see Section 2), we conclude. $\square$

Lemma 20.3. *Let k be a field. Let X be a smooth proper scheme over k equidimensional of dimension d . The map*

$$d : H^d(X, \Omega_{X/k}^{d-1}) \rightarrow H^d(X, \Omega_{X/k}^d)$$

is zero.

Proof. It is tempting to think this follows from a combination of Lemmas 20.2 and 20.1. However this doesn't work because the maps $\mathcal{O}_X \rightarrow \Omega_{X/k}^1$ and $\Omega_{X/k}^{d-1} \rightarrow \Omega_{X/k}^d$ are not $\mathcal{O}_X$ -linear and hence we cannot use the functoriality discussed in Duality for Schemes, Remark ?? to conclude the map in Lemma 20.2 is dual to the one in this lemma.

We may replace X by a connected component of X . Hence we may assume X is irreducible. By Varieties, Lemmas ?? and ?? we see that $k' = H^0(X, \mathcal{O}_X)$ is a finite separable extension k'/k . Since $\Omega_{k'/k} = 0$ (see for example Algebra, Lemma ??) we see that $\Omega_{X/k} = \Omega_{X/k'}$ (see Morphisms, Lemma ??). Thus we may replace k by k' and assume that $H^0(X, \mathcal{O}_X) = k$.

Assume $H^0(X, \mathcal{O}_X) = k$. We conclude that $\dim H^d(X, \Omega_{X/k}^d) = 1$ by Lemma 20.1. Assume first that the characteristic of k is a prime number p . Denote $F_{X/k} : X \rightarrow$

$X^{(p)}$ the relative Frobenius of X over k ; please keep in mind the facts proved about this morphism in Lemma 19.5. Consider the commutative diagram

$$\begin{array}{ccccc} H^d(X, \Omega_{X/k}^{d-1}) & \longrightarrow & H^d(X^{(p)}, F_{X/k,*} \Omega_{X/k}^{d-1}) & \xrightarrow[\Theta^{d-1}]{} & H^d(X^{(p)}, \Omega_{X^{(p)}/k}^{d-1}) \\ \downarrow & & \downarrow & & \downarrow \\ H^d(X, \Omega_{X/k}^d) & \longrightarrow & H^d(X^{(p)}, F_{X/k,*} \Omega_{X/k}^d) & \xrightarrow{\Theta^d} & H^d(X^{(p)}, \Omega_{X^{(p)}/k}^d) \end{array}$$

The left two horizontal arrows are isomorphisms as $F_{X/k}$ is finite, see Cohomology of Schemes, Lemma ???. The right square commutes as $\Theta_{X^{(p)}/X}$ is a morphism of complexes and Θ^{d-1} is zero. Thus it suffices to show that Θ^d is nonzero (because the dimension of the source of the map Θ^d is 1 by the discussion above). However, we know that

$$\Theta^d : F_{X/k,*} \Omega_{X/k}^d \rightarrow \Omega_{X^{(p)}/k}^d$$

is surjective and hence surjective after applying the right exact functor $H^d(X^{(p)}, -)$ (right exactness by the vanishing of cohomology beyond d as follows from Cohomology, Proposition ??). Finally, $H^d(X^{(d)}, \Omega_{X^{(d)}/k}^d)$ is nonzero for example it is dual to $H^0(X^{(d)}, \mathcal{O}_{X^{(p)}})$ by Lemma 20.1 applied to $X^{(p)}$ over k . This finishes the proof in this case.

Finally, assume the characteristic of k is 0. We can write k as the filtered colimit of its finite type $\mathbf{Z}$ -subalgebras R . For one of these we can find a cartesian diagram of schemes

$$\begin{array}{ccc} X & \longrightarrow & Y \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that $Y \rightarrow \mathrm{Spec}(R)$ is smooth of relative dimension d and proper. See Limits, Lemmas ??, ??, ??, and ??. The modules $M^{i,j} = H^j(Y, \Omega_{Y/R}^i)$ are finite R -modules, see Cohomology of Schemes, Lemma ??. Thus after replacing R by a localization we may assume all of these modules are finite free. We have $M^{i,j} \otimes_R k = H^j(X, \Omega_{X/k}^i)$ by flat base change (Cohomology of Schemes, Lemma ??). Thus it suffices to show that $M^{d-1,d} \rightarrow M^{d,d}$ is zero. This is a map of finite free modules over a domain, hence it suffices to find a dense set of primes $\mathfrak{p} \subset R$ such that after tensoring with $\kappa(\mathfrak{p})$ we get zero. Since R is of finite type over $\mathbf{Z}$, we can take the collection of primes $\mathfrak{p}$ whose residue field has positive characteristic (details omitted). Observe that

$$M^{d-1,d} \otimes_R \kappa(\mathfrak{p}) = H^d(Y_{\kappa(\mathfrak{p})}, \Omega_{Y_{\kappa(\mathfrak{p})}/\kappa(\mathfrak{p})}^{d-1})$$

for example by Limits, Lemma ??. Similarly for $M^{d,d}$. Thus we see that $M^{d-1,d} \otimes_R \kappa(\mathfrak{p}) \rightarrow M^{d,d} \otimes_R \kappa(\mathfrak{p})$ is zero by the case of positive characteristic handled above. $\square$

Proposition 20.4. *Let k be a field. Let X be a nonempty smooth proper scheme over k equidimensional of dimension d . There exists a k -linear map*

$$t : H_{dR}^{2d}(X/k) \longrightarrow k$$

unique up to precomposing by multiplication by a unit of $H^0(X, \mathcal{O}_X)$ with the following property: for all i the pairing

$$H_{dR}^i(X/k) \times H_{dR}^{2d-i}(X/k) \longrightarrow k, \quad (\xi, \xi') \longmapsto t(\xi \cup \xi')$$

is perfect.

Proof. By the Hodge-to-de Rham spectral sequence (Section 6), the vanishing of $\Omega_{X/k}^i$ for $i > d$, the vanishing in Cohomology, Proposition ?? and the results of Lemmas 20.2 and 20.3 we see that $H_{dR}^0(X/k) = H^0(X, \mathcal{O}_X)$ and $H^d(X, \Omega_{X/k}^d) = H_{dR}^{2d}(X/k)$. More precisely, these identifications come from the maps of complexes

$$\Omega_{X/k}^\bullet \rightarrow \mathcal{O}_X[0] \quad \text{and} \quad \Omega_{X/k}^d[-d] \rightarrow \Omega_{X/k}^\bullet$$

Let us choose $t : H_{dR}^{2d}(X/k) \rightarrow k$ which via this identification corresponds to a t as in Lemma 20.1. Then in any case we see that the pairing displayed in the lemma is perfect for $i = 0$.

Denote $\underline{k}$ the constant sheaf with value k on X . Let us abbreviate $\Omega^\bullet = \Omega_{X/k}^\bullet$. Consider the map (4.0.1) which in our situation reads

$$\wedge : \text{Tot}(\Omega^\bullet \otimes_{\underline{k}} \Omega^\bullet) \longrightarrow \Omega^\bullet$$

For every integer $p = 0, 1, \dots, d$ this map annihilates the subcomplex $\text{Tot}(\sigma_{>p}\Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p}\Omega^\bullet)$ for degree reasons. Hence we find that the restriction of $\wedge$ to the subcomplex $\text{Tot}(\Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p}\Omega^\bullet)$ factors through a map of complexes

$$\gamma_p : \text{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p}\Omega^\bullet) \longrightarrow \Omega^\bullet$$

Using the same procedure as in Section 4 we obtain cup products

$$H^i(X, \sigma_{\leq p}\Omega^\bullet) \times H^{2d-i}(X, \sigma_{\geq d-p}\Omega^\bullet) \longrightarrow H_{dR}^{2d}(X, \Omega^\bullet)$$

We will prove by induction on p that these cup products via t induce perfect pairings between $H^i(X, \sigma_{\leq p}\Omega^\bullet)$ and $H^{2d-i}(X, \sigma_{\geq d-p}\Omega^\bullet)$. For $p = d$ this is the assertion of the proposition.

The base case is $p = 0$. In this case we simply obtain the pairing between $H^i(X, \mathcal{O}_X)$ and $H^{d-i}(X, \Omega^d)$ of Lemma 20.1 and the result is true.

Induction step. Say we know the result is true for p . Then we consider the distinguished triangle

$$\Omega^{p+1}[-p-1] \rightarrow \sigma_{\leq p+1}\Omega^\bullet \rightarrow \sigma_{\leq p}\Omega^\bullet \rightarrow \Omega^{p+1}[-p]$$

and the distinguished triangle

$$\sigma_{\geq d-p}\Omega^\bullet \rightarrow \sigma_{\geq d-p-1}\Omega^\bullet \rightarrow \Omega^{d-p-1}[-d+p+1] \rightarrow (\sigma_{\geq d-p}\Omega^\bullet)[1]$$

Observe that both are distinguished triangles in the homotopy category of complexes of sheaves of $\underline{k}$ -modules; in particular the maps $\sigma_{\leq p}\Omega^\bullet \rightarrow \Omega^{p+1}[-p]$ and $\Omega^{d-p-1}[-d+p+1] \rightarrow (\sigma_{\geq d-p}\Omega^\bullet)[1]$ are given by actual maps of complexes, namely using the differential $\Omega^p \rightarrow \Omega^{p+1}$ and the differential $\Omega^{d-p-1} \rightarrow \Omega^{d-p}$. Consider

the long exact cohomology sequences associated to these distinguished triangles

$$\begin{array}{ccc}
H^{i-1}(X, \sigma_{\leq p} \Omega^\bullet) & & H^{2d-i+1}(X, \sigma_{\geq d-p} \Omega^\bullet) \\
\downarrow a & & \uparrow a' \\
H^i(X, \Omega^{p+1}[-p-1]) & & H^{2d-i}(X, \Omega^{d-p-1}[-d+p+1]) \\
\downarrow b & & \uparrow b' \\
H^i(X, \sigma_{\leq p+1} \Omega^\bullet) & & H^{2d-i}(X, \sigma_{\geq d-p-1} \Omega^\bullet) \\
\downarrow c & & \uparrow c' \\
H^i(X, \sigma_{\leq p} \Omega^\bullet) & & H^{2d-i}(X, \sigma_{\geq d-p} \Omega^\bullet) \\
\downarrow d & & \uparrow d' \\
H^{i+1}(X, \Omega^{p+1}[-p-1]) & & H^{2d-i-1}(X, \Omega^{d-p-1}[-d+p+1])
\end{array}$$

By induction and Lemma 20.1 we know that the pairings constructed above between the k -vectorspaces on the first, second, fourth, and fifth rows are perfect. By the 5-lemma, in order to show that the pairing between the cohomology groups in the middle row is perfect, it suffices to show that the pairs (a, a') , (b, b') , (c, c') , and (d, d') are compatible with the given pairings (see below).

Let us prove this for the pair (c, c') . Here we observe simply that we have a commutative diagram

$$\begin{array}{ccc}
\text{Tot}(\sigma_{\leq p} \Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p} \Omega^\bullet) & \longleftarrow & \text{Tot}(\sigma_{\leq p+1} \Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p} \Omega^\bullet) \\
\downarrow \gamma_p & & \downarrow \\
\Omega^\bullet & \xleftarrow{\gamma_{p+1}} & \text{Tot}(\sigma_{\leq p+1} \Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p-1} \Omega^\bullet)
\end{array}$$

Hence if we have $\alpha \in H^i(X, \sigma_{\leq p+1} \Omega^\bullet)$ and $\beta \in H^{2d-i}(X, \sigma_{\geq d-p} \Omega^\bullet)$ then we get $\gamma_p(\alpha \cup c'(\beta)) = \gamma_{p+1}(c(\alpha) \cup \beta)$ by functoriality of the cup product.

Similarly for the pair (b, b') we use the commutative diagram

$$\begin{array}{ccc}
\text{Tot}(\sigma_{\leq p+1} \Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p-1} \Omega^\bullet) & \longleftarrow & \text{Tot}(\Omega^{p+1}[-p-1] \otimes_{\underline{k}} \sigma_{\geq d-p-1} \Omega^\bullet) \\
\downarrow \gamma_{p+1} & & \downarrow \\
\Omega^\bullet & \xleftarrow{\wedge} & \Omega^{p+1}[-p-1] \otimes_{\underline{k}} \Omega^{d-p-1}[-d+p+1]
\end{array}$$

and argue in the same manner.

For the pair (d, d') we use the commutative diagram

$$\begin{array}{ccc}
\Omega^{p+1}[-p] \otimes_{\underline{k}} \Omega^{d-p-1}[-d+p] & \longleftarrow & \text{Tot}(\sigma_{\leq p} \Omega^\bullet \otimes_{\underline{k}} \Omega^{d-p-1}[-d+p]) \\
\downarrow & & \downarrow \\
\Omega^\bullet & \xleftarrow{\quad} & \text{Tot}(\sigma_{\leq p} \Omega^\bullet \otimes_{\underline{k}} \sigma_{\geq d-p} \Omega^\bullet)
\end{array}$$

and we look at cohomology classes in $H^i(X, \sigma_{\leq p} \Omega^\bullet)$ and $H^{2d-i}(X, \Omega^{d-p-1}[-d+p])$. Changing i to $i-1$ we get the result for the pair (a, a') thereby finishing the proof that our pairings are perfect.

We omit the argument showing the uniqueness of t up to precomposing by multiplication by a unit in $H^0(X, \mathcal{O}_X)$. $\square$

21. Chern classes

The results proved so far suffice to use the discussion in Weil Cohomology Theories, Section ?? to produce Chern classes in de Rham cohomology.

Lemma 21.1. *There is a unique rule which assigns to every quasi-compact and quasi-separated scheme X a total Chern class*

$$c^{dR} : K_0(\text{Vect}(X)) \longrightarrow \prod_{i \geq 0} H_{dR}^{2i}(X/\mathbf{Z})$$

with the following properties

- (1) *we have $c^{dR}(\alpha + \beta) = c^{dR}(\alpha)c^{dR}(\beta)$ for $\alpha, \beta \in K_0(\text{Vect}(X))$,*
- (2) *if $f : X \rightarrow X'$ is a morphism of quasi-compact and quasi-separated schemes, then $c^{dR}(f^*\alpha) = f^*c^{dR}(\alpha)$,*
- (3) *given $\mathcal{L} \in \text{Pic}(X)$ we have $c^{dR}([\mathcal{L}]) = 1 + c_1^{dR}(\mathcal{L})$*

The construction can easily be extended to all schemes, but to do so one needs to slightly upgrade the discussion in Weil Cohomology Theories, Section ??.

Proof. We will apply Weil Cohomology Theories, Proposition ?? to get this.

Let $\mathcal{C}$ be the category of all quasi-compact and quasi-separated schemes. This certainly satisfies conditions (1), (2), and (3) (a), (b), and (c) of Weil Cohomology Theories, Section ??.

As our contravariant functor A from $\mathcal{C}$ to the category of graded algebras will send X to $A(X) = \bigoplus_{i \geq 0} H_{dR}^{2i}(X/\mathbf{Z})$ endowed with its cup product. Functoriality is discussed in Section 3 and the cup product in Section 4. For the additive maps c_1^A we take c_1^{dR} constructed in Section 9.

In fact, we obtain commutative algebras by Lemma 4.1 which shows we have axiom (1) for A .

To check axiom (2) for A it suffices to check that $H_{dR}^*(X \amalg Y/\mathbf{Z}) = H_{dR}^*(X/\mathbf{Z}) \times H_{dR}^*(Y/\mathbf{Z})$. This is a consequence of the fact that de Rham cohomology is constructed by taking the cohomology of a sheaf of differential graded algebras (in the Zariski topology).

Axiom (3) for A is just the statement that taking first Chern classes of invertible modules is compatible with pullbacks. This follows from the more general Lemma 9.1.

Axiom (4) for A is the projective space bundle formula which we proved in Proposition 14.1.

Axiom (5). Let X be a quasi-compact and quasi-separated scheme and let $\mathcal{E} \rightarrow \mathcal{F}$ be a surjection of finite locally free $\mathcal{O}_X$ -modules of ranks $r+1$ and r . Denote $i : P' = \mathbf{P}(\mathcal{F}) \rightarrow \mathbf{P}(\mathcal{E}) = P$ the corresponding inclusion morphism. This is a morphism of smooth projective schemes over X which exhibits P' as an effective

Cartier divisor on P . Thus by Lemma 15.7 the complex of log poles for $P' \subset P$ over $\mathbf{Z}$ is defined. Hence for $a \in A(P)$ with $i^*a = 0$ we have $a \cup c_1^A(\mathcal{O}_P(P')) = 0$ by Lemma 15.6. This finishes the proof. $\square$

Remark 21.2. The analogues of Weil Cohomology Theories, Lemmas ?? (splitting principle) and ?? (chern classes of tensor products) hold for de Rham Chern classes on quasi-compact and quasi-separated schemes. This is clear as we've shown in the proof of Lemma 21.1 that all the axioms of Weil Cohomology Theories, Section ?? are satisfied.

Working with schemes over $\mathbf{Q}$ we can construct a Chern character.

Lemma 21.3. *There is a unique rule which assigns to every quasi-compact and quasi-separated scheme X over $\mathbf{Q}$ a “chern character”*

$$ch^{dR} : K_0(\text{Vect}(X)) \longrightarrow \prod_{i \geq 0} H_{dR}^{2i}(X/\mathbf{Q})$$

with the following properties

- (1) ch^{dR} is a ring map for all X ,
- (2) if $f : X' \rightarrow X$ is a morphism of quasi-compact and quasi-separated schemes over $\mathbf{Q}$, then $f^* \circ ch^{dR} = ch^{dR} \circ f^*$, and
- (3) given $\mathcal{L} \in \text{Pic}(X)$ we have $ch^{dR}([\mathcal{L}]) = \exp(c_1^{dR}(\mathcal{L}))$.

The construction can easily be extended to all schemes over $\mathbf{Q}$, but to do so one needs to slightly upgrade the discussion in Weil Cohomology Theories, Section ??.

Proof. Exactly as in the proof of Lemma 21.1 one shows that the category of quasi-compact and quasi-separated schemes over $\mathbf{Q}$ together with the functor $A^*(X) = \bigoplus_{i \geq 0} H_{dR}^{2i}(X/\mathbf{Q})$ satisfy the axioms of Weil Cohomology Theories, Section ??.

Moreover, in this case $A(X)$ is a $\mathbf{Q}$ -algebra for all X . Hence the lemma follows from Weil Cohomology Theories, Proposition ??. $\square$

22. A Weil cohomology theory

Let k be a field of characteristic 0. In this section we prove that the functor

$$X \longmapsto H_{dR}^*(X/k)$$

defines a Weil cohomology theory over k with coefficients in k as defined in Weil Cohomology Theories, Definition ??. We will proceed by checking the constructions earlier in this chapter provide us with data (D0), (D1), and (D2') satisfying axioms (A1) – (A9) of Weil Cohomology Theories, Section ??.

Throughout the rest of this section we fix the field k of characteristic 0 and we set $F = k$. Next, we take the following data

- (D0) For our 1-dimensional F vector space $F(1)$ we take $F(1) = F = k$.
- (D1) For our functor H^* we take the functor sending a smooth projective scheme X over k to $H_{dR}^*(X/k)$. Functoriality is discussed in Section 3 and the cup product in Section 4. We obtain graded commutative F -algebras by Lemma 4.1.
- (D2') For the maps $c_1^H : \text{Pic}(X) \rightarrow H^2(X)(1)$ we use the de Rham first Chern class introduced in Section 9.

We are going to show axioms (A1) – (A9) hold.

In this paragraph, we are going to reduce the checking of the axioms to the case where k is algebraically closed by using Weil Cohomology Theories, Lemma ?? . Denote k' the algebraic closure of k . Set $F' = k'$. We obtain data (D0), (D1), (D2') over k' with coefficient field F' in exactly the same way as above. By Lemma 3.5 there are functorial isomorphisms

$$H_{dR}^{2d}(X/k) \otimes_k k' \longrightarrow H_{dR}^{2d}(X_{k'}/k')$$

for X smooth and projective over k . Moreover, the diagrams

$$\begin{array}{ccc} \text{Pic}(X) & \xrightarrow{c_1^{dR}} & H_{dR}^2(X/k) \\ \downarrow & & \downarrow \\ \text{Pic}(X_{k'}) & \xrightarrow{c_1^{dR}} & H_{dR}^2(X_{k'}/k') \end{array}$$

commute by Lemma 9.1. This finishes the proof of the reduction.

Assume k is algebraically closed field of characteristic zero. We will show axioms (A1) – (A9) for the data (D0), (D1), and (D2') given above.

Axiom (A1). Here we have to check that $H_{dR}^*(X \amalg Y/k) = H_{dR}^*(X/k) \times H_{dR}^*(Y/k)$. This is a consequence of the fact that de Rham cohomology is constructed by taking the cohomology of a sheaf of differential graded algebras (in the Zariski topology).

Axiom (A2). This is just the statement that taking first Chern classes of invertible modules is compatible with pullbacks. This follows from the more general Lemma 9.1.

Axiom (A3). This follows from the more general Proposition 14.1.

Axiom (A4). This follows from the more general Lemma 15.6.

Already at this point, using Weil Cohomology Theories, Lemmas ?? and ??, we obtain a Chern character and cycle class maps

$$\gamma : \text{CH}^*(X) \longrightarrow \bigoplus_{i \geq 0} H_{dR}^{2i}(X/k)$$

for X smooth projective over k which are graded ring homomorphisms compatible with pullbacks between morphisms $f : X \rightarrow Y$ of smooth projective schemes over k .

Axiom (A5). We have $H_{dR}^*(\text{Spec}(k)/k) = k = F$ in degree 0. We have the Künneth formula for the product of two smooth projective k -schemes by Lemma 8.2 (observe that the derived tensor products in the statement are harmless as we are tensoring over the field k).

Axiom (A7). This follows from Proposition 17.3.

Axiom (A8). Let X be a smooth projective scheme over k . By the explanatory text to this axiom in Weil Cohomology Theories, Section ?? we see that $k' = H^0(X, \mathcal{O}_X)$ is a finite separable k -algebra. It follows that $H_{dR}^*(\text{Spec}(k')/k) = k'$ sitting in degree 0 because $\Omega_{k'/k} = 0$. By Lemma 20.2 we also have $H_{dR}^0(X, \mathcal{O}_X) = k'$ and we get the axiom.

Axiom (A6). Let X be a nonempty smooth projective scheme over k which is equidimensional of dimension d . Denote $\Delta : X \rightarrow X \times_{\text{Spec}(k)} X$ the diagonal morphism of X over k . We have to show that there exists a k -linear map

$$\lambda : H_{dR}^{2d}(X/k) \longrightarrow k$$

such that $(1 \otimes \lambda)\gamma([\Delta]) = 1$ in $H_{dR}^0(X/k)$. Let us write

$$\gamma = \gamma([\Delta]) = \gamma_0 + \dots + \gamma_{2d}$$

with $\gamma_i \in H_{dR}^i(X/k) \otimes_k H_{dR}^{2d-i}(X/k)$ the Künneth components. Our problem is to show that there is a linear map $\lambda : H_{dR}^{2d}(X/k) \rightarrow k$ such that $(1 \otimes \lambda)\gamma_0 = 1$ in $H_{dR}^0(X/k)$.

Let $X = \coprod X_i$ be the decomposition of X into connected and hence irreducible components. Then we have correspondingly $\Delta = \coprod \Delta_i$ with $\Delta_i \subset X_i \times X_i$. It follows that

$$\gamma([\Delta]) = \sum \gamma([\Delta_i])$$

and moreover $\gamma([\Delta_i])$ corresponds to the class of $\Delta_i \subset X_i \times X_i$ via the decomposition

$$H_{dR}^*(X \times X) = \prod_{i,j} H_{dR}^*(X_i \times X_j)$$

We omit the details; one way to show this is to use that in $\text{CH}^0(X \times X)$ we have idempotents $e_{i,j}$ corresponding to the open and closed subschemes $X_i \times X_j$ and to use that γ is a ring map which sends $e_{i,j}$ to the corresponding idempotent in the displayed product decomposition of cohomology. If we can find $\lambda_i : H_{dR}^{2d}(X_i/k) \rightarrow k$ with $(1 \otimes \lambda_i)\gamma([\Delta_i]) = 1$ in $H_{dR}^0(X_i/k)$ then taking $\lambda = \sum \lambda_i$ will solve the problem for X . Thus we may and do assume X is irreducible.

Proof of Axiom (A6) for X irreducible. Since k is algebraically closed we have $H_{dR}^0(X/k) = k$ because $H^0(X, \mathcal{O}_X) = k$ as X is a projective variety over an algebraically closed field (see Varieties, Lemma ?? for example). Let $x \in X$ be any closed point. Consider the cartesian diagram

$$\begin{array}{ccc} x & \longrightarrow & X \\ \downarrow & & \downarrow \Delta \\ X & \xrightarrow{x \times \text{id}} & X \times_{\text{Spec}(k)} X \end{array}$$

Compatibility of γ with pullbacks implies that $\gamma([\Delta])$ maps to $\gamma([x])$ in $H_{dR}^{2d}(X/k)$, in other words, we have $\gamma_0 = 1 \otimes \gamma([x])$. We conclude two things from this: (a) the class $\gamma([x])$ is independent of x , (b) it suffices to show the class $\gamma([x])$ is nonzero, and hence (c) it suffices to find any zero cycle α on X such that $\gamma(\alpha) \neq 0$. To do this we choose a finite morphism

$$f : X \longrightarrow \mathbf{P}_k^d$$

To see such a morphism exist, see Intersection Theory, Section ?? and in particular Lemma ?. Observe that f is finite syntomic (local complete intersection morphism by More on Morphisms, Lemma ?? and flat by Algebra, Lemma ??). By Proposition 19.3 we have a trace map

$$\Theta_f : f_* \Omega_{X/k}^\bullet \longrightarrow \Omega_{\mathbf{P}_k^d/k}^\bullet$$

whose composition with the canonical map

$$\Omega_{\mathbf{P}_k^d/k}^\bullet \longrightarrow f_* \Omega_{X/k}^\bullet$$

is multiplication by the degree of f . Hence we see that we get a map

$$\Theta : H_{dR}^{2d}(X/k) \rightarrow H_{dR}^{2d}(\mathbf{P}_k^d/k)$$

such that $\Theta \circ f^*$ is multiplication by a positive integer. Hence if we can find a zero cycle on $\mathbf{P}_k^d$ whose class is nonzero, then we conclude by the compatibility of γ with pullbacks. This is true by Lemma 11.4 and this finishes the proof of axiom (A6).

Below we will use the following without further mention. First, by Weil Cohomology Theories, Remark ?? the map $\lambda_X : H_{dR}^{2d}(X/k) \rightarrow k$ is unique. Second, in the proof of axiom (A6) we have seen that $\lambda_X(\gamma([x])) = 1$ when X is irreducible, i.e., the composition of the cycle class map $\gamma : \text{CH}^d(X) \rightarrow H_{dR}^{2d}(X/k)$ with λ_X is the degree map.

Axiom (A9). Let $Y \subset X$ be a nonempty smooth divisor on a nonempty smooth equidimensional projective scheme X over k of dimension d . We have to show that the diagram

$$\begin{array}{ccc} H_{dR}^{2d-2}(X/k) & \xrightarrow{c_1^{dR}(\mathcal{O}_X(Y)) \cap -} & H_{dR}^{2d}(X) \\ \text{restriction} \downarrow & & \downarrow \lambda_X \\ H_{dR}^{2d-2}(Y/k) & \xrightarrow{\lambda_Y} & k \end{array}$$

commutes where λ_X and λ_Y are as in axiom (A6). Above we have seen that if we decompose $X = \coprod X_i$ into connected (equivalently irreducible) components, then we have correspondingly $\lambda_X = \sum \lambda_{X_i}$. Similarly, if we decompose $Y = \coprod Y_j$ into connected (equivalently irreducible) components, then we have $\lambda_Y = \sum \lambda_{Y_j}$. Moreover, in this case we have $\mathcal{O}_X(Y) = \otimes_j \mathcal{O}_X(Y_j)$ and hence

$$c_1^{dR}(\mathcal{O}_X(Y)) = \sum_j c_1^{dR}(\mathcal{O}_X(Y_j))$$

in $H_{dR}^{2d}(X/k)$. A straightforward diagram chase shows that it suffices to prove the commutativity of the diagram in case X and Y are both irreducible. Then $H_{dR}^{2d-2}(Y/k)$ is 1-dimensional as we have Poincaré duality for Y by Weil Cohomology Theories, Lemma ??. By axiom (A4) the kernel of restriction (left vertical arrow) is contained in the kernel of cupping with $c_1^{dR}(\mathcal{O}_X(Y))$. This means it suffices to find one cohomology class $a \in H_{dR}^{2d-2}(X)$ whose restriction to Y is nonzero such that we have commutativity in the diagram for a . Take any ample invertible module $\mathcal{L}$ and set

$$a = c_1^{dR}(\mathcal{L})^{d-1}$$

Then we know that $a|_Y = c_1^{dR}(\mathcal{L}|_Y)^{d-1}$ and hence

$$\lambda_Y(a|_Y) = \deg(c_1(\mathcal{L}|_Y)^{d-1} \cap [Y])$$

by our description of λ_Y above. This is a positive integer by Chow Homology, Lemma ?? combined with Varieties, Lemma ??. Similarly, we find

$$\lambda_X(c_1^{dR}(\mathcal{O}_X(Y)) \cap a) = \deg(c_1(\mathcal{O}_X(Y)) \cap c_1(\mathcal{L})^{d-1} \cap [X])$$

Since we know that $c_1(\mathcal{O}_X(Y)) \cap [X] = [Y]$ more or less by definition we have an equality of zero cycles

$$(Y \rightarrow X)_* (c_1(\mathcal{L}|_Y)^{d-1} \cap [Y]) = c_1(\mathcal{O}_X(Y)) \cap c_1(\mathcal{L})^{d-1} \cap [X]$$

on X . Thus these cycles have the same degree and the proof is complete.

Proposition 22.1. *Let k be a field of characteristic zero. The functor that sends a smooth projective scheme X over k to $H_{dR}^*(X/k)$ is a Weil cohomology theory in the sense of Weil Cohomology Theories, Definition ??.*

Proof. In the discussion above we showed that our data (D0), (D1), (D2') satisfies axioms (A1) – (A9) of Weil Cohomology Theories, Section ??. Hence we conclude by Weil Cohomology Theories, Proposition ??.

Please don't read what follows. In the proof of the assertions we also used Lemmas 3.5, 9.1, 15.6, 8.2, 20.2, and 11.4, Propositions 14.1, 17.3, and 19.3, Weil Cohomology Theories, Lemmas ??, ??, ??, and ??, Weil Cohomology Theories, Remark ??, Varieties, Lemmas ?? and ??, Intersection Theory, Section ?? and Lemma ??, More on Morphisms, Lemma ??, Algebra, Lemma ??, and Chow Homology, Lemma ??.

Remark 22.2. In exactly the same manner as above one can show that Hodge cohomology $X \mapsto H_{Hodge}^*(X/k)$ equipped with c_1^{Hodge} determines a Weil cohomology theory. If we ever need this, we will precisely formulate and prove this here. This leads to the following amusing consequence: If the betti numbers of a Weil cohomology theory are independent of the chosen Weil cohomology theory (over our field k of characteristic 0), then the Hodge-to-de Rham spectral sequence degenerates at E_1 ! Of course, the degeneration of the Hodge-to-de Rham spectral sequence is known (see for example [DI87] for a marvelous algebraic proof), but it is by no means an easy result! This suggests that proving the independence of betti numbers is a hard problem as well and as far as we know is still an open problem. See Weil Cohomology Theories, Remark ?? for a related question.

23. Gysin maps for closed immersions

In this section we define the Gysin map for closed immersions.

Remark 23.1. Let $X \rightarrow S$ be a morphism of schemes. Let $f_1, \dots, f_c \in \Gamma(X, \mathcal{O}_X)$. Let $Z \subset X$ be the closed subscheme cut out by $f_1, \dots, f_c$. Below we will study the *Gysin map*

$$(23.1.1) \quad \gamma_{f_1, \dots, f_c}^p : \Omega_{Z/S}^p \longrightarrow \mathcal{H}_Z^c(\Omega_{X/S}^{p+c})$$

defined as follows. Given a local section ω of $\Omega_{Z/S}^p$ which is the restriction of a section $\tilde{\omega}$ of $\Omega_{X/S}^p$ we set

$$\gamma_{f_1, \dots, f_c}^p(\omega) = c_{f_1, \dots, f_c}(\tilde{\omega}|_Z) \wedge df_1 \wedge \dots \wedge df_c$$

where $c_{f_1, \dots, f_c} : \Omega_{X/S}^p \otimes \mathcal{O}_Z \rightarrow \mathcal{H}_Z^c(\Omega_{X/S}^p)$ is the map constructed in Derived Categories of Schemes, Remark ??. This is well defined: given ω we can change our choice of $\tilde{\omega}$ by elements of the form $\sum f_i \omega'_i + \sum d(f_i) \wedge \omega''_i$ which are mapped to zero by the construction.

Lemma 23.2. *The Gysin map (23.1.1) is compatible with the de Rham differentials on $\Omega_{X/S}^\bullet$ and $\Omega_{Z/S}^\bullet$.*

Proof. This follows from an almost trivial calculation once we correctly interpret this. First, we recall that the functor $\mathcal{H}_Z^c$ computed on the category of $\mathcal{O}_X$ -modules agrees with the similarly defined functor on the category of abelian sheaves on X ,

see Cohomology, Lemma ?? . Hence, the differential $d : \Omega_{X/S}^p \rightarrow \Omega_{X/S}^{p+1}$ induces a map $\mathcal{H}_Z^c(\Omega_{X/S}^p) \rightarrow \mathcal{H}_Z^c(\Omega_{X/S}^{p+1})$. Moreover, the formation of the extended alternating Čech complex in Derived Categories of Schemes, Remark ?? works on the category of abelian sheaves. The map

$$\text{Coker} \left(\bigoplus \mathcal{F}_{1 \dots \hat{i} \dots c} \rightarrow \mathcal{F}_{1 \dots c} \right) \longrightarrow i_* \mathcal{H}_Z^c(\mathcal{F})$$

used in the construction of $c_{f_1, \dots, f_c}$ in Derived Categories of Schemes, Remark ?? is well defined and functorial on the category of all abelian sheaves on X . Hence we see that the lemma follows from the equality

$$d \left(\frac{\tilde{\omega} \wedge df_1 \wedge \dots \wedge df_c}{f_1 \dots f_c} \right) = \frac{d(\tilde{\omega}) \wedge df_1 \wedge \dots \wedge df_c}{f_1 \dots f_c}$$

which is clear. $\square$

Lemma 23.3. *Let $X \rightarrow S$ be a morphism of schemes. Let $Z \rightarrow X$ be a closed immersion of finite presentation whose conormal sheaf $\mathcal{C}_{Z/X}$ is locally free of rank c . Then there is a canonical map*

$$\gamma^p : \Omega_{Z/S}^p \rightarrow \mathcal{H}_Z^c(\Omega_{X/S}^{p+c})$$

which is locally given by the maps $\gamma_{f_1, \dots, f_c}^p$ of Remark 23.1.

Proof. The assumptions imply that given $x \in Z \subset X$ there exists an open neighbourhood U of x such that Z is cut out by c elements $f_1, \dots, f_c \in \mathcal{O}_X(U)$. Thus it suffices to show that given $f_1, \dots, f_c$ and $g_1, \dots, g_c$ in $\mathcal{O}_X(U)$ cutting out $Z \cap U$, the maps $\gamma_{f_1, \dots, f_c}^p$ and $\gamma_{g_1, \dots, g_c}^p$ are the same. To do this, after shrinking U we may assume $g_j = \sum a_{ji} f_i$ for some $a_{ji} \in \mathcal{O}_X(U)$. Then we have $c_{f_1, \dots, f_c} = \det(a_{ji}) c_{g_1, \dots, g_c}$ by Derived Categories of Schemes, Lemma ?? . On the other hand we have

$$d(g_1) \wedge \dots \wedge d(g_c) \equiv \det(a_{ji}) d(f_1) \wedge \dots \wedge d(f_c) \bmod (f_1, \dots, f_c) \Omega_{X/S}^c$$

Combining these relations, a straightforward calculation gives the desired equality. $\square$

Lemma 23.4. *Let $X \rightarrow S$ and $i : Z \rightarrow X$ be as in Lemma 23.3. The Gysin map γ^p is compatible with the de Rham differentials on $\Omega_{X/S}^\bullet$ and $\Omega_{Z/S}^\bullet$.*

Proof. We may check this locally and then it follows from Lemma 23.2. $\square$

Lemma 23.5. *Let $X \rightarrow S$ and $i : Z \rightarrow X$ be as in Lemma 23.3. Given $\alpha \in H^q(X, \Omega_{X/S}^p)$ we have $\gamma^p(\alpha|_Z) = i^{-1}\alpha \wedge \gamma^0(1)$ in $H^q(Z, \mathcal{H}_Z^c(\Omega_{X/S}^{p+c}))$. Please see proof for notation.*

Proof. The restriction $\alpha|_Z$ is the element of $H^q(Z, \Omega_{Z/S}^p)$ given by functoriality for Hodge cohomology. Applying functoriality for cohomology using $\gamma^p : \Omega_{Z/S}^p \rightarrow \mathcal{H}_Z^c(\Omega_{X/S}^{p+c})$ we get $\gamma^p(\alpha|_Z)$ in $H^q(Z, \mathcal{H}_Z^c(\Omega_{X/S}^{p+c}))$. This explains the left hand side of the formula.

To explain the right hand side, we first pullback by the map of ringed spaces $i : (Z, i^{-1}\mathcal{O}_X) \rightarrow (X, \mathcal{O}_X)$ to get the element $i^{-1}\alpha \in H^q(Z, i^{-1}\Omega_{X/S}^p)$. Let $\gamma^0(1) \in H^0(Z, \mathcal{H}_Z^c(\Omega_{X/S}^c))$ be the image of $1 \in H^0(Z, \mathcal{O}_Z) = H^0(Z, \Omega_{Z/S}^0)$ by γ^0 . Using cup product we obtain an element

$$i^{-1}\alpha \cup \gamma^0(1) \in H^{q+c}(Z, i^{-1}\Omega_{X/S}^p \otimes_{i^{-1}\mathcal{O}_X} \mathcal{H}_Z^c(\Omega_{X/S}^c))$$

Using Cohomology, Remark ?? and wedge product there are canonical maps

$$i^{-1}\Omega_{X/S}^p \otimes_{i^{-1}\mathcal{O}_X}^{\mathbf{L}} R\mathcal{H}_Z(\Omega_{X/S}^c) \rightarrow R\mathcal{H}_Z(\Omega_{X/S}^p \otimes_{\mathcal{O}_X}^{\mathbf{L}} \Omega_{X/S}^c) \rightarrow R\mathcal{H}_Z(\Omega_{X/S}^{p+c})$$

By Derived Categories of Schemes, Lemma ?? the objects $R\mathcal{H}_Z(\Omega_{X/S}^j)$ have vanishing cohomology sheaves in degrees $> c$. Hence on cohomology sheaves in degree c we obtain a map

$$i^{-1}\Omega_{X/S}^p \otimes_{i^{-1}\mathcal{O}_X} \mathcal{H}_Z^c(\Omega_{X/S}^c) \longrightarrow \mathcal{H}_Z^c(\Omega_{X/S}^{p+c})$$

The expression $i^{-1}\alpha \wedge \gamma^0(1)$ is the image of the cup product $i^{-1}\alpha \cup \gamma^0(1)$ by the functoriality of cohomology.

Having explained the content of the formula in this manner, by general properties of cup products (Cohomology, Section ??), it now suffices to prove that the diagram

$$\begin{array}{ccc} i^{-1}\Omega_X^p \otimes \Omega_Z^0 & \xrightarrow{\text{id} \otimes \gamma^0} & i^{-1}\Omega_X^p \otimes \mathcal{H}_Z^c(\Omega_X^c) \\ \downarrow & & \downarrow \wedge \\ \Omega_Z^p \otimes \Omega_Z^0 & \xrightarrow{\wedge} \Omega_Z^p \xrightarrow{\gamma^p} & \mathcal{H}_Z^c(\Omega_X^{p+c}) \end{array}$$

is commutative in the category of sheaves on Z (with obvious abuse of notation). This boils down to a simple computation for the maps $\gamma_{f_1, \dots, f_c}^j$ which we omit; in fact these maps are chosen exactly such that this works and such that 1 maps to $\frac{df_1 \wedge \dots \wedge df_c}{f_1 \dots f_c}$. $\square$

Lemma 23.6. *Let $c \geq 0$ be an integer. Let*

$$\begin{array}{ccccc} Z' & \longrightarrow & X' & \longrightarrow & S' \\ h \downarrow & & g \downarrow & & \downarrow \\ Z & \longrightarrow & X & \longrightarrow & S \end{array}$$

be a commutative diagram of schemes. Assume

- (1) $Z \rightarrow X$ and $Z' \rightarrow X'$ satisfy the assumptions of Lemma 23.3,
- (2) the left square in the diagram is cartesian, and
- (3) $h^*\mathcal{C}_{Z/X} \rightarrow \mathcal{C}_{Z'/X'}$ (Morphisms, Lemma ??) is an isomorphism.

Then the diagram

$$\begin{array}{ccc} h^*\Omega_{Z/S}^p & \xrightarrow{h^{-1}\gamma^p} & \mathcal{O}_{X'|Z'} \otimes_{h^{-1}\mathcal{O}_X|Z} h^{-1}\mathcal{H}_Z^c(\Omega_{X/S}^{p+c}) \\ \downarrow & & \downarrow \\ \Omega_{Z'/S'}^p & \xrightarrow{\gamma^p} & \mathcal{H}_{Z'}^c(\Omega_{X'/S'}^{p+c}) \end{array}$$

is commutative. The left vertical arrow is functoriality of modules of differentials and the right vertical arrow uses Cohomology, Remark ??.

Proof. More precisely, consider the composition

$$\begin{aligned} \mathcal{O}_{X'|Z'} \otimes_{h^{-1}\mathcal{O}_X|Z}^{\mathbf{L}} h^{-1}R\mathcal{H}_Z(\Omega_{X/S}^{p+c}) &\rightarrow R\mathcal{H}_{Z'}(Lg^*\Omega_{X/S}^{p+c}) \\ &\rightarrow R\mathcal{H}_{Z'}(g^*\Omega_{X/S}^{p+c}) \\ &\rightarrow R\mathcal{H}_{Z'}(\Omega_{X'/S'}^{p+c}) \end{aligned}$$

where the first arrow is given by Cohomology, Remark ?? and the last one by functoriality of differentials. Since we have the vanishing of cohomology sheaves in degrees $> c$ by Derived Categories of Schemes, Lemma ?? this induces the right vertical arrow. We can check the commutativity locally. Thus we may assume Z is cut out by $f_1, \dots, f_c \in \Gamma(X, \mathcal{O}_X)$. Then Z' is cut out by $f'_i = g^\sharp(f_i)$. The maps $c_{f_1, \dots, f_c}$ and $c_{f'_1, \dots, f'_c}$ fit into the commutative diagram

$$\begin{array}{ccc} h^* i^* \Omega_{X/S}^p & \xrightarrow{h^{-1} c_{f_1, \dots, f_c}} & \mathcal{O}_{X'|Z'} \otimes_{h^{-1} \mathcal{O}_X|Z} h^{-1} \mathcal{H}_Z^c(\Omega_{X/S}^p) \\ \downarrow & & \downarrow \\ (i')^* \Omega_{X'/S'}^p & \xrightarrow{c_{f'_1, \dots, f'_c}} & \mathcal{H}_{Z'}^c(\Omega_{X'/S'}^p) \end{array}$$

See Derived Categories of Schemes, Remark ?. Recall given a p -form ω on Z we define $\gamma^p(\omega)$ by choosing (locally on X and Z) a p -form $\tilde{\omega}$ on X lifting ω and taking $\gamma^p(\omega) = c_{f_1, \dots, f_c}(\tilde{\omega}) \wedge df_1 \wedge \dots \wedge df_c$. Since the form $df_1 \wedge \dots \wedge df_c$ pulls back to $df'_1 \wedge \dots \wedge df'_c$ we conclude. $\square$

Remark 23.7. Let $X \rightarrow S$, $i : Z \rightarrow X$, and $c \geq 0$ be as in Lemma 23.3. Let $p \geq 0$ and assume that $\mathcal{H}_Z^i(\Omega_{X/S}^{p+c}) = 0$ for $i = 0, \dots, c-1$. This vanishing holds if $X \rightarrow S$ is smooth and $Z \rightarrow X$ is a Koszul regular immersion, see Derived Categories of Schemes, Lemma ?. Then we obtain a map

$$\gamma^{p,q} : H^q(Z, \Omega_{Z/S}^p) \longrightarrow H^{q+c}(X, \Omega_{X/S}^{p+c})$$

by first using $\gamma^p : \Omega_{Z/S}^p \rightarrow \mathcal{H}_Z^c(\Omega_{X/S}^{p+c})$ to map into

$$H^q(Z, \mathcal{H}_Z^c(\Omega_{X/S}^{p+c})) = H^q(Z, R\mathcal{H}_Z(\Omega_{X/S}^{p+c})[c]) = H^q(X, i_* R\mathcal{H}_Z(\Omega_{X/S}^{p+c})[c])$$

and then using the adjunction map $i_* R\mathcal{H}_Z(\Omega_{X/S}^{p+c}) \rightarrow \Omega_{X/S}^{p+c}$ to continue on to the desired Hodge cohomology module.

Lemma 23.8. Let $X \rightarrow S$ and $i : Z \rightarrow X$ be as in Lemma 23.3. Assume $X \rightarrow S$ is smooth and $Z \rightarrow X$ Koszul regular. The Gysin maps $\gamma^{p,q}$ are compatible with the de Rham differentials on $\Omega_{X/S}^\bullet$ and $\Omega_{Z/S}^\bullet$.

Proof. This follows immediately from Lemma 23.4. $\square$

Lemma 23.9. Let $X \rightarrow S$, $i : Z \rightarrow X$, and $c \geq 0$ be as in Lemma 23.3. Assume $X \rightarrow S$ smooth and $Z \rightarrow X$ Koszul regular. Given $\alpha \in H^q(X, \Omega_{X/S}^p)$ we have $\gamma^{p,q}(\alpha|_Z) = \alpha \cup \gamma^{0,0}(1)$ in $H^{q+c}(X, \Omega_{X/S}^{p+c})$ with $\gamma^{a,b}$ as in Remark 23.7.

Proof. This lemma follows from Lemma 23.5 and Cohomology, Lemma ?. We suggest the reader skip over the more detailed discussion below.

We will use without further mention that $R\mathcal{H}_Z(\Omega_{X/S}^j) = \mathcal{H}_Z^c(\Omega_{X/S}^j)[-c]$ for all j as pointed out in Remark 23.7. We will also silently use the identifications $H_Z^{q+c}(X, \Omega_{X/S}^j) = H^{q+c}(Z, R\mathcal{H}_Z(\Omega_{X/S}^j)) = H^q(Z, \mathcal{H}_Z^c(\Omega_{X/S}^j))$, see Cohomology, Lemma ?? for the first one. With these identifications

- (1) $\gamma^0(1) \in H_Z^c(X, \Omega_{X/S}^c)$ maps to $\gamma^{0,0}(1)$ in $H^c(X, \Omega_{X/S}^c)$,
- (2) the right hand side $i^{-1}\alpha \wedge \gamma^0(1)$ of the equality in Lemma 23.5 is the (image by wedge product of the) cup product of Cohomology, Remark ?? of the elements α and $\gamma^0(1)$, in other words, the constructions in the proof of Lemma 23.5 and in Cohomology, Remark ?? match,

- (3) by Cohomology, Lemma ?? this maps to $\alpha \cup \gamma^{0,0}(1)$ in $H^{q+c}(X, \Omega_{X/S}^p \otimes \Omega_{X/S}^c)$, and
 (4) the left hand side $\gamma^p(\alpha|_Z)$ of the equality in Lemma 23.5 maps to $\gamma^{p,q}(\alpha|_Z)$.

This finishes the proof. $\square$

Lemma 23.10. *Let $c \geq 0$ and*

$$\begin{array}{ccccc} Z' & \longrightarrow & X' & \longrightarrow & S' \\ h \downarrow & & g \downarrow & & \downarrow \\ Z & \longrightarrow & X & \longrightarrow & S \end{array}$$

satisfy the assumptions of Lemma 23.6 and assume in addition that $X \rightarrow S$ and $X' \rightarrow S'$ are smooth and that $Z \rightarrow X$ and $Z' \rightarrow X'$ are Koszul regular immersions. Then the diagram

$$\begin{array}{ccc} H^q(Z, \Omega_{Z/S}^p) & \xrightarrow{\gamma^{p,q}} & H^{q+c}(X, \Omega_{X/S}^{p+c}) \\ \downarrow & & \downarrow \\ H^q(Z', \Omega_{Z'/S'}^p) & \xrightarrow{\gamma^{p,q}} & H^{q+c}(X', \Omega_{X'/S'}^{p+c}) \end{array}$$

is commutative where $\gamma^{p,q}$ is as in Remark 23.7.

Proof. This follows on combining Lemma 23.6 and Cohomology, Lemma ?? $\square$

Lemma 23.11. *Let k be a field. Let X be an irreducible smooth proper scheme over k of dimension d . Let $Z \subset X$ be the reduced closed subscheme consisting of a single k -rational point x . Then the image of $1 \in k = H^0(Z, \mathcal{O}_Z) = H^0(Z, \Omega_{Z/k}^0)$ by the map $H^0(Z, \Omega_{Z/k}^0) \rightarrow H^d(X, \Omega_{X/k}^d)$ of Remark 23.7 is nonzero.*

Proof. The map $\gamma^0 : \mathcal{O}_Z \rightarrow \mathcal{H}_Z^d(\Omega_{X/k}^d) = R\mathcal{H}_Z(\Omega_{X/k}^d)[d]$ is adjoint to a map

$$g^0 : i_* \mathcal{O}_Z \longrightarrow \Omega_{X/k}^d[d]$$

in $D(\mathcal{O}_X)$. Recall that $\Omega_{X/k}^d = \omega_X$ is a dualizing sheaf for X/k , see Duality for Schemes, Lemma ???. Hence the k -linear dual of the map in the statement of the lemma is the map

$$H^0(X, \mathcal{O}_X) \rightarrow \text{Ext}_X^d(i_* \mathcal{O}_Z, \omega_X)$$

which sends 1 to g^0 . Thus it suffices to show that g^0 is nonzero. This we may do in any neighbourhood U of the point x . Choose U such that there exist $f_1, \dots, f_d \in \mathcal{O}_X(U)$ vanishing only at x and generating the maximal ideal $\mathfrak{m}_x \subset \mathcal{O}_{X,x}$. We may assume $U = \text{Spec}(R)$ is affine. Looking over the construction of γ^0 we find that our extension is given by

$$k \rightarrow (R \rightarrow \bigoplus_{i_0} R_{f_{i_0}} \rightarrow \bigoplus_{i_0 < i_1} R_{f_{i_0} f_{i_1}} \rightarrow \dots \rightarrow R_{f_1 \dots f_d})[d] \rightarrow R[d]$$

where 1 maps to $1/f_1 \dots f_d$ under the first map. This is nonzero because $1/f_1 \dots f_d$ is a nonzero element of local cohomology group $H_{(f_1, \dots, f_d)}^d(R)$ in this case, $\square$

24. Relative Poincaré duality

In this section we prove Poincaré duality for the relative de Rham cohomology of a proper smooth scheme over a base. We strongly urge the reader to look at Section 20 first.

Situation 24.1. Here S is a quasi-compact and quasi-separated scheme and $f : X \rightarrow S$ is a proper smooth morphism of schemes all of whose fibres are nonempty and equidimensional of dimension n .

Lemma 24.2. *In Situation 24.1 the pushforward $f_*\mathcal{O}_X$ is a finite étale $\mathcal{O}_S$ -algebra and locally on S we have $Rf_*\mathcal{O}_X = f_*\mathcal{O}_X \oplus P$ in $D(\mathcal{O}_S)$ with P perfect of tor amplitude in $[1, \infty)$. The map $d : f_*\mathcal{O}_X \rightarrow f_*\Omega_{X/S}$ is zero.*

Proof. The first part of the statement follows from Derived Categories of Schemes, Lemma ?? . Setting $S' = \underline{\mathrm{Spec}}_S(f_*\mathcal{O}_X)$ we get a factorization $X \rightarrow S' \rightarrow S$ (this is the Stein factorization, see More on Morphisms, Section ??, although we don't need this) and we see that $\Omega_{X/S} = \Omega_{X/S'}$ for example by Morphisms, Lemma ?? and ??. This of course implies that $d : f_*\mathcal{O}_X \rightarrow f_*\Omega_{X/S}$ is zero. $\square$

Lemma 24.3. *In Situation 24.1 there exists an $\mathcal{O}_S$ -module map*

$$t : Rf_*\Omega_{X/S}^n[n] \longrightarrow \mathcal{O}_S$$

unique up to precomposing by multiplication by a unit of $H^0(X, \mathcal{O}_X)$ with the following property: for all p the pairing

$$Rf_*\Omega_{X/S}^p \otimes_{\mathcal{O}_S}^{\mathbf{L}} Rf_*\Omega_{X/S}^{n-p}[n] \longrightarrow \mathcal{O}_S$$

given by the relative cup product composed with t is a perfect pairing of perfect complexes on S .

Proof. We first assume S is a Noetherian scheme. Let $\omega_{X/S}^\bullet$ be the relative dualizing complex of X over S as in Duality for Schemes, Remark ?? and let $Rf_*\omega_{X/S}^\bullet \rightarrow \mathcal{O}_S$ be its trace map. By Duality for Schemes, Lemma ?? (here we use that S is Noetherian) there exists an isomorphism $\omega_{X/S}^\bullet \cong \Omega_{X/S}^n[n]$ and using this isomorphism we obtain t . The complexes $Rf_*\Omega_{X/S}^p$ are perfect by Lemma 3.5. Since $\Omega_{X/S}^p$ is locally free and since $\Omega_{X/S}^p \otimes_{\mathcal{O}_X} \Omega_{X/S}^{n-p} \rightarrow \Omega_{X/S}^n$ exhibits an isomorphism $\Omega_{X/S}^p \cong \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/S}^{n-p}, \Omega_{X/S}^n)$ we see that the pairing induced by the relative cup product is perfect by Duality for Schemes, Remark ??.

Proof of existence in the general case. By absolute Noetherian approximation we can find a cartesian diagram

$$\begin{array}{ccc} X & \longrightarrow & X_0 \\ \downarrow f & & \downarrow f_0 \\ S & \xrightarrow{g} & S_0 \end{array}$$

with S_0 Noetherian and f_0 a proper and smooth morphism all of whose fibres are nonempty and equidimensional of dimension n . See Limits, Lemmas ??, ??, ??, ??, and ??. Choose a map

$$t_0 : Rf_{0,*}\Omega_{X_0/S_0}^n[n] \longrightarrow \mathcal{O}_{S_0}$$

as in the statement of the lemma. By cohomology and base change the complexes $Rf_*(\Omega_{X/S}^p)$ are equal to $Lg^*(Rf_{0,*}\Omega_{X_0/S_0}^p)$ for all $p \geq 0$, see Lemma 3.5. In particular the pullback of t_0 by g gives us a map t as in the statement of the lemma. Namely, by Cohomology, Lemma ??, the cup product map

$$Rf_*\Omega_{X/S}^p \otimes_{\mathcal{O}_S}^{\mathbf{L}} Rf_*\Omega_{X/S}^{n-p}[n] \longrightarrow Rf_*\Omega_{X/S}^n$$

is the pullback by g of the corresponding cup product map for f_0 . Hence the fact that t_0 produces perfect pairings, pulls back to the same property for t on S .

Uniqueness of t . Choose a distinguished triangle $f_*\mathcal{O}_X \rightarrow Rf_*\mathcal{O}_X \rightarrow P \rightarrow f_*\mathcal{O}_X[1]$. By Lemma 24.2 the object P is perfect of tor amplitude in $[1, \infty)$ and the triangle is locally on S split. Thus $R\mathcal{H}om_{\mathcal{O}_X}(P, \mathcal{O}_X)$ is perfect of tor amplitude in $(-\infty, -1]$. Hence duality (above) shows that locally on S we have

$$Rf_*\Omega_{X/S}^n[n] \cong R\mathcal{H}om_{\mathcal{O}_S}(f_*\mathcal{O}_X, \mathcal{O}_S) \oplus R\mathcal{H}om_{\mathcal{O}_X}(P, \mathcal{O}_X)$$

This shows that $R^n f_*\Omega_{X/S}^n$ is finite locally free and that we obtain a perfect $\mathcal{O}_S$ -bilinear pairing

$$f_*\mathcal{O}_X \times R^n f_*\Omega_{X/S}^n \longrightarrow \mathcal{O}_S$$

using t . This implies that any $\mathcal{O}_S$ -linear map $t' : R^n f_*\Omega_{X/S}^n \rightarrow \mathcal{O}_S$ is of the form $t' = t \circ g$ for some $g \in \Gamma(S, f_*\mathcal{O}_X) = \Gamma(X, \mathcal{O}_X)$. In order for t' to still determine a perfect pairing g will have to be a unit. This finishes the proof. $\square$

Lemma 24.4. *In Situation 24.1 the map $d : R^n f_*\Omega_{X/S}^{n-1} \rightarrow R^n f_*\Omega_{X/S}^n$ is zero.*

As we mentioned in the proof of Lemma 20.3 this lemma is not an easy consequence of Lemmas 24.3 and 24.2.

Proof in case S is reduced. Assume S is reduced. Observe that $d : R^n f_*\Omega_{X/S}^{n-1} \rightarrow R^n f_*\Omega_{X/S}^n$ is an $\mathcal{O}_S$ -linear map of (quasi-coherent) $\mathcal{O}_S$ -modules. The $\mathcal{O}_S$ -module $R^n f_*\Omega_{X/S}^n$ is finite locally free (as the dual of the finite locally free $\mathcal{O}_S$ -module $f_*\mathcal{O}_X$ by Lemmas 24.3 and 24.2). Since S is reduced it suffices to show that the stalk of d in every generic point $\eta \in S$ is zero; this follows by looking at sections over affine opens, using that the target of d is locally free, and Algebra, Lemma ?? part (2). Since S is reduced we have $\mathcal{O}_{S,\eta} = \kappa(\eta)$, see Algebra, Lemma ?. Thus d_η is identified with the map

$$d : H^n(X_\eta, \Omega_{X_\eta/\kappa(\eta)}^{n-1}) \longrightarrow H^n(X_\eta, \Omega_{X_\eta/\kappa(\eta)}^n)$$

which is zero by Lemma 20.3. $\square$

Proof in the general case. Observe that the question is flat local on S : if $S' \rightarrow S$ is a surjective flat morphism of schemes and the map is zero after pullback to S' , then the map is zero. Also, formation of the map commutes with base change by flat morphisms by flat base change (Cohomology of Schemes, Lemma ??).

Consider the Stein factorization $X \rightarrow S' \rightarrow S$ as in More on Morphisms, Theorem ?. By Lemma 24.2 the morphism $\pi : S' \rightarrow S$ is finite étale. The morphism $f : X \rightarrow S'$ is proper (by the theorem), smooth (by More on Morphisms, Lemma ??) with geometrically connected fibres by the theorem on Stein factorization. In the proof of Lemma 24.2 we saw that $\Omega_{X/S} = \Omega_{X/S'}$ because $S' \rightarrow S$ is étale. Hence $\Omega_{X/S}^\bullet = \Omega_{X/S'}^\bullet$. We have

$$R^q f_*\Omega_{X/S}^p = \pi_* R^q f'_*\Omega_{X/S'}^p$$

for all p, q by the Leray spectral sequence (Cohomology, Lemma ??), the fact that π is finite hence affine, and Cohomology of Schemes, Lemma ?? (of course we also use that $R^q f'_* \Omega_{X'/S}^p$ is quasi-coherent). Thus the map of the lemma is π_* applied to $d : R^n f'_* \Omega_{X'/S}^{n-1} \rightarrow R^n f'_* \Omega_{X/S}^n$. In other words, in order to prove the lemma we may replace $f : X \rightarrow S$ by $f' : X \rightarrow S'$ to reduce to the case discussed in the next paragraph.

Assume f has geometrically connected fibres and $f_* \mathcal{O}_X = \mathcal{O}_S$. For every $s \in S$ we can choose an étale neighbourhood $(S', s') \rightarrow (S, s)$ such that the base change $X' \rightarrow S'$ of S has a section. See More on Morphisms, Lemma ??. By the initial remarks of the proof this reduces us to the case discussed in the next paragraph.

Assume f has geometrically connected fibres, $f_* \mathcal{O}_X = \mathcal{O}_S$, and we have a section $s : S \rightarrow X$ of f . We may and do assume $S = \text{Spec}(A)$ is affine. The map $s^* : R\Gamma(X, \mathcal{O}_X) \rightarrow R\Gamma(S, \mathcal{O}_S) = A$ is a splitting of the map $A \rightarrow R\Gamma(X, \mathcal{O}_X)$. Thus we can write

$$R\Gamma(X, \mathcal{O}_X) = A \oplus P$$

where P is the “kernel” of s^* . By Lemma 24.2 the object P of $D(A)$ is perfect of tor amplitude in $[1, n]$. As in the proof of Lemma 24.3 we see that $H^n(X, \Omega_{X/S}^n)$ is a locally free A -module of rank 1 (and in fact dual to A so free of rank 1 – we will soon choose a generator but we don’t want to check it is the same generator nor will it be necessary to do so).

Denote $Z \subset X$ the image of s which is a closed subscheme of X by Schemes, Lemma ??. Observe that $Z \rightarrow X$ is a regular (and a fortiori Koszul regular by Divisors, Lemma ??) closed immersion by Divisors, Lemma ??. Of course $Z \rightarrow X$ has codimension n . Thus by Remark 23.7 we can consider the map

$$\gamma^{0,0} : H^0(Z, \Omega_{Z/S}^0) \longrightarrow H^n(X, \Omega_{X/S}^n)$$

and we set $\xi = \gamma^{0,0}(1) \in H^n(X, \Omega_{X/S}^n)$.

We claim ξ is a basis element. Namely, since we have base change in top degree (see for example Limits, Lemma ??) we see that $H^n(X, \Omega_{X/S}^n) \otimes_A k = H^n(X_k, \Omega_{X_k/k}^n)$ for any ring map $A \rightarrow k$. By the compatibility of the construction of ξ with base change, see Lemma 23.10, we see that the image of ξ in $H^n(X_k, \Omega_{X_k/k}^n)$ is nonzero by Lemma 23.11 if k is a field. Thus ξ is a nowhere vanishing section of an invertible module and hence a generator.

Let $\theta \in H^n(X, \Omega_{X/S}^{n-1})$. We have to show that $d(\theta)$ is zero in $H^n(X, \Omega_{X/S}^n)$. We may write $d(\theta) = a\xi$ for some $a \in A$ as ξ is a basis element. Then we have to show $a = 0$.

Consider the closed immersion

$$\Delta : X \rightarrow X \times_S X$$

This is also a section of a smooth morphism (namely either projection) and hence a regular and Koszul immersion of codimension n as well. Thus we can consider the maps

$$\gamma^{p,q} : H^q(X, \Omega_{X/S}^p) \longrightarrow H^{q+n}(X \times_S X, \Omega_{X \times_S X/S}^{p+n})$$

of Remark 23.7. Consider the image

$$\gamma^{n-1,n}(\theta) \in H^{2n}(X \times_S X, \Omega_{X \times_S X/S}^{2n-1})$$

By Lemma 8.1 we have

$$\Omega_{X \times_S X}^{2n-1} = \Omega_{X/S}^{n-1} \boxtimes \Omega_{X/S}^n \oplus \Omega_{X/S}^n \boxtimes \Omega_{X/S}^{n-1}$$

By the Künneth formula (either Derived Categories of Schemes, Lemma ?? or Derived Categories of Schemes, Lemma ??) we see that

$$H^{2n}(X \times_S X, \Omega_{X/S}^{n-1} \boxtimes \Omega_{X/S}^n) = H^n(X, \Omega_{X/S}^{n-1}) \otimes_A H^n(X, \Omega_{X/S}^n)$$

and

$$H^{2n}(X \times_S X, \Omega_{X/S}^n \boxtimes \Omega_{X/S}^{n-1}) = H^n(X, \Omega_{X/S}^n) \otimes_A H^n(X, \Omega_{X/S}^{n-1})$$

Namely, since we are looking in top degree there no higher tor groups that intervene. Combined with the fact that ξ is a generator this means we can write

$$\gamma^{n-1,n}(\theta) = \theta_1 \otimes \xi + \xi \otimes \theta_2$$

with $\theta_1, \theta_2 \in H^n(X, \Omega_{X/S}^{n-1})$. Arguing in exactly the same manner we can write

$$\gamma^{n,n}(\xi) = b\xi \otimes \xi$$

in $H^{2n}(X \times_S X, \Omega_{X \times_S X/S}^{2n}) = H^n(X, \Omega_{X/S}^n) \otimes_A H^n(X, \Omega_{X/S}^n)$ for some $b \in H^0(S, \mathcal{O}_S)$.

Claim: $\theta_1 = \theta$, $\theta_2 = \theta$, and $b = 1$. Let us show that the claim implies the desired result $a = 0$. Namely, by Lemma 23.8 we have

$$\gamma^{n,n}(d(\theta)) = d(\gamma^{n-1,n}(\theta))$$

By our choices above this gives

$$a\xi \otimes \xi = \gamma^{n,n}(a\xi) = d(\theta \otimes \xi + \xi \otimes \theta) = a\xi \otimes \xi + (-1)^n a\xi \otimes \xi$$

The right most equality comes from the fact that the map $d : \Omega_{X \otimes_S X/S}^{2n-1} \rightarrow \Omega_{X \times_S X/S}^{2n}$ by Lemma 8.1 is the sum of the differential $d \boxtimes 1 : \Omega_{X/S}^{n-1} \boxtimes \Omega_{X/S}^n \rightarrow \Omega_{X/S}^n \boxtimes \Omega_{X/S}^n$ and the differential $(-1)^n 1 \boxtimes d : \Omega_{X/S}^n \boxtimes \Omega_{X/S}^{n-1} \rightarrow \Omega_{X/S}^n \boxtimes \Omega_{X/S}^n$. Please see discussion in Section 8 and Derived Categories of Schemes, Section ?? for more information. Since $\xi \otimes \xi$ is a basis for the rank 1 free A -module $H^n(X, \Omega_{X/S}^n) \otimes_A H^n(X, \Omega_{X/S}^n)$ we conclude

$$a = a + (-1)^n a \Rightarrow a = 0$$

as desired.

In the rest of the proof we prove the claim above. Let us denote $\eta = \gamma^{0,0}(1) \in H^n(X \times_S X, \Omega_{X \times_S X/S}^n)$. Since $\Omega_{X \times_S X/S}^n = \bigoplus_{p+p'=n} \Omega_{X/S}^p \boxtimes \Omega_{X/S}^{p'}$ we may write

$$\eta = \eta_0 + \eta_1 + \dots + \eta_n$$

where η_p is in $H^n(X \times_S X, \Omega_{X/S}^p \boxtimes \Omega_{X/S}^{n-p})$. For $p = 0$ we can write

$$\begin{aligned} H^n(X \times_S X, \mathcal{O}_X \boxtimes \Omega_{X/S}^n) &= H^n(R\Gamma(X, \mathcal{O}_X) \otimes_A^L R\Gamma(X, \Omega_{X/S}^n)) \\ &= A \otimes_A H^n(X, \Omega_{X/S}^n) \oplus H^n(P \otimes_A^L R\Gamma(X, \Omega_{X/S}^n)) \end{aligned}$$

by our previously given decomposition $R\Gamma(X, \mathcal{O}_X) = A \oplus P$. Consider the morphism $(s, \text{id}) : X \rightarrow X \times_S X$. Then $(s, \text{id})^{-1}(\Delta) = Z$ scheme theoretically. Hence we see that $(s, \text{id})^* \eta = \xi$ by Lemma 23.10. This means that

$$\xi = (s, \text{id})^* \eta = (s^* \otimes \text{id})(\eta_0)$$

This means exactly that the first component of η_0 in the direct sum decomposition above is ξ . In other words, we can write

$$\eta_0 = 1 \otimes \xi + \eta'_0$$

with $\eta'_0 \in H^n(P \otimes_A^{\mathbf{L}} R\Gamma(X, \Omega_{X/S}^n))$. In exactly the same manner for $p = n$ we can write

$$\begin{aligned} H^n(X \times_S X, \Omega_{X/S}^n \boxtimes \mathcal{O}_X) &= H^n(R\Gamma(X, \Omega_{X/S}^n) \otimes_A^{\mathbf{L}} R\Gamma(X, \mathcal{O}_X)) \\ &= H^n(X, \Omega_{X/S}^n) \otimes_A A \oplus H^n(R\Gamma(X, \Omega_{X/S}^n) \otimes_A^{\mathbf{L}} P) \end{aligned}$$

and we can write

$$\eta_n = \xi \otimes 1 + \eta'_n$$

with $\eta'_n \in H^n(R\Gamma(X, \Omega_{X/S}^n) \otimes_A^{\mathbf{L}} P)$.

Observe that $\mathrm{pr}_1^* \theta = \theta \otimes 1$ and $\mathrm{pr}_2^* \theta = 1 \otimes \theta$ are Hodge cohomology classes on $X \times_S X$ which pull back to θ by Δ . Hence by Lemma 23.9 we have

$$\theta_1 \otimes \xi + \xi \otimes \theta_2 = \gamma^{n-1, n}(\theta) = (\theta \otimes 1) \cup \eta = (1 \otimes \theta) \cup \eta$$

in the Hodge cohomology ring of $X \times_S X$ over S . In terms of the direct sum decomposition on the modules of differentials of $X \times_S X/S$ we obtain

$$\theta_1 \otimes \xi = (\theta \otimes 1) \cup \eta_0 \quad \text{and} \quad \xi \otimes \theta_2 = (1 \otimes \theta) \cup \eta_n$$

Looking at the formula $\eta_0 = 1 \otimes \xi + \eta'_0$ we found above, we see that to show that $\theta_1 = \theta$ it suffices to prove that

$$(\theta \otimes 1) \cup \eta'_0 = 0$$

To do this, observe that cupping with $\theta \otimes 1$ is given by the action on cohomology of the map

$$(P \otimes_A^{\mathbf{L}} R\Gamma(X, \Omega_{X/S}^n))[-n] \xrightarrow{\theta \otimes 1} R\Gamma(X, \Omega_{X/S}^{n-1}) \otimes_A^{\mathbf{L}} R\Gamma(X, \Omega_{X/S}^n)$$

in the derived category, see Cohomology, Remark ???. This map is the derived tensor product of the two maps

$$\theta : P[-n] \rightarrow R\Gamma(X, \Omega_{X/S}^{n-1}) \quad \text{and} \quad 1 : R\Gamma(X, \Omega_{X/S}^n) \rightarrow R\Gamma(X, \Omega_{X/S}^n)$$

by Derived Categories of Schemes, Remark ???. However, the first of these is zero in $D(A)$ because it is a map from a perfect complex of tor amplitude in $[n+1, 2n]$ to a complex with cohomology only in degrees $0, 1, \dots, n$, see More on Algebra, Lemma ???. A similar argument works to show the vanishing of $(1 \otimes \theta) \cup \eta'_n$. Finally, in exactly the same manner we obtain

$$b\xi \otimes \xi = \gamma^{n, n}(\xi) = (\xi \otimes 1) \cup \eta_0$$

and we conclude as before by showing that $(\xi \otimes 1) \cup \eta'_0 = 0$ in the same manner as above. This finishes the proof. $\square$

Proposition 24.5. *Let S be a quasi-compact and quasi-separated scheme. Let $f : X \rightarrow S$ be a proper smooth morphism of schemes all of whose fibres are nonempty and equidimensional of dimension n . There exists an $\mathcal{O}_S$ -module map*

$$t : R^{2n} f_* \Omega_{X/S}^\bullet \longrightarrow \mathcal{O}_S$$

unique up to precomposing by multiplication by a unit of $H^0(X, \mathcal{O}_X)$ with the following property: the pairing

$$Rf_* \Omega_{X/S}^\bullet \otimes_{\mathcal{O}_S}^{\mathbf{L}} Rf_* \Omega_{X/S}^\bullet[2n] \longrightarrow \mathcal{O}_S, \quad (\xi, \xi') \longmapsto t(\xi \cup \xi')$$

is a perfect pairing of perfect complexes on S .

Proof. The proof is exactly the same as the proof of Proposition 20.4.

By the relative Hodge-to-de Rham spectral sequence

$$E_1^{p,q} = R^q f_* \Omega_{X/S}^p \Rightarrow R^{p+q} f_* \Omega_{X/S}^\bullet$$

(Section 6), the vanishing of $\Omega_{X/S}^i$ for $i > n$, the vanishing in for example Limits, Lemma ?? and the results of Lemmas 24.2 and 24.4 we see that $R^0 f_* \Omega_{X/S} = R^0 f_* \mathcal{O}_X$ and $R^n f_* \Omega_{X/S}^n = R^{2n} f_* \Omega_{X/S}^\bullet$. More precisely, these identifications come from the maps of complexes

$$\Omega_{X/S}^\bullet \rightarrow \mathcal{O}_X[0] \quad \text{and} \quad \Omega_{X/S}^n[-n] \rightarrow \Omega_{X/S}^\bullet$$

Let us choose $t : R^{2n} f_* \Omega_{X/S} \rightarrow \mathcal{O}_S$ which via this identification corresponds to a t as in Lemma 24.3.

Let us abbreviate $\Omega^\bullet = \Omega_{X/S}^\bullet$. Consider the map (4.0.1) which in our situation reads

$$\wedge : \text{Tot}(\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \Omega^\bullet) \longrightarrow \Omega^\bullet$$

For every integer $p = 0, 1, \dots, n$ this map annihilates the subcomplex $\text{Tot}(\sigma_{>p} \Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p} \Omega^\bullet)$ for degree reasons. Hence we find that the restriction of $\wedge$ to the subcomplex $\text{Tot}(\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p} \Omega^\bullet)$ factors through a map of complexes

$$\gamma_p : \text{Tot}(\sigma_{\leq p} \Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p} \Omega^\bullet) \longrightarrow \Omega^\bullet$$

Using the same procedure as in Section 4 we obtain relative cup products

$$Rf_* \sigma_{\leq p} \Omega^\bullet \otimes_{\mathcal{O}_S}^L Rf_* \sigma_{\geq n-p} \Omega^\bullet \longrightarrow Rf_* \Omega^\bullet$$

We will prove by induction on p that these cup products via t induce perfect pairings between $Rf_* \sigma_{\leq p} \Omega^\bullet$ and $Rf_* \sigma_{\geq n-p} \Omega^\bullet[2n]$. For $p = n$ this is the assertion of the proposition.

The base case is $p = 0$. In this case we have

$$Rf_* \sigma_{\leq p} \Omega^\bullet = Rf_* \mathcal{O}_X \quad \text{and} \quad Rf_* \sigma_{\geq n-p} \Omega^\bullet[2n] = Rf_*(\Omega^n[-n])[2n] = Rf_* \Omega^n[n]$$

In this case we simply obtain the pairing between $Rf_* \mathcal{O}_X$ and $Rf_* \Omega^n[n]$ of Lemma 24.3 and the result is true.

Induction step. Say we know the result is true for p . Then we consider the distinguished triangle

$$\Omega^{p+1}[-p-1] \rightarrow \sigma_{\leq p+1} \Omega^\bullet \rightarrow \sigma_{\leq p} \Omega^\bullet \rightarrow \Omega^{p+1}[-p]$$

and the distinguished triangle

$$\sigma_{\geq n-p} \Omega^\bullet \rightarrow \sigma_{\geq n-p-1} \Omega^\bullet \rightarrow \Omega^{n-p-1}[-n+p+1] \rightarrow (\sigma_{\geq n-p} \Omega^\bullet)[1]$$

Observe that both are distinguished triangles in the homotopy category of complexes of sheaves of $f^{-1}\mathcal{O}_S$ -modules; in particular the maps $\sigma_{\leq p} \Omega^\bullet \rightarrow \Omega^{p+1}[-p]$ and $\Omega^{n-p-1}[-n+p+1] \rightarrow (\sigma_{\geq n-p} \Omega^\bullet)[1]$ are given by actual maps of complexes, namely using the differential $\Omega^p \rightarrow \Omega^{p+1}$ and the differential $\Omega^{n-p-1} \rightarrow \Omega^{n-p}$.

Consider the distinguished triangles associated gotten from these distinguished triangles by applying Rf_*

$$\begin{array}{ccc}
 Rf_*\sigma_{\leq p}\Omega^\bullet & & Rf_*\sigma_{\geq n-p}\Omega^\bullet \\
 \downarrow a & & \uparrow a' \\
 Rf_*\Omega^{p+1}[-p-1] & & Rf_*\Omega^{n-p-1}[-n+p+1] \\
 \downarrow b & & \uparrow b' \\
 Rf_*\sigma_{\leq p+1}\Omega^\bullet & & Rf_*\sigma_{\geq n-p-1}\Omega^\bullet \\
 \downarrow c & & \uparrow c' \\
 Rf_*\sigma_{\leq p}\Omega^\bullet & & Rf_*\sigma_{\geq n-p}\Omega^\bullet \\
 \downarrow d & & \uparrow d' \\
 Rf_*\Omega^{p+1}[-p-1] & & Rf_*\Omega^{n-p-1}[-n+p+1]
 \end{array}$$

We will show below that the pairs (a, a') , (b, b') , (c, c') , and (d, d') are compatible with the given pairings. This means we obtain a map from the distinguished triangle on the left to the distinguished triangle obtained by applying $R\mathcal{H}om(-, \mathcal{O}_S)$ to the distinguished triangle on the right. By induction and Lemma 20.1 we know that the pairings constructed above between the complexes on the first, second, fourth, and fifth rows are perfect, i.e., determine isomorphisms after taking duals. By Derived Categories, Lemma ?? we conclude the pairing between the complexes in the middle row is perfect as desired.

Let $e : K \rightarrow K'$ and $e' : M' \rightarrow M$ be maps of objects of $D(\mathcal{O}_S)$ and let $K \otimes_{\mathcal{O}_S}^{\mathbf{L}} M \rightarrow \mathcal{O}_S$ and $K' \otimes_{\mathcal{O}_S}^{\mathbf{L}} M' \rightarrow \mathcal{O}_S$ be pairings. Then we say these pairings are compatible if the diagram

$$\begin{array}{ccc}
 K' \otimes_{\mathcal{O}_S}^{\mathbf{L}} M' & \xleftarrow{e \otimes 1} & K \otimes_{\mathcal{O}_S}^{\mathbf{L}} M' \\
 \downarrow & & \downarrow 1 \otimes e' \\
 \mathcal{O}_S & \xleftarrow{\quad} & K \otimes_{\mathcal{O}_S}^{\mathbf{L}} M
 \end{array}$$

commutes. This indeed means that the diagram

$$\begin{array}{ccc}
 K & \longrightarrow & R\mathcal{H}om(M, \mathcal{O}_S) \\
 e \downarrow & & \downarrow R\mathcal{H}om(e', -) \\
 K' & \longrightarrow & R\mathcal{H}om(M', \mathcal{O}_S)
 \end{array}$$

commutes and hence is sufficient for our purposes.

Let us prove this for the pair (c, c') . Here we observe simply that we have a commutative diagram

$$\begin{array}{ccc}
 \text{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p}\Omega^\bullet) & \xleftarrow{\quad} & \text{Tot}(\sigma_{\leq p+1}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p}\Omega^\bullet) \\
 \gamma_p \downarrow & & \downarrow \\
 \Omega^\bullet & \xleftarrow{\gamma_{p+1}} & \text{Tot}(\sigma_{\leq p+1}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p-1}\Omega^\bullet)
 \end{array}$$

By functoriality of the cup product we obtain commutativity of the desired diagram. Similarly for the pair (b, b') we use the commutative diagram

$$\begin{array}{ccc} \mathrm{Tot}(\sigma_{\leq p+1}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p-1}\Omega^\bullet) & \longleftarrow & \mathrm{Tot}(\Omega^{p+1}[-p-1] \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p-1}\Omega^\bullet) \\ \gamma_{p+1} \downarrow & & \downarrow \\ \Omega^\bullet & \xleftarrow{\quad \wedge \quad} & \Omega^{p+1}[-p-1] \otimes_{f^{-1}\mathcal{O}_S} \Omega^{n-p-1}[-n+p+1] \end{array}$$

For the pairs (d, d') and (a, a') we use the commutative diagram

$$\begin{array}{ccc} \Omega^{p+1}[-p] \otimes_{f^{-1}\mathcal{O}_S} \Omega^{n-p-1}[-n+p] & \longleftarrow & \mathrm{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \Omega^{n-p-1}[-n+p]) \\ \downarrow & & \downarrow \\ \Omega^\bullet & \xleftarrow{\quad} & \mathrm{Tot}(\sigma_{\leq p}\Omega^\bullet \otimes_{f^{-1}\mathcal{O}_S} \sigma_{\geq n-p}\Omega^\bullet) \end{array}$$

We omit the argument showing the uniqueness of t up to precomposing by multiplication by a unit in $H^0(X, \mathcal{O}_X)$. $\square$

25. Other chapters

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(86) Duality for Spaces	
(87) Formal Algebraic Spaces	

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